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2023 K-Innovation Partnership Program with Ghana

Establishment and Operation of Ghana Innovation & Research Commercialisation (GIRC) Centre (3rd Year)

Strengthening R&I Capacity and Cooperative
Partnership with Korea

2023 K-Innovation Partnership Program with Ghana

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Establishment and Operation of Ghana Innovation & Research Commercialisation (GIRC) Centre (3rd Year): Strengthening R&I Capacity and Cooperative Partnership with Korea

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Executive Summary

From 2021 to 2023, STEPI and MESTI jointly implemented the K-Innovation Partnership Program under the following title: “Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre”. The third year of the programme (2023) was aimed at exploring the cooperative partnership between Korea and Ghana in order to establish and operate the GIRC Centre.

To achieve the aforementioned objective, the project has been carried out in two parts: i) development of R&I capacity and ii) exploration of a cooperative partnership. First, the Capacity Development Workshop was held to strengthen the GIRC Centre’s operational capacities, which was then followed by the Field Study where participants had opportunities to visit STI-related agencies in Korea. In addition, the Cooperative Partnership Workshop was held to explore future partnerships between Korea and Ghana. As part of this workshop, the participants also attended the Field Meeting to discuss concrete measures for promoting cooperation between Korean STI agencies and their Ghanaian counterparts. During the project, five ‘Project Concept Notes’ were proposed for future cooperation between five potential Korean STI agencies and their Ghanaian counterparts, and concrete measures for enhancing cooperation between the Korean and Ghanaian partner agencies were discussed. Five project agendas proposed in the process of the project are expected to serve as a foundation for operating the GIRC Centre and building up the national innovation system in Ghana.

This project, which falls within the last year of the three-year programme (2021-2023), concludes this three-year-long effort. For the successful establishment and operation of the GIRC Centre, the project suggests the following initiatives. First, the governance structure of the GIRC Centre (‘GIA’ in the future) and the legal systems required to support its governance need to be mapped out and established as quickly as possible. For this purpose,

a so-called ‘Stakeholders’ Exchange Meeting’ needs to be formed and operated to discuss and implement the GIRC Centre’s governance system. Furthermore, political efforts need to be made through this channel to persuade the Ghanaian government, while legislative efforts are required to support the operation of the GIRC Centre once it has been established as an independent incorporated entity.

Second, successful operation of the GIRC Centre requires stable budgetary support. This implies the need to institutionalise funding from the National Research Fund (NRF) so as to financially support the establishment and operation of the GIRC Centre, which has already been approved by the National Assembly. To secure a stable supply of funds from the NRF for operation of the GIRC Centre, extra efforts need to be made to persuade the Ghanaian government of its necessity.

Lastly, the full-time employees who will practically run the GIRC Centre need to be secured and their competencies developed accordingly. Currently, MESTI has secured the physical infrastructure required to establish the GIRC Centre, including its construction site and online website; however, full-time employees, including the director of the centre responsible for its daily operation, have yet to be secured. Therefore, full-time employees including the director need to be immediately recruited, and their skills and capacities for building and running the centre’s operational programmes need to be developed and strengthened as soon as the centre’s governance system and funding mechanism are finalised.

In parallel with these initiatives, efforts need to be made to secure global funding for establishing and operating the GIRC Centre. First of all, ODA grants and concessional loans need to be secured through intensive efforts aimed at building close cooperative networks with various international development agencies and banks (e.g. UNDP, AfDB, World Bank) and/or ODA support groups in other countries (e.g. SGCI, DFID, GIZ, and KOICA).

Second, efforts need to be exerted to acquire the practical knowledge required to establish and operate the GIRC Centre. For example, the operational know-how of leading global STI agencies, such as the IIA in Israel, the TIA in South Africa and the NRF in Korea, needs to be learned through networking with these institutions. In addition, partners who can support

the successful operation of the centre need to be secured through networking with global U-R-I institutions such as the NST and SEWB in Korea.

Lastly, one of the most important factors in ensuring the successful operation of the centre is the active exploration of opportunities for cooperative partnerships with five tentative partners (e.g. STEPI, NRF, TIPA, BGTP, and JDCC) encountered in the third year of the project. As part of the project, five “Project Concept Notes” have been proposed and concrete cooperative measures between the centre and its cooperation partners have been discussed. As such, extensive partnership efforts need to be exerted so as not to nullify or waste the cooperative opportunities identified by the project.

This third-year project was launched with the Kick-off Meeting and Field Visit held in March 2023 in Accra, Ghana, followed shortly thereafter by the 1st Cooperative Partnership Workshop. From April to June, five tentative partnership institutions developed the “Project Concept Notes” and exchanged their views on them with their Ghanaian counterparts. Based on these exchanges of opinions, the 2nd Cooperative Partnership Workshop was held in July. Then, in August of the same year, the Ghanaian project team visited Korea to attend the Capacity Development Workshop and Field Study. During their stay in Korea, the Ghanaian team visited the potential partnership institutions and discussed possible cooperative measures with their Korean counterparts. From September to November, the project team developed the final report on the third-year project. Finally, in December, a wrap-up meeting was held online to conclude the project activities of the past three years.

The project team would like to offer special thanks to Dr. Kwaku Afriyie, Minister of METSI, who has provided extensive support for the team throughout the entire project. Our appreciation also goes to Mr. Oliver Boachie, a special advisor to the Minister of MESTI, and other experts who have visited Korea. It is our ardent wish that this project serve as a landmark achievement in establishing the GIRC Centre (‘GIA’ in the future) which will lead the future of Ghana.

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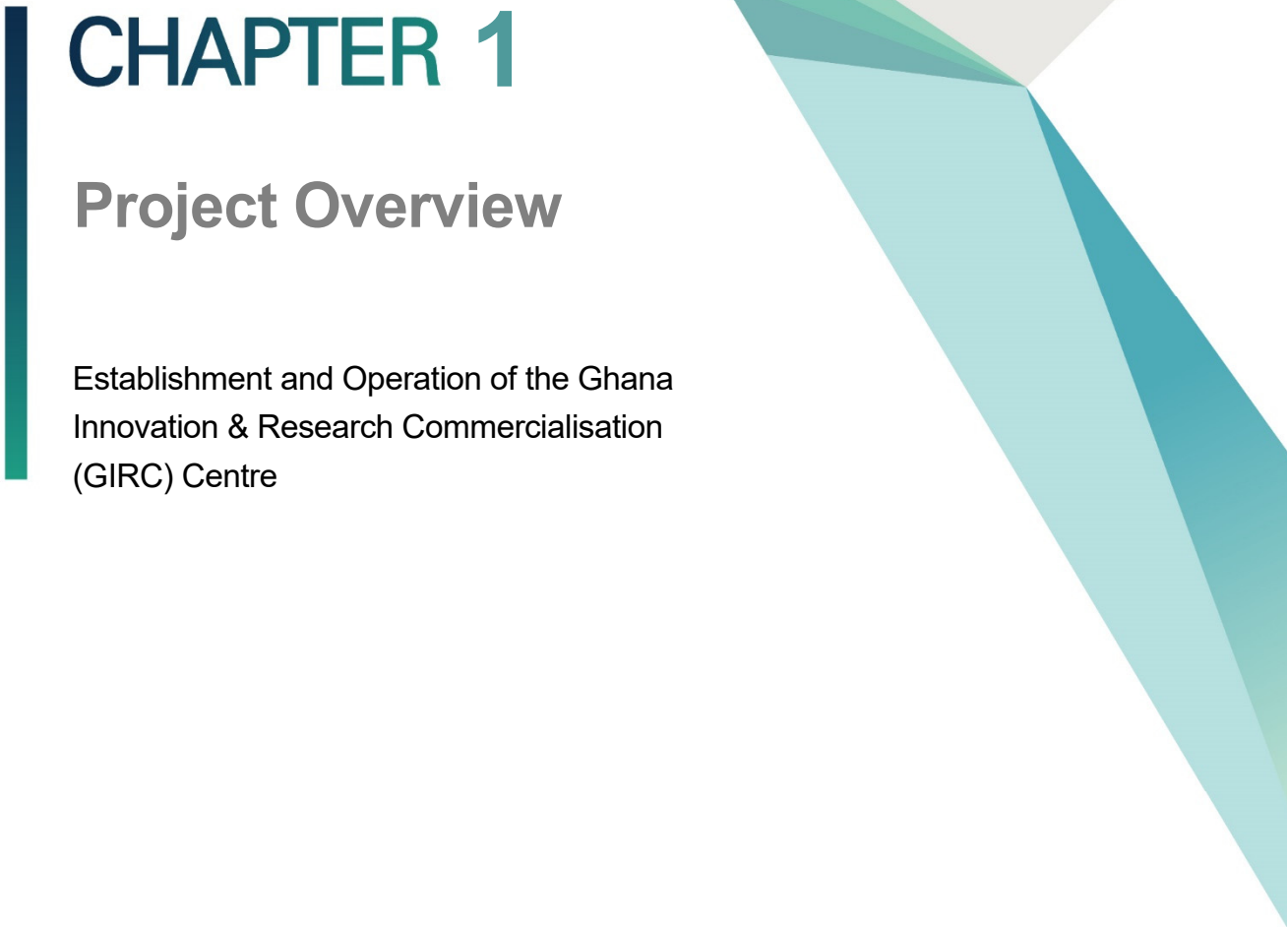
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CHAPTER 1

Project Overview

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre

Chapter 1. Project Overview

1. Project Background

As of 2023, Ghana is placed in the rank of the lower-middle-income countries (LMICs) with a per capita GDP of USD 2,002. From the inauguration of the new government in 2017 up until 2023, Ghana's economy maintained a CAGR of 3.9% (IMF, 2023)¹. The Nana Akufo-Addo government announced "Ghana, Beyond Aid" as its key national agenda, seeking economic growth without resorting to foreign aid while pursuing a vision of transforming the country from an LMIC into a UMIC (upper-middle-income country). As an STI framework to support its vision, the Ghanaian government has come up with seven components and is now pursuing STI-based sustainable development under a mid-to-long-term plan called the "Decade of Innovation".

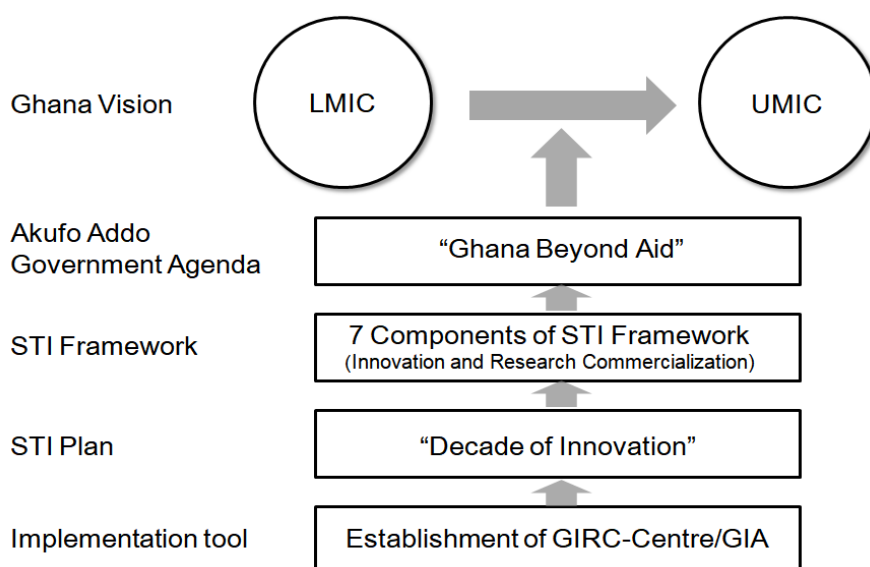
However, Ghana's capacity in certain key components of its national science, technology, and innovation (STI) system has long been extremely limited, for example, in such areas as university-industry-research institute cooperation, R&D, and technology transfer and commercialisation. To overcome such weaknesses, the Ministry of Environment, Science, Technology & Innovation (hereinafter referred to as 'MESTI') officially announced a plan to establish the Ghana Innovation & Research Commercialisation Centre (hereinafter referred to as 'GIRC Centre') in July 2018. In 2020, the Ghanaian government asked for help from the K-Innovation Partnership Program hosted by the Science and Technology Policy Institute (hereinafter referred to as 'STEPI') out of its aspiration to learn from Korea's development

¹ <https://www.imf.org/external/datamapper/profile/GHA> (Accessed Dec 18, 2023)

experience in the commercialisation of research innovations and to develop a master plan for the GIRC Centre.

Against this backdrop, STEPI and MESTI finally launched a three-year project (2021-2023) called the “Establishment and Operation of Ghana Innovation & Research Commercialisation (GIRC) Centre” based on the approval of the governments of the two republics.

[Figure 1-1] Akufo Addo Government's STI-led Growth



Source: STEPI & MESTI (2022), Master Plan for the GIRC Centre.

2. Project Objectives

The project was designed as a three-year programme with specific project objectives for each year. In the first year (2021), the programme focused on analysing Ghana's Technological Innovation Support Programmes (TISPs) and sharing South Korea's experience. In the second year (2022), the programme focused on developing a master plan for establishing and operating the GIRC Centre based on the analyses of the first year. Finally, in the third year (2023), the project team aimed to develop the capacities required for successful operation of the GIRC Centre and to explore cooperative partnerships between Korea and Ghana.

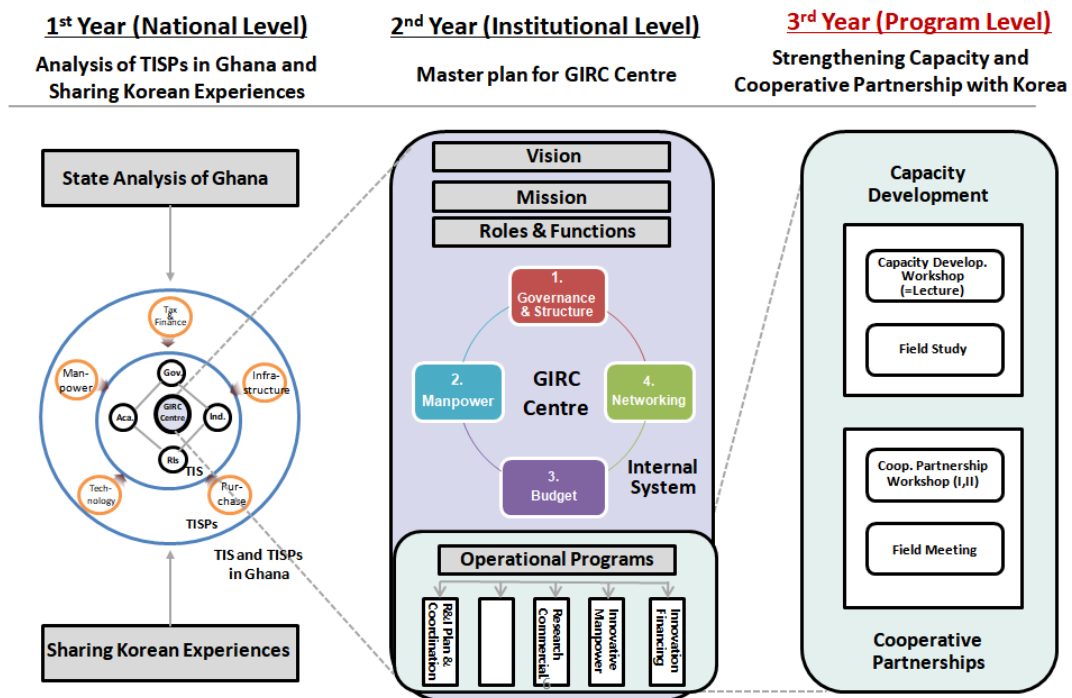
To be more specific, the 1st year of the project was focused on analysing Ghana's TISPs at the national level and sharing Korea's experiences, as shown in the Figure below. TISPs extensively cover the following nine themes: i) tax & finance, ii) manpower, iii) technology development and information, iv) technology transfer and commercialisation, v) start-up and incubation, vi) intellectual property management, vii) purchase and public procurement, viii) accreditation and certification, and ix) U-I-R (university-industry-research institute) linkage. The project team shared Korea's experiences in each of these areas and raised the respective implications for Ghana.

In 2022, the second year of the project, the project team developed a master plan for successfully establishing and operating the GIRC Centre at the institutional level, comprising three key components: strategic directions, internal systems, and operational programmes. First, as regards the strategic directions to be pursued by the GIRC Centre, the master plan presents the centre's vision, mission, roles, and functions. As for the centre's internal systems, the master plan reviews the direction for designing its governance structure and explores a series of measures for securing its budget, manpower, and networking. Regarding the centre's operational programmes, the master plan suggests programmes for enabling the centre to play the roles of an STI agency, an R&I secretariat, and a hub that facilitates the innovation ecosystem.

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In 2023, the third and last year of the project, the main focus consisted in strengthening the capacities of the personnel of the GIRC Centre so as to ensure the successful implementation of the master plan developed in the second year. In 2023, the project also concentrated on exploring cooperative partnerships between Korea and Ghana to enhance the centre's sustainability. As regards the capacity-building aspect of the centre, the Capacity Development Workshop was held in the form of lectures by inviting Ghanaian experts to Korea and providing them with opportunities to visit Korean partner institutions through the Field Study. In addition, Cooperative Partnership Workshops were held twice, and Field Meetings with Korean partner institutions were conducted to explore cooperative partnerships between the two countries.

[Figure 1-2] Analytical Framework by Year



3. Project Team

In the third year of the project, the collaboration between the Korean and Ghanaian teams, led respectively by STEPI and MESTI, continued to thrive. Dr. Wangdong Kim, a senior research fellow at STEPI, headed the Korean team, with Dr. Myung-Do Oh, ex-Vice-President of the University of Seoul, acting as a senior advisor. The Korean team included experts from significant STI-related agencies such as the Gyeongbuk Technopark (GBTP), the National Research Foundation of Korea (NRF), the Jeonbuk Digital Convergence Centre (JDCC), and the Korea Technology and Information Promotion Agency for SMEs (TIPA). On the Ghanaian side, MESTI played a crucial role, with Mr. Oliver Boachie, special advisor to the Minister, leading the project.

[Table 1-1] List of Korean Team Members

Name	Organisation	Position	Area of Expertise
Dr. Myung-Do Oh	University of Seoul	Former Vice-President	Senior Advisor
Dr. Wangdong Kim	STEPI	Senior Research Fellow	Project Management
Dr. Inkyoung Sun	STEPI	Research Fellow	Ghana-STEPI Cooperation Project
Dr. Yang Jin Kim	Gyeongbuk Technopark	Director	Ghana-GBTP Cooperation Project
Dr. Jong-Deok Kim	National Research Foundation of Korea (NRF)	Director	Ghana-NRF Cooperation Project
Ms. Hyun Young Yoon	Korea Technopark Association	Team Leader	Ghana-KTPA Cooperation Project
Dr. Sang-Ho Choi	Jeonbuk Technopark	Director	Ghana-JDCC Cooperation Project
Ms. May Cho	Korea Technology and Information Promotion Agency for SMEs (TIPA)	Team Leader	Ghana-TIPA Cooperation Project
Ms. Seohee Yoon	STEPI	Researcher	Coordinator

4. Project Activities

4.1 Activity 1: Preparation of the Project (January-March)

Following the successful introduction of the GIRC Centre's Master Plan in December 2022, MESTI broadened its efforts to spread awareness about the Master Plan both within the ministry and to the related stakeholders. The third year's focus shifted to building the capacity of GIRC Centre stakeholders in order to execute the operational programmes outlined in the Master Plan. STEPI forwarded the 2023 project proposal to Mr. Oliver Boachie at MESTI. With the easing of COVID-19 restrictions, the teams decided to conduct the "Kick-off Workshop" and "Field Interview" in Ghana in late March 2023. MESTI concurred with this direction for the third year, and both parties agreed to conduct these events onsite.

4.2 Activity 2: Kick-off Workshop, Korea-Ghana Science and Technology Innovation Cooperation Partnership Workshop and Field Interview (March)

In March 2023, the highly significant Kick-off Workshop and Korea-Ghana Science and Technology Innovation Cooperation Partnership Workshop were held in Ghana. This event included a presentation on the progress and outcomes of the 2021-2022 project and an introduction to and finalisation of the 2023 project plan. The Korea-Ghana STI Cooperation Partnership Workshop also explored potential areas for Korea-Ghana cooperation, introducing six prospective Korean developmental partner institutions, namely, STEPI, NRF, TIPa, KTPA, GBTP, and JBTP-CTC. Field interviews with Ghanaian partner institutions such as MESTI, the GIRC Centre, and CSIR were also conducted in an effort to grasp their current status and identify areas for institutional collaboration. High-level meetings with Ghanaian officials, including the Minister of MESTI, were held to gather opinions and discuss the direction and schedule of future projects, including the next phase of the STEPI-MESTI project for 2024-2026.

[Figure 1-3] Kick-off Workshop and Korea-Ghana STI Cooperation Partnership Workshop



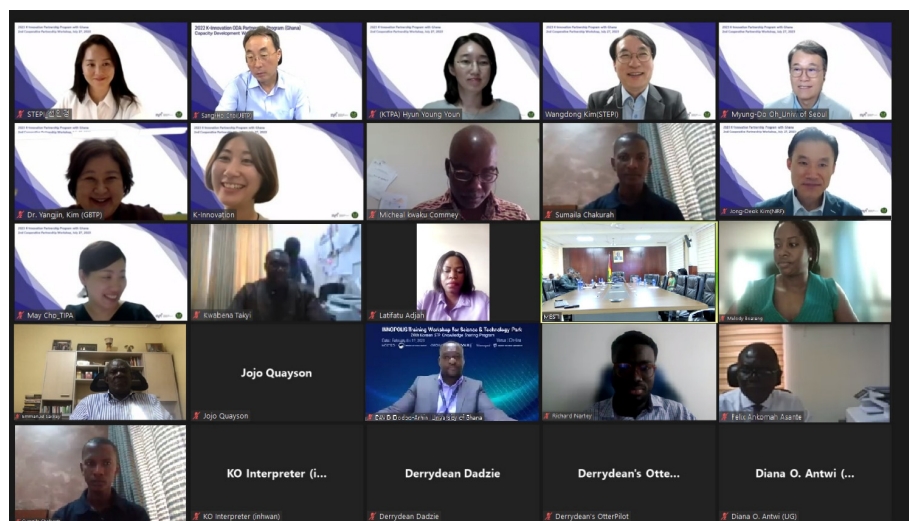
[Figure 1-4] Field Interview



4.3 Activity 3: The 2nd Korea-Ghana Science and Technology Innovation Cooperation Partnership Workshop (July, Online)

This workshop aimed to delve into specific themes and programmes for potential cooperation between Korea and Ghana, focusing on fortifying the collaborative network for future endeavours. The primary participants in this initiative were representatives from MESTI and other relevant sectors and institutions in Ghana, signifying a targeted approach towards enhancing bilateral relations in science and technology innovation.

[Figure 1-5] The 2nd Korea-Ghana Science and Technology Innovation Cooperation Partnership Workshop



4.4 Activity 4: Invitational Training Programme (September, Korea)

The objective of this initiative was to strengthen the skills of officials from Ghana's Environmental Science and Technology Innovation Ministry (MESTI) and the GIRC Centre, along with professionals from the industrial, academic and research sectors, with the focus on Korea-Ghana cooperation projects. The targeted participants were from MESTI and other relevant Ghanaian sectors and institutions. This comprehensive programme encompassed, among other things, ways of enhancing institutional capabilities and identifying future collaboration opportunities, visits to key Korean research and innovation

commercialisation entities, specialised lectures and workshops for operating the GIRC Centre, and the establishment of a network between Korean and Ghanaian experts in innovation and research commercialisation.

[Figure 1-6] Invitational Training Programme



2023 K-Innovation Partnership Program with Ghana



4.5 Activity 5: Writing the Final Report (October-November)

The objective was to create a draft report on the GIRC Centre's operational programme, using insights derived from both the capacity-building and Korea-Ghana cooperation workshops. This was achieved by refining the initial draft, incorporating feedback from

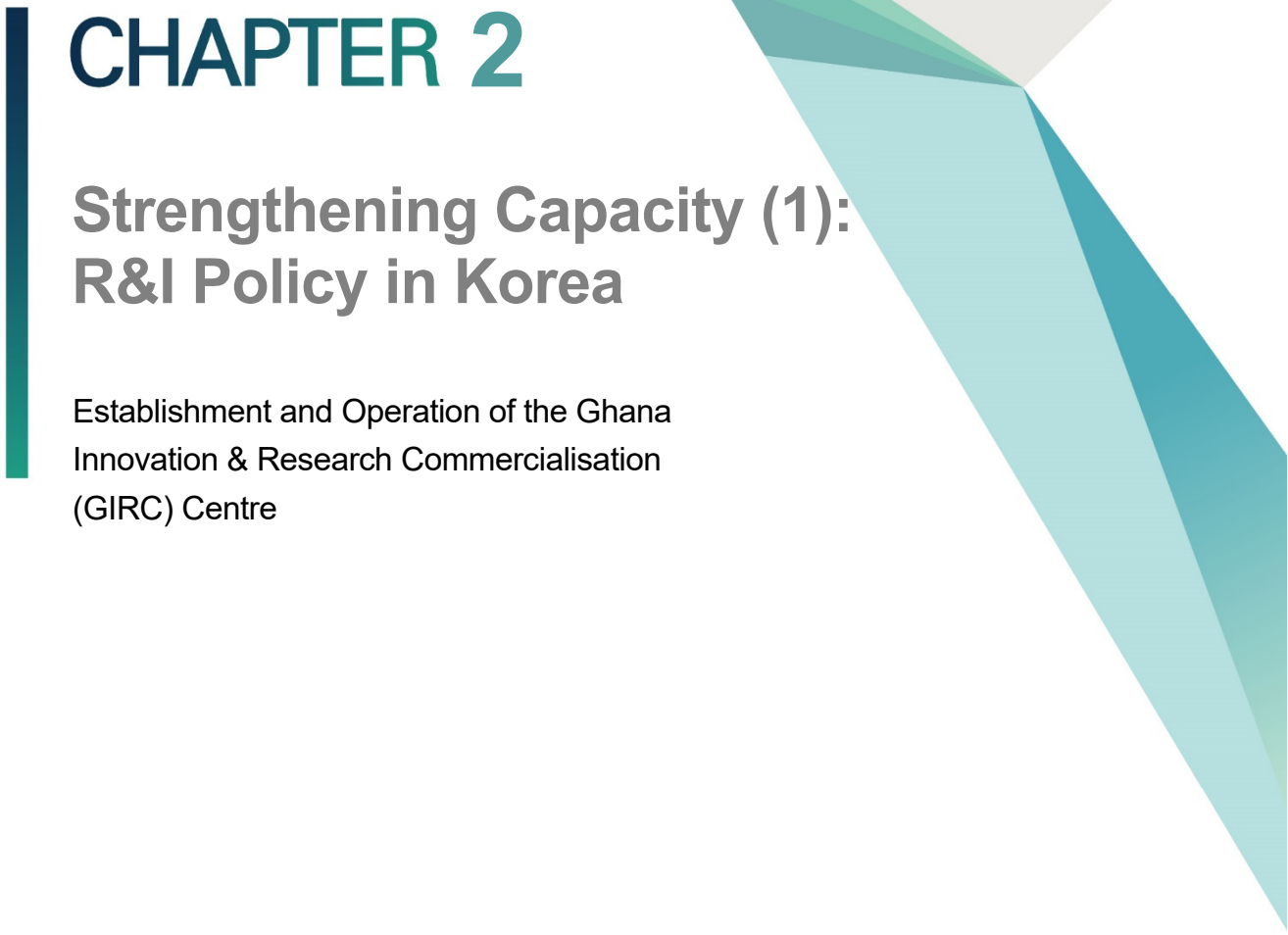
Ghanaian experts on the concept notes developed by Korean experts. This collaborative revision process ensured that the draft report effectively represented the shared perspectives and contributions from the specialists of both countries.

4.6 Activity 6: Dissemination Workshop (December, Online)

For the purpose of disseminating the policy consulting outcomes of this project, STEPI held the Dissemination Workshop virtually on December 13. The goal of the workshop was to disseminate the Korea-Ghana cooperation project plan, a key outcome of the third year, to stakeholders at MESTI, the GIRC Centre, and other relevant parties in Ghana. This approach aimed to ensure widespread awareness and engagement among all key stakeholders in Ghana.

4.7 Activity 7: Final Reporting

After incorporating the feedback from the Dissemination Workshop, STEPI planned to send an electronic version of the final report to MESTI by the end of the year. Following this, a printed copy of the report would also be provided, ensuring that MESTI received both digital and physical records of the project's conclusions and insights.

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CHAPTER 2

Strengthening Capacity (1): R&I Policy in Korea

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre

Chapter 2. Strengthening Capacity (1): R&I Policy in Korea

1. STI Development and Policy in Korea

1.1 Introduction

Korea is a small peninsula country in the Far East, hinging upon the Chinese continent, sharing a narrow border with Russia in the north, and surrounded by the Japanese archipelago in the south. Being in the neighborhood of three superpowers and geographically remote from the Western world, Korea's international interactions -- political, economic, cultural and others -- at one time seldom reached beyond the region. In a sense, therefore, Korea's history up until the end of World War II was more a reflection of the political, economic and military tensions and cooperative relations among the regional powers, rather than its own efforts to secure political stability and promote its economic development. Korea could only enjoy stability when the regional powers were on good terms, while any tensions among the big powers were translated into hardships for the Korean people. It is these geopolitical constraints that enabled Korea to remain a homogeneous society, both ethnically and culturally, and that preserved Korean society from external influences, including those of the Industrial Revolution among others. It took almost two centuries for the waves of the Industrial Revolution to reach the Korean peninsula. In addition to these geopolitical disadvantages, Korea, geographically speaking, is a small mountainous land, only one-quarter of which is suitable as arable land, and contains almost no economically meaningful natural resources.² Even the meagre industrial

² It was also the geo-economic disadvantages of the Korean peninsula that enabled Korea to survive for thousands of years as a distinct nation.

base that Korea had inherited from the Japanese colonial period (1909-1945) was almost totally destroyed during the Korean war (1950-53).

The above brief description is sufficient to explain how miserable the economic situation was back in the 1960s when Korea launched its drive for economic development. At that time it was one of the poorest economies in the world, with a per capita GDP of \$82(1960), relying on the primary sectors for more than 60% of its employment. A such, it suffered all the problems commonly found in the poor countries in those days. However, Korea differed from those poor countries in that it had a large pool of well-educated and highly motivated but underutilised workers with an average level of education as high as that expected of a country more than twice as rich as Korea then was.

It was in this context that Korea launched its drive for industrialisation in 1961. Since then, this drive for industrialisation has brought about an astonishing transformation of Korea from a poor agrarian society into a dynamic industrial economy within less than half a century. Today it is world's tenth largest economy with a per capita GDP of over US\$32,000 (Nominal value, 2021, UN Statistics Division, National Account) ³ . Underlying this phenomenal growth are South Korea's unparalleled efforts to promote science, technology and innovation, which can be explained in part by the resources Korea allocates for R&D – it is currently the fifth largest spender on R&D in the world and tops among the OECD countries in terms of R&D intensity (i.e. gross expenditures on R&D as a share of GDP, Korea: over 4.5%).⁴ The rapid growth of R&D investment has translated into the rapid growth of GDP and a dramatic transformation of the country's economic structure from a low-mid-technology-based one into a high-tech-based one. Indeed, no country matches Korea in its efforts to harness the potential of science and technology for economic development, and in this sense, one may refer to Korean' growth as "S&T-based" growth.

The questions one may ask about SouthKorea's remarkable growth include:

³ United Nations Department of Economic and Social Affairs(2023)

⁴ National Science & Technology Information Service, <https://www.ntis.go.kr> (Accessed 18 Dec, 2023)

- A. Why did Korea adopt an S&T-based growth strategy among many other possible options?
- B. What are the key features of Korea's S&T-based development strategy?
- C. What are the main social, economic and political factors that have made the strategy work?
- D. What are the policy implications of Korea's experience for other countries that aspire to promote S&T and innovation for social and economic development?

This note aims to provide answers to these questions by analysing South Korea's S&T policy efforts over the past five decades.

1.2 Why did South Korea adopt an S&T-based growth strategy?

National development strategy refers to a set of schemes, according to which the government and industries are to leverage the nation's endowments – both tangible and intangible - for social and economic development, under the given social, political and economic conditions. Different countries choose different strategies for development for the following reasons:

- 1) Because they are endowed with different resource that they can exploit for economic development;
- 2) Because they have different attitudes towards development and well-being, in other words, differences in motivation and desire for change and development; and/or
- 3) Because they differ in terms of their capacity to initiate, carry out and deal with changes in an effective and efficient manner.

South Korea's choice of S&T-based development strategy can be explained by the above logic. As mentioned in the previous section, Korea was one of the poorest economies in the world as late as the early 1960s, and thus the key challenge for Korea as a nation was how to develop the economy and liberate its people from chronic poverty. But in tackling that

challenge, Korea was bound by diverse constraints: First of all, Korea was extremely poor in natural resources and had a weak agricultural base, which excluded a resource-based strategy as an option. The only possible option, then, was industrialisation, for which capital and technologies are essential requirements. Many countries rely on foreign direct investment (FDI) for capital and technologies, but even that was not a feasible option for Korea back then because (1) the Korean people in the early 1960s had very strong feelings against foreign investors due to concerns about the possibility of being recolonised economically by foreign capital, and also because (2) foreign investors did not want to invest in Korea because of the political uncertainties and economic risks associated with investing in a country situated on an ideologically divided, cold-war frontier.

Korea could meet the capital requirement for industrialisation by borrowing from international financial institutions, but technology was not something that could be simply borrowed and utilised; rather, it required a process of absorption and assimilation before it could be applied to domestic industries. Fortunately, Korea had a large pool of well-trained human resources, which means that Korea also had a relatively strong technological absorptive capacity. So, Korea resorted to non-FDI channels for technology acquisition, such as turn-key plant importation, foreign licensing, OEM (Own Equipment Manufacturing) production arrangements, and so on. In addition, it could acquire technologies by learning how to operate the technologies brought in from foreign sources in the early stages of development. In the 1980s, Korea began to switch the mode of technology acquisition from learning to domestic R&D, and launched national R&D programmes for the development of key technologies required for the development of strategic industries such as electronic and electric technologies, machinery, etc. Since then, Korea has pursued a development strategy based on domestic “R&D and innovation”. Thus, “S&T-based development strategy” was not in fact a strategy that Korea opted for but rather one that was imposed upon Korea.

1.3 Korea's S&T policy for development: Key features of the S&T-based development strategy

1.3.1 The role of the government: Setting the stage for S&T-based growth

Innovation is not a natural phenomenon but a result of human interventions in the form of new ideas, research, development, and other factors that are costly in the economic sense. When people face competition, new economic opportunities, or external threats or have an internal desire for changes for the better, they seek ways of dealing with the situation, for which they need to have certain capabilities. In other words, innovation results from human efforts to deal with economic, social, military or other challenges. And innovation can flourish in an environment that is conducive to changes for the better. In sum, for innovation to take place within a given society or nation, it has to press for it, be capable of dealing with it, and keen to change.

Unfortunately, very often in many developing countries, efforts to promote innovation for development get nowhere because of the lack of either pressure for innovation or innovation capability, or a lack of both. Korea in the early 1960s was such a case. So, the government had to initiate the efforts by trying to create or generate demand for innovation while simultaneously building innovation capability, all of which required tremendous policy capacity. Therefore, one of the early measures taken by the government in this regard consisted in acquiring S&T policy capacity. The actions included the following:

- A. Construction of an S&T information system
- B. Acquisition of analytical capacity: It was necessary to build a policy research system in order to understand the nature of the policy issues – diagnosis and prescriptions.
- C. Acquisition of the capacity to formulate and implement policy measures: Creation of a government agency responsible for S&T policy.

Based on the policy capacity, the government played multiple roles in S&T in the early stage of development in Korea. The government generated demand for innovation through export-oriented industrial development programmes, while building S&T capacity by

creating government-funded R&D institutes, instituting new technical training systems, and increasing funds for R&D. Along with this, the government designed and offered various programmes designed to encourage and support industrial innovation activities.

1.3.2 Strategy for S&T-based development

Korea's strategy for ST-based development can be summarised as follows:

- A. Step-by-step approach: Korea started with learning by imitating foreign technologies, and gradually moved towards innovation based on indigenous R&D.
- B. Pragmatic approach: Korea's S&T policy has been strictly linked to industrial development and therefore has followed a kind of 'technology first, science later' approach.
- C. Heavy investment in domestic R&D: Instead of relying on foreign direct investment for technology, Korea chose to attain technological self-sufficiency by investing heavily in domestic R&D.
- D. Strategic focus on R&D investment: Being small in economic terms, Korea has pursued efficiency in R&D activities by concentrating R&D resources on strategic technology areas.

1.3.2.1 Step-by-step approach

Korea's growth has been a process of transition from learning to innovation: In the early stage (1960s-1970s), Korea learned by operating and imitating mature technologies brought in from foreign sources. It was a period of learning by simply duplicating foreign technologies, and therefore R&D was not at the heart of the S&T policy. The R&D activities in those days, if any, were geared to assimilating foreign technologies for local application.

The period of imitation was extended into the subsequent decades, but now the imitation was different in nature in that it was based on domestic R&D. During the 1980s to 1990s, Korea brought in newly emerging pre-commercial technologies, and then tried to create new value-added by conducting R&D to extend the technologies brought in from foreign

sources. As such, this period can be dubbed “a stage of creative imitation”. During this period, Korea tried to catch up with advanced technologies and transform its mid-low technology-based economy into one based on high technologies. It was also during this period that R&D investment was energised and began to grow rapidly.

Beginning in the 2000s, R&D became the major source of technology acquisition, and it has increasingly been geared to creative research. Since then, South Korea has emerged as a major player in the world of R&D, not just in terms of the amount of R&D expenditure but also the number of patents and scientific publications.

1.3.2.2 Pragmatic approach

Korea pursued a pragmatic approach to S&T development by strictly linking S&T to industrial development goals. For maximum utility of the S&T policy, the government identified target technologies based on the needs of the country’s industries and focused S&T policy resources on acquiring those technologies. In prioritising technologies, the government conducted inter-ministerial consultations as well as public-private collaboration. Reflecting this, more than 50% of the Korean government’s R&D is geared to economic development (2020), the highest figure among the OECD countries. Also, high-tech industries account for over 53% of GERD (gross expenditure on R&D) in Korea (2015), which again is the highest figure in the OECD, according to the NTIS.

1.3.2.3 Heavy investment in R&D

In order to promote R&D in both the private and the public sector, the government has been setting numerical targets for its R&D expenditures every five years since the mid-1970s. The first target set by the government was to raise the total R&D expenditure to 1% of GDP by 1976, which was far higher than what was actually achieved, i.e. 0.42%. Along with setting numerical targets, the government offered various policy incentives to promote and encourage private industrial investment in R&D and innovation, while simultaneously placing high priority on R&D in the allocation of the government budget. Over the past two

decades, however, S&T has been receiving 4-5.5% of the government budget: R&D had the fifth largest share (4.9%) of the government budget in 2022, after health and welfare (36%), education (16%), and defense (9.1%). The target for R&D expenditure for the year 2012 was set at 5.0% of GDP, while actual R&D expenditure rose to 4.12% in 2012, 4.3% in 2014, and over 4.5% in 2020, recording the highest growth rate of GERD in the world. Korea is now one of the leading countries in terms of both the amount of R&D expenditure and R&D intensity. Still, it should be noted that the share of foreign R&D funds in the GERD of Korea remains at only about 1%, the lowest figure among the OECD countries, according to data released by the NTIS. This anomaly results from the domestic R&D-oriented S&T strategy, and shows that Korea's R&D system is one of the least internationalised in the OECD.

1.3.2.4 Strategic allocation of R&D resources

Being a small country, Korea has been constrained by its limited resources, and thus has remained very focused on the efficient use of limited resources. To ensure an efficient use of limited resources, Korea has had no option but to focus its resources on the strategic field of technology, the selection of which has been dictated by industrial development goals. During the 1970s, Korea concentrated its resources on six strategic industries, including steel, shipbuilding, heavy machinery, electronics, and non-ferrous metals, all of which are capital and technology intensive. This contributed to Korea's structural transition towards an economy based on the heavy machinery and chemical industries.

Beginning in the 1990s, the priority was switched to the newly emerging high-technology areas, including ICT, bio-engineering, and others. In the early 1990s, the government launched "the Highly Advanced National Program" or HAN project, with the aim of facilitating another structural transformation of the economy. This programme was formulated and implemented through public-private collaboration and inter-ministerial consultation. The HAN project was a ten-year programme, focused on a set of priority technologies that included semi-conductors, ISDN, and so forth. In the mid-1990s, the Korean government launched the "informatisation drive" based on the "National

Informatization Promotion Plan,” directing the lion’s share of national resources into ICT R&D.

Korea chose the ICT industry as the leading sector for growth and development for the following reasons:

- A. Knowledge intensity: The ICT industry is more knowledge-intensive and requires fewer resources, and therefore is more favorable for Korea, which is poor in natural resources but relatively rich in human resources.
- B. Technological opportunity: ICT offers an opportunity for Korea to “leap-frog” technologically.
- C. Potential for application: ICT offers great potential for diverse industrial applications.
- D. Social and economic potential: ICT has enormous potential to enhance productivity and efficiency in public services, industries, and other areas.

For effective implementation of this strategy, the Korean government concentrated its R&D resources on ICT, and the private sector followed suit by investing heavily in ICT as a way to take advantage of the new business opportunities offered by the government’s plan. Since the launch of this strategy, Korean industries have been spending about 40% of their R&D funds on ICT, and over 20% of the government’s R&D budget has been allocated to ICT. In pursuing ICT development, the government also has been taking a pragmatic approach, linking ICT R&D to social and economic development plans and programmes. Korea’s R&D has been and still is extremely biased in favor of hardware, to which more than 90% of ICT R&D has been devoted. In particular, Korea’s ICT R&D has been very much focused on such key technologies as semiconductors, ISDN, mobile communication systems, optics, multimedia devices, etc. Thus, about 72% of business expenditures on R&D is allocated to computers, electronics, optical products, electric equipment, machinery, motor vehicles, and other transport equipment. This represents a remarkable concentration of R&D resources that is rarely observed in other OECD countries (OECD ANBERD data).

1.4 Scientific and technological achievements

1.4.1 S&T and economic achievements

First of all, Korea has grown to become the fifth largest investor in R&D in the world – well ahead of traditional leaders such as France, the United Kingdom, and other advanced nations. In terms of R&D intensity, Korea leads the OECD countries, suggesting that Korea spends a larger share of its GDP on R&D than any other OECD country. The growth of R&D expenditure has translated into the growth of S&T outputs, i.e. the number of scientific publications and patents. From seemingly out of nowhere, Korea has grown to become, in the field of S&T, the ninth largest producer of scientific publications,⁵ and the third largest producer of US patents, trailing only the US and Japan (Chung, 2023).

[Table 2-1] GERD of major countries, 2020

Rank	Country	Amount (Billion US\$)
1	United States	720.9
2	China	582.8
3	Japan	174.1
4	Germany	143.4
5	Korea	112.9
6	France	74.6
7	United Kingdom	56.0

Source: Sargent, J. F. Jr. (2022); OECD(2017)

Korea's achievement has been more visible in the areas of ICT. The OECD (2017) analysis of patents in ICT-related technology shows that Korea belongs to the top three countries in terms of overall patenting performance in the area of ICT. This analysis indicates that Korea excels in such areas as information communication devices, imaging and sound technology, human interface, mobile communication, and large capacity and high-speed storage. These

⁵ <https://nces.nsf.gov/pubs/nsb20214/publication-output-by-country-region-or-economy-and-scientific-field> (Accessed Dec 18, 2023)

achievements have made it possible for Korea to make a remarkable transformation from a poor agrarian society into one of the world's most dynamic industrial economies. Korea today is a major trading country exporting such high-technology commodities as semiconductors, automobiles, ships and ocean structures, mobile communication devices, petroleum products, flat displays, auto parts, synthetic resins, steel plate and plastic products. These technological achievements have contributed to Korea's transformation into a high-tech-based dynamic economy from a very impoverished and stagnant agrarian society. In addition, in terms of GDP per capita, Korea has surpassed the EU average and risen near to the level of Japan.

[Table 2-2] Shares of the key global players in ICT-related IP5 patents: 2012-15, %

Technology areas	JPN	KOR	USA	CHN	TWN	DEU
Information communication device	33.5	21.0	11.9	9.8	11.5	N/A
Imaging and sound technology	37.3	19.1	13.2	9.0	7.7	N/A
High-speed network	19.3	13.6	23.3	13.7	6.6	N/A
Human interface	22.2	23.8	14.6	12.3	14.7	N/A
Mobile communication	15.9	18.5	1.8	17.2	N/A	N/A
Large-capacity information analysis	20.2	10.9	33.0	7.0	N/A	5.2
High-speed computing	22.4	11.9	32.6	7.9	8.5	N/A
Large-capacity high-speed storage	28.4	25.8	21.9	4.5	9.3	N/A
Security	17.4	10.6	28.0	9.0	N/A	5.5
Cognition and meaning understanding	28.9	14.9	20.8	7.8	6.1	N/A
Others	36.3	10.9	17.8	6.2	19.0	N/A
Electronic measurement	19.6	10.3	25.0	N/A	N/A	14.7
Sensor and device network	20.7	12.7	21.1	N/A	9.0	10.1

Source: OECD (2017)

1.4.2 Key factors responsible for Korea's achievements

There are diverse factors that have contributed to Korean S&T and economic achievements, of which the following are the most noteworthy:

- A. The patronage and support for S&T and development of the Korean people, well-educated with a strong motivation and desire for change for the better.
- B. Political leadership that could translate the motivations and desires of the people into energy for development.
- C. The government's policy capacity to develop and implement a strategy to effectively leverage the potential of S&T and innovation for social and economic development.
- D. Geopolitical tensions that have kept Korea and its people constantly alert – a sense of urgency about the need to become stronger.
- E. The homogeneity of Korean society, which has made it easier and less costly to reach a national consensus on the direction of national development.
- F. The international environment which, during the period of the early stages of Korea's development (1960s and 1970s), was generous to developing countries in trade and technology transfer.

1.4.3 The state of the national system of innovation of Korea and the challenges lying ahead

The key features of the current system of innovation are an outgrowth from the strategies of the past. As a consequence of the technology policy based on domestic R&D, Korea's reliance on foreign R&D resources is lowest among the OECD countries, which is one of the weaknesses of the Korean system of innovation.

Second, the government deserves great credit for all that it has done for the development of S&T and innovation, although it should be pointed out that the Korean government has failed in adjusting its role in the innovation system in response to the changing policy

environments, which means that the dominance of the government still persists and hinders spontaneous and active interactions among innovation players.

Third, Korea's pragmatic approach emphasising close linkages between S&T and industrial development still persists, leaving fundamental sciences and creative research more or less unattended. Even though the pragmatic policy was initially very effective in boosting industrialisation in the process of economic development, it now works as an obstacle to Korea's transition towards a bona fide innovation-based economy. Another fallout of the pragmatic policy is the relative weakness of universities, whose role cannot be overemphasised for Korea's further growth.

Fourth, Korea's innovation system is ahead of that of other countries in terms of human and financial resources, at least quantitatively. But Korea's innovation potential is heavily constrained by such factors as low levels of trust, a vertical social order, risk aversion, a regulatory system that is unfriendly to innovation, and the dominance of the government, etc., all of which determine the relational capital of Korean society. Lastly, as mentioned earlier, Korea is one of the least internationalised countries in terms of research and development, which causes Korea to forego the diverse opportunities that international interactions have to offer.

1.4.4 Lessons for others

There is no doubt whatsoever that the Korean experience is a real-life testimony to how R&D and innovation can contribute to national development. The lessons and policy suggestions that the Korean experience offers can be summarised as the following:

- A. Korea's S&T-based growth strategy has been successful because it has been (1) consistently linked to industrialisation and national development goals; (2) blessed with the strong support of both political leaders and the general public; and (3) blessed with high-quality human resources.
- B. Korea has been able to sustain a high growth rate of R&D investment largely due to

(1) the support of taxpayers for the vision for national development and the role of S&T in realising that vision; (2) the policy actions and measures for promoting and encouraging private R&D and innovation; (3) the export-based development strategy that exposed domestic enterprises to international competition; and (4) a “domestic R&D-oriented” strategy that stresses internal R&D for technology acquisition rather than relying on foreign sources, such as FDI.

- C. Government intervention may be highly effective in promoting R&D and innovation in the early stages of development, but to sustain that level of effectiveness, the role of government has to be adjusted to changes in the policy environment.
- D. The S&T-based growth strategy might have worked better with more international interactions for R&D and innovation.

There is no such thing as the best innovation system. A national innovation system inevitably evolves within the national context, which is itself determined by various factors, including culture, history, political systems, natural environments, and others that affect human behavior. South Korea, as a country poor in natural resources but relatively rich in human resources, has pursued an S&T-based strategy, which has been successful thanks to its human resources and the way they have been used.

2. STI Governance

2.1 STI Committees

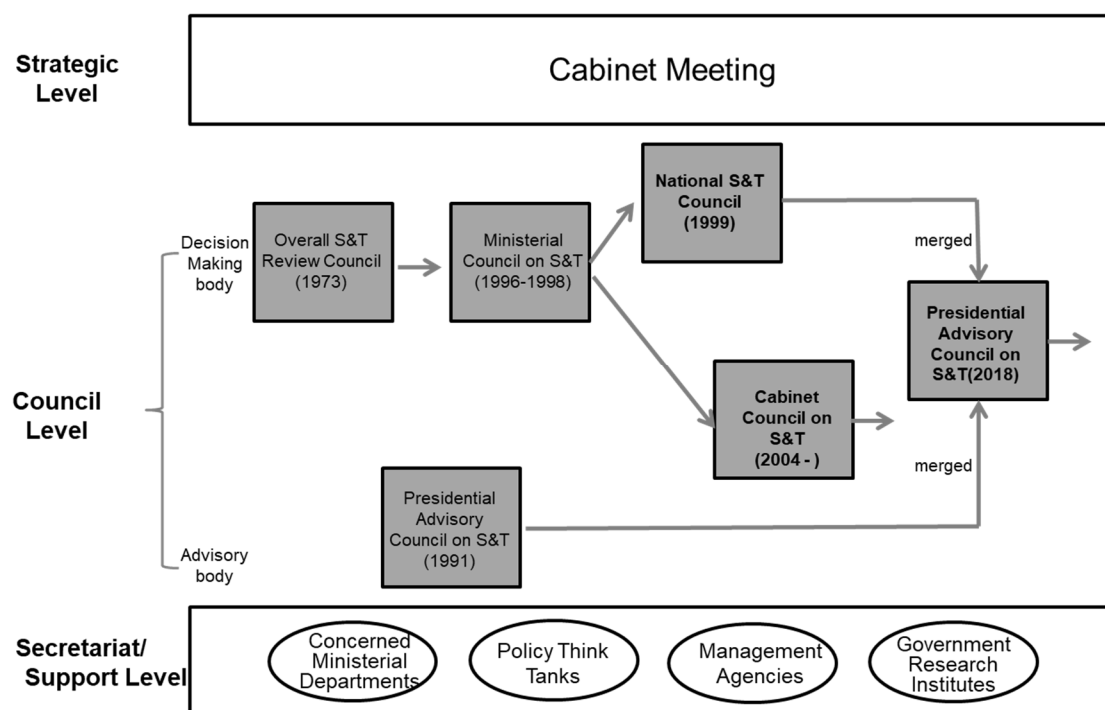
South Korea's STI governance is often characterised by STI-related committees representing three different hierarchical levels. At the uppermost strategic level is the Cabinet Meeting, which is presided over by the President of the Republic of Korea. Then, there is a slew of councils at the middle level, which are assisted by the group of secretariats and supporting organisations positioned at the bottom level. These three levels are closely interconnected, affecting STI policy planning and decision-making.

The history of the first Korean STI committee dates back to the early 1970s. The Korean government, led by late former president Park, Jung-hee, established the Overall S&T Review Council in 1973 to effectively implement policies for promoting the heavy and chemical industries. The council coordinated STI policies and projects by appointing the country's prime minister as its chair and the deputy prime minister for economic affairs as its vice-chair. The members of the council also included related ministers and experts from the private sector. After twenty years of active operation, this council was reshuffled into the Ministerial Council on S&T in 1996, presided over by the Minister of Finance and Economy. One of the key missions of the renewed council represented by STI-related ministries was to coordinate STI policies and projects.

As the scope of the National R&D Programme initiated in the early 1980s started to expand to involve multiple ministries, there rose a need for a council at the highest level capable of inter-ministerial coordination. To meet such needs, the National S&T Council (NSTC) was launched in 1999 as the highest-ranked STI-related decision-making body and remained operative until 2018. In addition, the Cabinet Council on S&T (CCST) was formed in 1996 to effectively address working-level issues among S&T-related ministries. Aside from these decision-making bodies, the Presidential Advisory Council on S&T (PACST) was established in 1991. Since then, the council has faithfully fulfilled its roles as the presidential advisory

body. Finally, in 2018, the decision-making and advisory functions of these councils were merged into the Presidential Advisory Council on S&T.

[Figure 2-1] Evolutionary Path of STI Councils



Source: Revised from Kim (2016)

The following are the key characteristics of the aforementioned three representative councils. In principle, PACST is an advisory body, whereas the NSTC and CCST are decision-making bodies. Since 2018, however, PACST has effectively carried out the NSTC's deliberation and decision-making functions by merging these functions.

A closer look into the functions of the three councils reveals their differences. While the NSTC served as the highest-ranked decision-making body presided over by the national president, CCST was hosted by the S&T Minister. Further, whereas the NSTC was focused on long-term macro-level policies, the CCST was committed to addressing short-term micro-level issues causing inter-ministerial conflicts.

In terms of their organisational composition, a key difference between the NSTC and the CCST was the chairmanship, as the latter was chaired by the S&T Minister, whereas the NSTC was presided over by the national president. However, they shared a similarity in that both councils included government officials and private sector experts as members. Before 2018, however, PACST differentiated itself from the NSTC and the CCST in that its members did not include any government officials, although it did include experts from more diverse backgrounds such as industries, academia, and research institutes.

2.2 STI Ministries

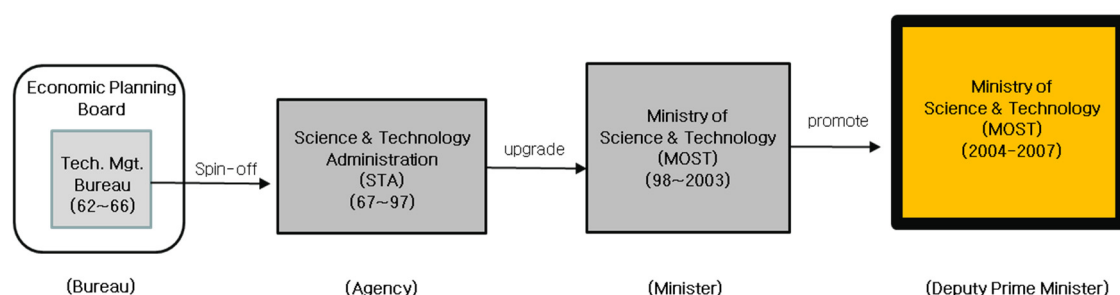
In Korea, the first STI ministries were established in the 1960s. In the early 1960s, the Korean government considered science and technology as a subordinate function of economic policy, and thus established the Technology Management Bureau under the umbrella of the Economic Planning Board (EPB). From the mid-1960s, however, Korea started to recognise STI as a critical means of economic growth and therefore established the Science and Technology Administration (STA) as a government body charged with coordinating such functions. The STA, which laid the foundations for Korea's S&T development, became the first government ministry in the world to be dedicated to S&T-related matters.

In the 1980s, the intensification of trade barriers and tightened controls over intellectual property rights by the developed countries highlighted the importance of independent S&T development. During the early period after its inception in 1982, the National R&D Programme was mainly led by the STA. Then, the programme started to involve other ministries like the Ministry of Commerce, Industry and Energy (MOCIE) and the Ministry of Information and Communication (MIC), which in turn increased the need for STI policy coordination. As a result, the Korean government promoted the STA to the Ministry of Science and Technology (MOST) in 1988 and established the National Science and Technology Committee (NSTC) as a government body to coordinate the ministries responsible for conducting STI-related R&D.

In recognition of the increased importance of S&T in the 2000s, the Korean government went even further and promoted the Minister of Science and Technology to the level of deputy prime minister. This implies that the status of STI policies was raised higher than that of other policies, and that MOST was able to secure a leadership position among STI-related ministries. The increasing number of national R&D programmes carried out by STI-related ministries also highlighted the importance of coordination and control. Against such a backdrop, the Korean government launched the Office of Science and Technology Innovation (OSTI) under the Ministry of Science and Technology and made OSTI function as an NSTC secretariat.

Afterwards, successive governments devoted their efforts to strengthening STI, which led to various changes in policy governance as well as experiments. In its first year in 2008, the Lee Myung-bak administration launched the Ministry of Education, Science, and Technology by merging the Ministry of Education with the Ministry of Science and Technology. Then, in its first year in 2013, the Park Geun-hye administration created the Ministry of Science, ICT, and Future Planning by merging the Ministry of Science and Technology with the Ministry of Information and Communication and designated the newly merged ministry as a key organ for leading national development. Under the subsequent Moon Jae-in administration inaugurated in 2017, the Ministry of Science, ICT, and Future Planning was given its current title of Ministry of Science and ICT.

[Figure 2-2] Evolution of STI-related Ministries



Source: Revised from Kim (2017)

2.3 STI Agencies

STI agencies are the key actors that support various councils and government ministries. In general, government ministries are responsible for carrying out such functions as planning, implementing, and evaluating policies. In the process of implementing and evaluating policies, however, government ministries often face the challenge of inefficiencies in carrying out their responsibilities as a result of the increased scale of government programmes or projects. In response, the government established a number of management agencies under its supervision, from which it receives necessary support.

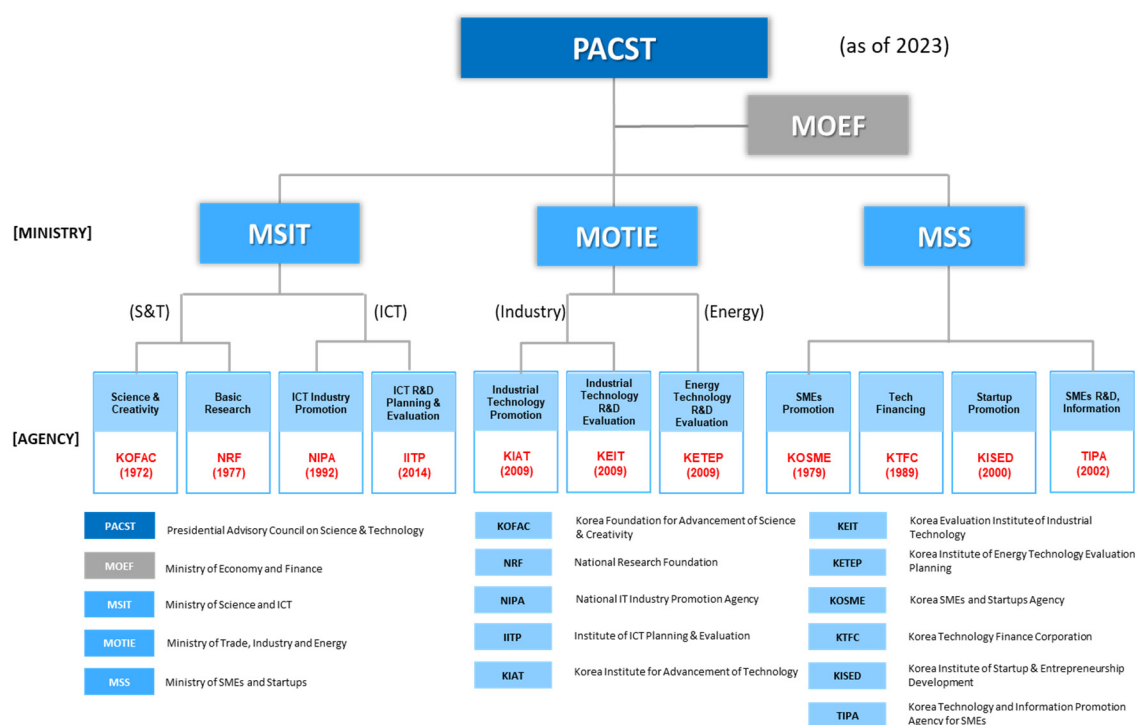
The Korean government is now operating various R&D planning, evaluation, and management agencies under the STI-related ministries. As of 2023, for example, the Ministry of Science and ICT (MSIT) is operating such agencies as the Korea Foundation for the Advancement of Science & Creativity (KOFAC), the National Research Foundation (NRF), the National IT Industry Promotion Agency (NIPA), and the Institute of Information & Communications Technology Planning & Evaluation (IITP). KOFAC is responsible for promoting science and technology, while the NRF is in charge of supporting basic research. Meanwhile, NIPA is dedicated to promoting the ICT industry, whereas IITP is devoted to R&D planning and evaluation in ICT.

Likewise, the Ministry of Trade, Industry, and Energy (MOTIE) also operates supporting agencies, including the Korea Institute for the Advancement of Technology (KIAT), the Korea Planning & Evaluation Institute of Industrial Technology (KEIT), and the Korea Institute of Energy Technology Evaluation and Planning (KETEP). KIAT is committed to promoting industrial technology, while KEIT is in charge of R&D evaluation and KETEP supports the research and development of energy technology.

Under the Ministry of SMEs and Startups (MSS), there also exist various agencies like the Korea SMEs and Startups Agency (KOSME), the Korea Technology Finance Corporation (KTFC), the Korea Institute of Startup & Entrepreneurship Development (KISED), and the Korea Technology and Information Promotion Agency for SMEs (TIPA), which are dedicated to supporting SMEs and startups.

The key roles of Korean STI agencies can be summarised as follows. First of all, STI agencies provide innovation actors (e.g. universities, companies, and research institutes) with diverse forms of support including subsidies, research funds, scholarships, and facilities and equipment. These agencies also play an important role in the so-called “PIMEF” (i.e. planning-implementation-monitoring-evaluation-feedback) cycle of the government R&D programme. In addition, STI agencies also support government officials of the relevant ministries or committee organisations by writing short-term policy reports or carrying out collaborative works.

[Figure 2-3] STI Agencies (as of 2023)



Source: Revised from Kim (2016)

2.4 STI Policy Think Tanks

STI policy think tanks constitute another important group of actors supporting the policies driven by various STI committees and government ministries. These policy think tanks play an important role in identifying the needs of the general public, conveying the identified needs to policymakers, providing them with a variety of policy suggestions, and disseminating the policy contents to the public. STI policy think tanks started to emerge to meet the increasing demand for expertise due to the increasingly complex nature of the policy environment resulting from massive economic development.

The history of Korean STI think tanks dates back to the 1970s. Ever since the Park Jung-hee government launched the first Five-Year Economic Development Plan in 1962 to revive the impoverished Korean economy, successive Korean governments have consistently pursued the implementation of five-year economic development plans. As the five-year cycle repeats itself over the years, however, the complexity of the plans has increased greatly, raising the need to secure expertise in economic policies and policy options. In the early stages, Korea tended to rely on foreign brains from the developed countries as an important source of expertise. However, the country soon ran up against the limitations of using foreign expertise in developing policies tailored to the Korean context, which increased the need to develop and secure internal expertise.

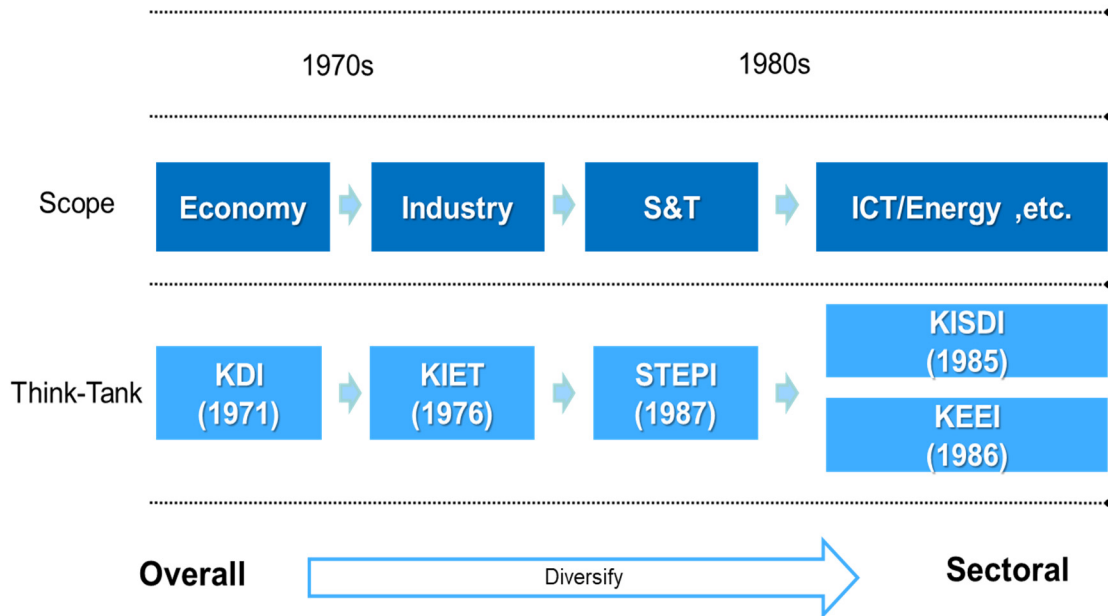
Recognising such needs, the Park Jung-hee administration established the Korea Development Institute (KDI), the first Korean policy think-tank, in 1971. The KDI made a significant contribution to the country's development by actively engaging in the development of the Fourth Five-Year Economic Development Plan and playing a key role in drafting subsequent plans. In 1973, when the Korean government announced a policy for promoting the heavy and chemical industries, the country needed industrial expertise and policy options across a wide range of industries. To meet such needs, the Korean government created the Korea Institute for Industrial Economics and Trade (KIET) in 1976, whose key mission was to support the government's industrial policies. In the early 1980s, the increasingly high levels of technology protectionism adopted by the developed

countries also highlighted the importance of independent technology development. Against this backdrop, the Korean government initiated the National R&D Programme for independent technology development. To better support such efforts, the Science and Technology Policy Institute (STEPI) was launched in 1987.

The research areas of Korean STI think tanks have steadily expanded over the years. Initially, resources were concentrated on research and studies on the economy under the lead of the KDI. Then, the scope of their policy research was further expanded to cover such areas as industry (e.g. KIET) and S&T (e.g. STEPI). More recently, certain industrial areas like ICT and energy were added to the research scope of the Korean STI think tanks. In summary, the research scope of Korean policy think tanks was not specified in their early stage, so they tended to carry out overall policy research functions. Later, however, the scope of their research was diversified to include certain industrial sectors and their sectoral research was strengthened.

Korean STI policy think tanks play a set of key roles. First of all, their most important role is to forecast future trends in the relevant fields and develop mid- and long-term plans accordingly. They also forecast the demand and supply of human resources in the relevant fields, reflect these forecasts in the policies, and carry out planning, management, and evaluation of national R&D programmes. In addition, they support various government ministries and committees. In this process, they create and disseminate an array of STI-related information and data.

[Figure 2-4] Evolution of STI Think Tanks



- KDI: Korea Development Institute
- KIET : Korea Institute for Industrial Economics and Trade
- STEPI: Science and Technology Policy Institute
- KISDI : Korea Information Society Development Institute
- KEEI : Korea Energy Economics Institute

Source: Kim (2016)

3. National R&D Programme

3.1 National S&T Management System

3.1.1 Process of National S&T Management

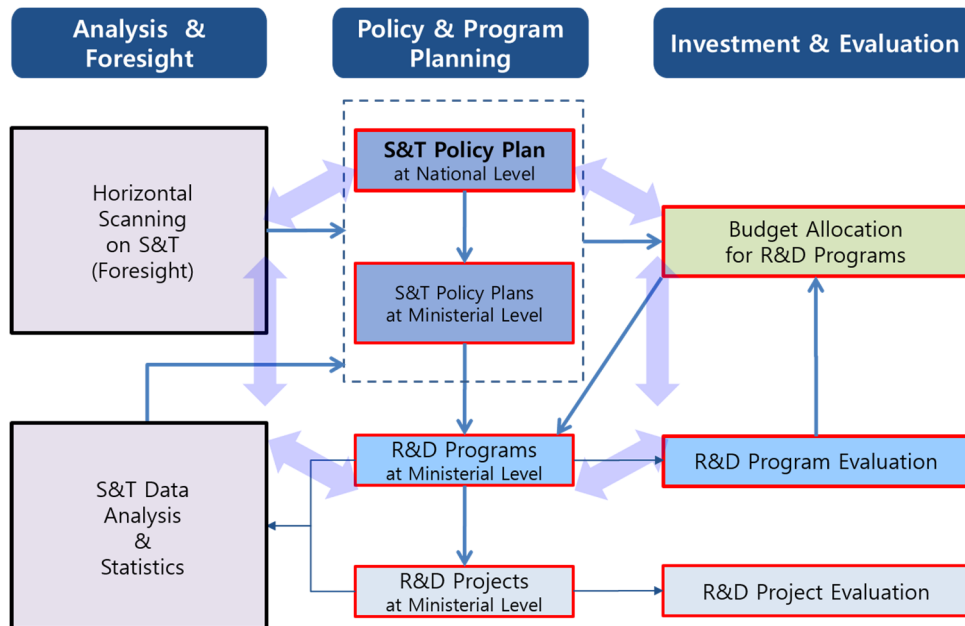
The overall process of National S&T Management is as follows.

Before planning S&T policy and R&D programmes, it is necessary to conduct analysis and foresight. This involves performing horizontal scanning of S&T to forecast which promising future technologies will be realised and analysing S&T data in order to understand our current position.

Using the results of such analysis and foresight, national-level S&T policy is planned first. Each ministry then plans S&T policy and R&D programmes at the ministerial level with the assistance of funding agencies and other experts. The funding agency plans the R&D projects that constitute the R&D programme and the related management system.

If it is feasible to launch the programme, it can be funded through budget allocation. Subsequently, evaluations are conducted at both the programme and project levels. Thereafter, the results of the evaluations of the R&D programmes provide feedback for allocation of the budget for the next year's R&D programme.

[Figure 2-5] The Overall Process of National S&T Management



Source: Kim (2016)

3.2 Coordination of National R&D Programmes

3.2.1 Key Features of Current R&D Investment

In 2021, the total national R&D investment in Korea, including both the government and the private sector, amounted to approximately US\$ 89.3 billion, ranking 5th in the world. National R&D investment as a proportion of GDP stood at 4.96%, ranking 2nd in the world.

The government's R&D budget for FY 2021 amounted to US\$ 23.8 billion, accounting for 4.9% of total government expenditure.

3.2.2 Historical Overview of R&D Investment and S&T Activities

In 1963, Korea invested a mere US\$ 1million in national R&D, whereas some thirty years later, in 1991, we invested US\$ 3.8 billion. Fast forward another thirty years, to 2021, and we invested US\$ 89.3 billion in national R&D. Such a rate of increase is indeed remarkable.

Now, let's look at the trend in the government R&D budget and major related activities.

The first government-funded research institute, KIST, was established in 1966. The first government R&D programme was launched in 1982, with an R&D budget of US\$ 192 million. In 2021, the government R&D budget reached US\$ 23.8 billion. As such, we can say that the Korean government has been focusing on R&D investment for our future.

3.2.3 Needs for Coordination on National R&D Programmes

As the government R&D budget has increased, the country has recorded remarkable S&T achievements, including scientific outputs, industrial products, and global contributions. However, Korea has also encountered unexpected side effects.

Managing an ever growing number of projects and programmes, and involving many ministries in government R&D programmes, has made efficiency and effectiveness in budget allocation far more critical. Furthermore, considering the specialisation in the area of science and technology for budgeting is crucial because disruptive innovation occurs more frequently.

In the early 2000s, government R&D investment continued to increase, leading to the diversification of R&D activities across diverse ministries and agencies. At that time, there were more than 200 R&D programmes involving 20 ministries and agencies. Consequently, there was a growing focus on the efficient and effective allocation of government R&D budgets, leading to the establishment of the NSTC (National S&T Council) in 1999. The NSTC was established to develop new initiatives for the evaluation and coordination of national R&D programmes.

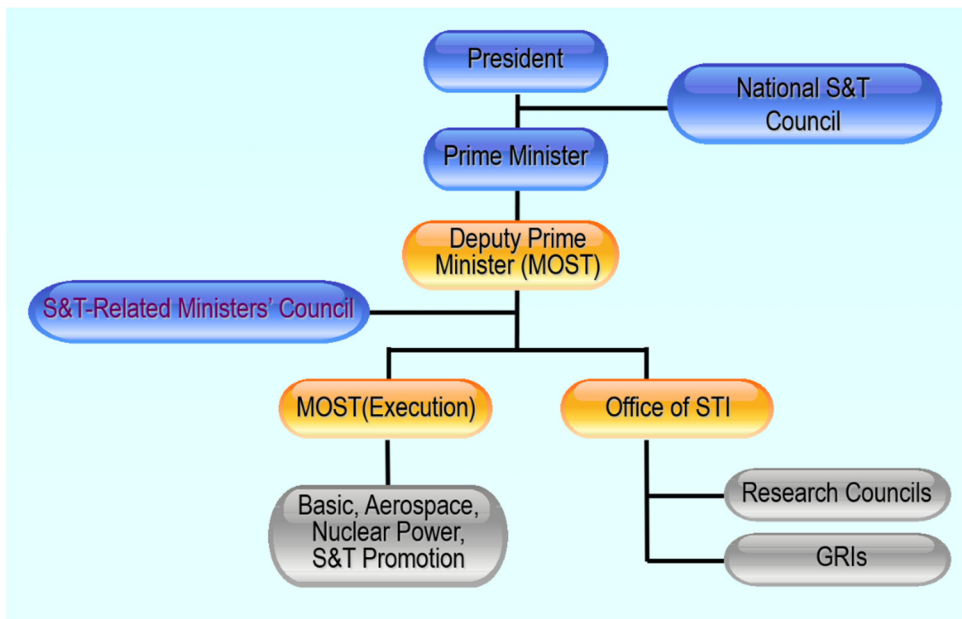
3.2.4 S&T Governance Structure

S&T governance was restructured in 2004. The status of the Minister of the Ministry of Science and Technology (MOST) was raised to that of Deputy Prime Minister. Additionally,

the Office of S&T Innovation (OSTI) was newly established under MOST, with the responsibility for evaluating and allocating the budget for all ministerial R&D programmes.

Established in 1999 and chaired by the President of Korea, the NSCT was composed of fourteen S&T-related cabinet ministers, the chairperson of the Presidential Advisory Council for Science and Technology (PACST), and nine non-government members appointed by the President. MOST served as the NSTC's secretariat. The Korea Institute of Science and Technology Evaluation and Planning (KISTEP), also established in 1999, played a role in national-level S&T planning, assisting in the evaluation and coordination of national R&D programmes, among other responsibilities. Therefore, one of the main missions of KISTEP is to support the NSTC.

[Figure 2-6] S&T Governance Structure in 2004



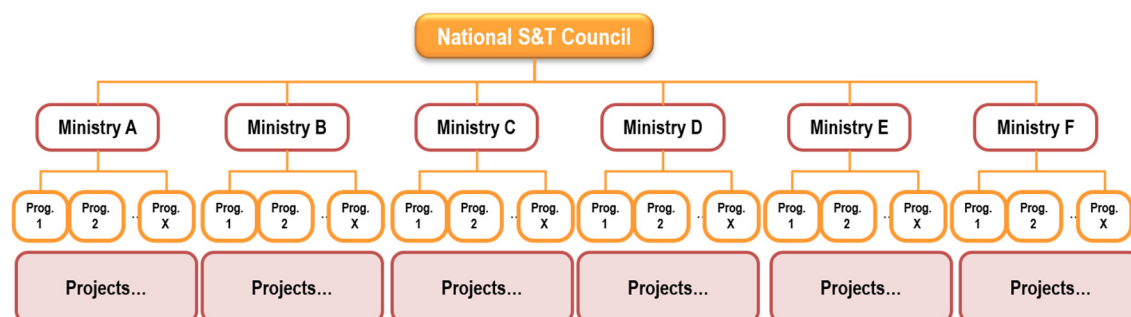
Source: Hwang(2023)

3.2.5 Coordination of R&D Programmes

In 1999, the NSTC was in charge of programme coordination, while various ministries were responsible for managing the R&D programmes, and R&D funding agencies were in charge

of project execution. Meanwhile, research institutes, universities, and enterprises performed the R&D projects.

[Figure 2-7] Entities Responsible for Managing R&D Programmes



Source: Hwang(2023)

3.3 National R&D Management

3.3.1 Government R&D Budget Allocation

The general budgeting process is as follows.

At the beginning of each year, the Ministry of Economy and Finance (MOEF), responsible for budget management, establishes a budget ceiling for each ministry. All ministries submit their budget proposals for the next year to MOEF, which in turn reviews and allocates budgets for programmes across all ministries, and then submits the budget bill to the National Assembly. This constitutes the general budget process.

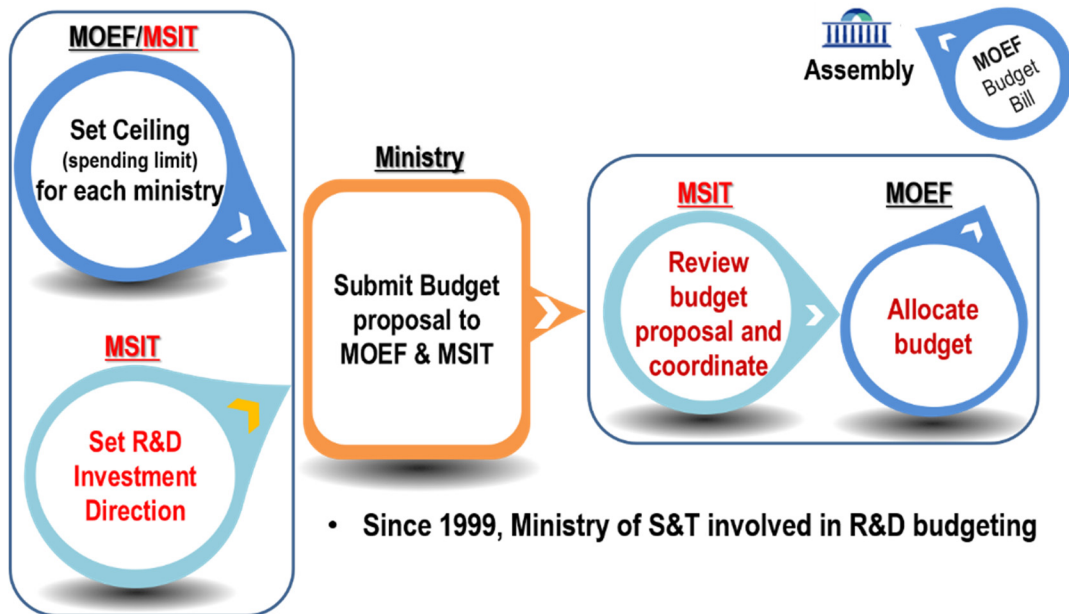
However, the R&D budget flow differs from that of other sectors so as to accommodate the specialisation of S&T. Since 1999, the Ministry of Science and ICT (MSIT) has been involved in R&D budgeting.

At the start of the year, MOEF, with assistance from MSIT, sets the budget ceiling for each ministry, while MSIT establishes the direction of R&D investment for the upcoming year. All ministries submit their R&D budget proposals for the next year to MOEF and MSIT, which then review them. MSIT distributes and coordinates the budget for the next year's principal R&D programmes and submits the results to MOEF. Then, MOEF allocates the budget for all

R&D programmes, including the principal R&D programmes, and submits the budget bill to the National Assembly. The R&D budget for the next year is finally deliberated and confirmed by the National Assembly.

Government R&D programmes consist of principal and general programs. Principal R&D programmes account for over 80% of all R&D programmes and include basic research, applied research, development research, and construction of large-scale facilities and equipment, etc. General R&D programmes encompass policy research, and the operating expenses of research institutes and universities among other things.

[Figure 2-8] Government R&D Budget Flow



Source: Hwang(2023)

We have been operating Expert Committees within the NSTC to facilitate more scientific, rational decision-making.

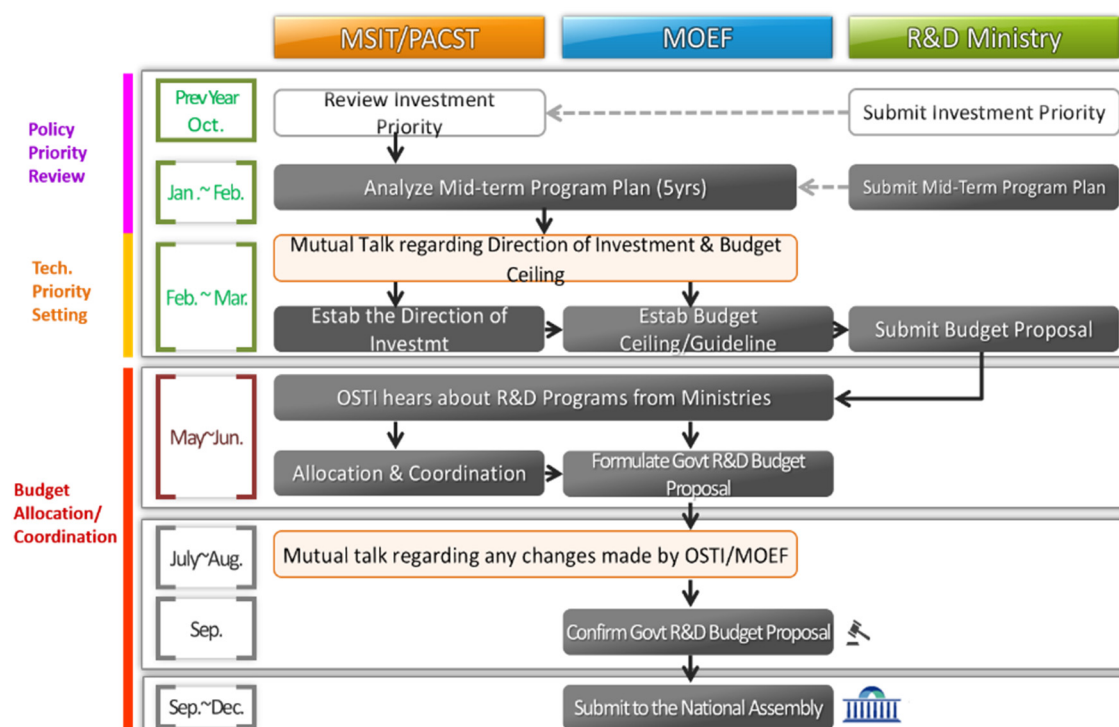
The major role of the Expert Committees is to produce review reports of budget requests from ministries that perform R&D. These reports are used primarily to assess the validity of each programme, the clarity of its goals, the adequacy of the requested budget amount, and so forth.

There are seven Expert Committees, each composed of sixteen experts. Since 2004, many experts from industry have participated in the committees, serving as a bridge between the government and the private sector.

During the R&D budget process, the Korean government also considers the Science and Technology Basic Plan, which outlines the fundamental direction of the national science and technology policy for the next five years. Additionally, to incorporate technological trends and changes, MSIT presents the next year's Investment Direction and Technology Priority in March each year.

Figure 2-9 illustrates the budget flow process of R&D programmes in Korea, providing a time schedule and outlining the roles of various bodies in the allocation of the next year's R&D budget, as explained earlier.

[Figure 2-9] Government R&D Budget Flow at a Glance

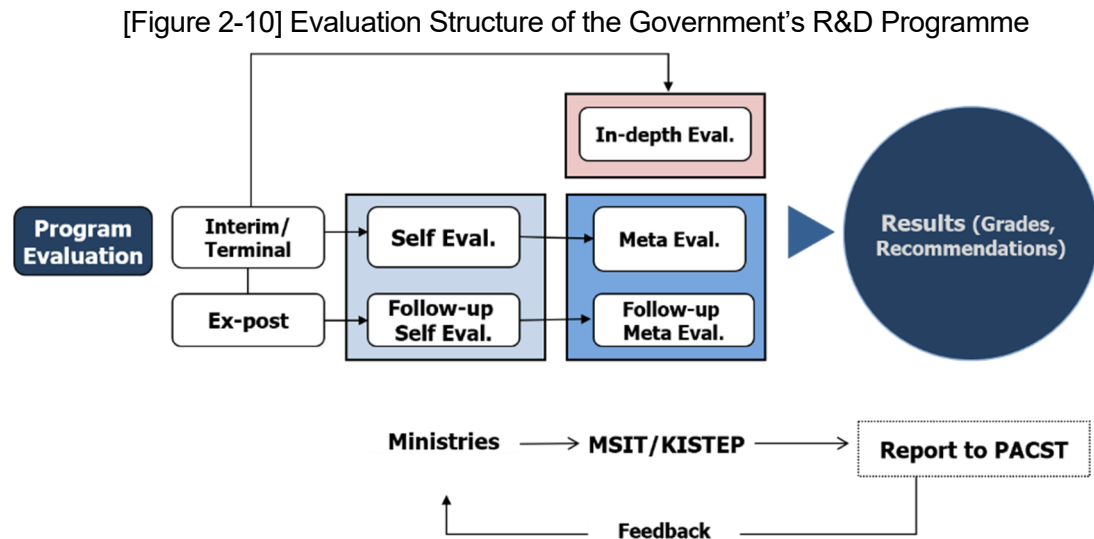


Source: Hwang(2023)

3.3.2 Government R&D Programme Evaluation

Figure 2-10 shows a schematic diagram of the R&D evaluation structure. Regarding the lifecycle of R&D programmes, the types of evaluation can be classified into interim, terminal, and ex-post evaluation.

First, line ministries conduct a self-evaluation and report the results to MSIT and KISTEP. Then, MSIT and KISTEP conduct a meta-evaluation and finalise the results with specific scores or grades. In contrast to this self- and meta-evaluation, an in-depth programme evaluation is directly performed by MSIT and KISTEP, the results of which are reported to the NSTC.



Source: Hwang (2023).

In-depth evaluation is a crucial step in the process, and one of its most important aspects is the selection of the programmes to be evaluated. The selection criteria for in-depth evaluation include long-term R&D programmes with large budgets, R&D programmes suspected of duplicating each other, and R&D programmes that are of national concern.

There are no specific guidelines for in-depth evaluation because the issues to be addressed can vary from programme to programme. Nevertheless, an in-depth evaluation generally

follows four steps. After selecting the programmes for evaluation, the project manager (PM) establishes an evaluation strategy, including a logic model, identification of evaluation issues, a data collection plan, and methodologies. Then, the PM collects data from the ministries responsible for running the programmes and analyses and interprets the data. Next, the PM reviews and consolidates the results of the previous interim and final evaluations, and concludes the evaluation report after considering the opinions of the ministries involved. Finally, the results of the evaluation can be used to improve the programmes, so the ministry formulates a programme management action plan based on the evaluation results.

4. NST and Government Research Institutes

4.1 Evolution of Korean Government-funded Research Institutes

4.1.1 Creation and Evolution of Government-funded Research Institutes in Korea

Public sector research organisations (PROs) in Korea fall into three types: government-funded research institutes (GRIs), national/local research institutes (N/LRIs), and non-profit research institutes (NPRIs). PROs accounted for around 26% of total national R&D investment as of 2022. Among them, GRIs accounted for nearly 20 %, while N/LRIs and NPRIs accounted for around 3% each. Among all PROs, GRIs have been the core and major R&D performers in Korea.

NPRIs are PROs sponsored by the central government owned/managed companies or by majorly Ministry of Trade, Industry and Energy (MOTIE) for regional innovation across nation. They were established under the Act for Catalysing Innovation of Industrial Technology, and, currently, fourteen NPRIs are being operated in the fields of ICT, robotics, machinery, automobile, shipbuilding, dye and textiles, etc.

N/LRIs are part of the central government departments or local governments based on the relevant laws and regulations of local governments. The first ones were established between 1910 and 1945, during the period of Japanese colonial rule, and they continued to be formed until there were seventy-two of them by the early 1960s. Although the number of researchers and staff came to less than 2,000, they served as the main public research organisations until the mid-1970s. Of the total government expenditure on R&D, N/LRIs accounted for 82.1% in 1967 and 59.3% in 1975. N/LRIs conducted testing, investigation, and research in the fields of agriculture, engineering, medicine and physical science. However, they were characterised by their weak R&D capability, small size, poor facilities, and heavily bureaucratic culture. Consequently, the Korean government decided to establish highly qualified public research organisations to support industry, policy, energy

and defense, etc. That is why the Korea Institute of Science and Technology (KIST) was established in 1966 as the first GRI with the support of US President Lyndon Johnson.

After the creation of KIST as a non-profit private contractual research institute operated by grant-in-aid and research contracts with government and industry, GRIs proliferated throughout the 1970s and 1980s with the support and supervision of related departments and budgetary control by the central government until 1998. In 1999, five Research Councils were established and most GRIs became members of those research councils. The Framework Act accords every GRI the status of an independent legal entity with institutional distinctiveness. The Board of each Research Council appoints a president with a three-year term to each GRI after an open competition. Each president works full-time and is responsible for managing a GRI. As of 2023, twenty-five GRIs belong to the National Research Council of Science and Technology (NST), which was created as the only research council for these GRIs through a merger of the research councils in 2014.

4.1.2 Historical Characteristics of the GRI Regime

KIST played an important role as the backbone of Korea's Innovation model. KIST was established pursuant to the KIST Assistance Act as an independent legal entity and guaranteed an annual endowment from the government to be used for its basic operations and as capital expenditure for its construction. The objectives, strategy, governance, funding and management style of KIST were very different from those of the N/LRIs and formed a significant part of Korea's innovation model, thereby strengthening national R&D and innovation capacity in Korea. KIST led Korea's early stage of technological innovation and made a huge contribution to solving problems during the process of adapting and applying imported technologies. KIST's main objective was to develop technologies and policies for industrial development with strong support from the then President Park Jung-Hee. KIST was characterised by a powerful president- and project manager-centred governance, and was managed based on contractual funding from both the government and industry, with a high degree of autonomous management and high salaries.

In the 1970s as Korea's labor-intensive industries were gradually replaced by the heavy and chemical industries, S&T infrastructure capable of modifying and internalising foreign technology was essential to meet the demand for the greater technological sophistication required by the machinery, steel, chemical engineering, shipbuilding and electronics industries. A number of GRIs were established in the 1970s in such fields as standards, chemicals, machinery, shipbuilding, marine science, electronics, and electricity in order to satisfy various industrial needs - mostly in Daedeok Innopolis which began to develop in 1973. Those GRIs were similar to the KIST regime in the sense that they developed specialised technologies for each industry, conducted more and more national R&D programmes, favoured project-manager centred governance, and had a dual funding system comprising basic institutional funding from the government and contractual funding mainly from industry. The main principle of GRI management was autonomy and stable financial support from the government.

Until 1980, the number of GRIs expanded and duplicated their role with increasing inefficiency. Then, the government discarded the principle of autonomy and introduced strong regulatory guidelines for the GRIs' organisational management and a more bureaucratic culture aimed at weakening the power of the project managers.

Gross domestic expenditure on R&D increased by 0.58% in 1980 to 1.91% in 1990, with the annual rate of increase averaging 30.8%. The proportion of R&D performed by GRIs declined by 53.5% of GERD in 1981 to 19.4% in 1990 due to the growth of the industrial sector and the higher education sector. By 1990, industry accounted for twenty times as much R&D activity as it had in 1981, while higher education sector accounted for nine times more, and the GRIs only four times as much. During the 1980s, R&D activities spread rapidly from the government sector, including GRIs, to the industry and higher education sectors. The strong regulations of the central government departments were not working effectively in the 1980s, which is why the GRIs experienced significant regime changes throughout the 1990s. The system for evaluating organisational management and research performance was set in 1991. In 1996, a new project base system was introduced, comprising contracts for

projects based on full cost pricing, including the cost of staff wages and operational costs, while the dual funding system was abolished in order to give more responsibility for R&D activities to project managers with greater transparency and better performance with regard to R&D.

Another crucial change to the GRI regime came about in the 1990s with the creation of five research councils and the abolition of the GRIs' boards of directors. GRIs in the field of natural science and engineering were regrouped into three research councils. Each GRI had the characteristic features of different legal entities, but they all had the same type of board of directors. In addition, the GRI regime in the 1990s was significantly different to how it had been in the 1980s: They were forced to contribute more to industry-led innovation and achieve greater performance in national R&D programmes; to perform more collaborative R&D with industry and universities; and to adapt to the newly introduced project base system and the system of evaluation by research councils.

Throughout the 2000s and 2010s, the GRIs evolved further due to changes in the research councils, which were merged from three into two and, ultimately, into just one research council called the National Research Council of Science and Technology (NST). As of 2022, there were twenty-five GRIs under the NST, with an the annual budget of US\$ 5 billion and 23,000 personnel.

4.2 Role of the National Research Council of Science & Technology

4.2.1 Historical development of the NST

The history of the NST began on 15 March 1999 with the establishment of three research councils categorised into fundamental, public, and industrial technologies pursuant to promulgation of the Act on the Establishment, Operation and Fostering of Government-Funded Research Institutions. The three research councils were the Korea Research Council of Fundamental Science and Technology, the Korea Research Council of Public Science and Technology, and the Korea Research Council of Industrial Science and Technology, all of

which were supervised by the Prime Minister's Office. In 2004, these Research Councils were moved to the Ministry of Science and Technology to strengthen the more scientific management than PMO.

Among the three research councils, the public research council was merged into the other two research councils in 2008, and these two councils were moved into new government departments at the Ministry of Education, Science and Technology and the Ministry of Knowledge Economy. Meanwhile, the supervisory office of the two research councils became the new Ministry of Science, ICT, and Future Planning in 2013.

The two-research-council system resulted in less collaborative relations between GRIs amid an era characterised by a shift towards technological convergence. That is why the government replaced the two-research-council system with the National Research Council of Science and Technology, which benefitted from strong support from the scientific and technological community.

4.2.2 Characteristics and Roles of the NST Regime

The main purpose of the NST is to support national research projects and policies and lead the development of the knowledge industry by supporting, fostering and managing GRIs systematically and promoting more collaboration between them.

The main functions of the NST are as follows: to plan research and policy for the development of the GRIs; to adjust and align their functions (including the establishment, integration and dissolution of institutes); to evaluate their research performance and management (including the management of corporations invested in by the GRIs); to support cooperative research between the GRIs and assist them in improving their research performance and disseminating the results; to propose policies for innovating national science and technology and strengthening its competitiveness; and to support the self-auditing of the GRIs.

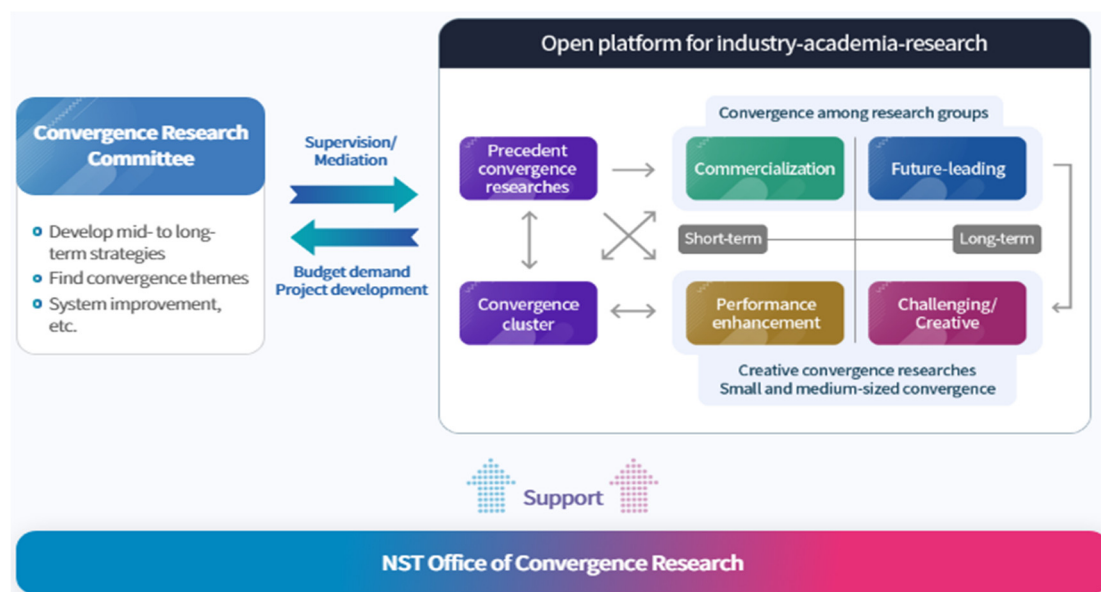
The NST established crucial operational strategies to make GRIs open collaborative platform, as follows:

1. Stimulate convergence research between the GRIs.
2. Plan a better direction of management for the GRIs.
3. Assist the GRIs with enhancing their research performance and disseminating the results.
4. Resolve common difficulties faced by the GRIs.
5. Plan and conduct GRI-customised self-audits.

The NST Board is the highest decision-making body regarding the operation of the NST and the twenty-five GRIs and, as of 2022, consisted of up to twenty directors, with seven fixed vice-ministers from seven central government departments and eleven part-time positions from industry, academia, and research institutes.

The crucial mission of the NST is to build a convergence research platform with a specially planned budget of approximately US\$ 100 million, managed by the NST's Office of Convergence Research.

[Figure 2-11] Implementation System of the NST's Convergence Research Platform



Source: NST Homepage⁶

⁶ <https://www.nst.re.kr/eng/contents.do?key=141#n> (Accessed 18 Dec, 2023)

Another important function of the NST is to disseminate and commercialise research results supporting technology licensing office in GRIs with more collaborative technology transfers, spin-off companies, and deep tech commercialisation.

[Figure 2-12] Implementation System of NST' Performance Dissemination



Source: NST Homepage⁷

In view of the rising trend towards regional innovation and globalisation, the NST has been compelled to contribute to the regional innovation ecosystems and globalisation since the mid-2000s. The majority of local governments and society have wanted more GRI branches in the cities and provinces. As of 2023, more than 100 local branches of GRIs are operating for technology-base regional innovation.

Amid the globalisation trend, twelve global branches for global research and innovation collaboration have been established.

⁷ <https://www.nst.re.kr/eng/contents.do?key=145> (Accessed 18 Dec, 2023)

4.3 Major Issues of the NST & GRIs

The impending issues surrounding the NST and GRIs are outlined as follows:

1. The NST requires more funding power and budget adjustment capacity to promote more open collaboration within the GRIs.
2. It is necessary to reform the project base system (PBS) with a larger rate of institutional funding .
3. More places and programmes are required for collaboration on research and innovation.
4. It is necessary to nurture world-class research groups within the GRIs in order to come up with original, world's best technologies.
5. Greater focus on future fundamental research and solutions to social problems is required.
6. A more open collaborative platform with universities and industries is required.

4.4 Characteristics of and Lessons Learned from the Korean Model of the NST and GRIs

The GRIs have gone through several distinct periods: The 1960s and 1970s, during which many evolutionary achievements were accumulated with a high level of autonomous management; the 1980s and 1990s, when the GRIs changed non-incrementally; and the 2000s-2020s, during which time the GRIs changed incrementally in line with the PBS and the Research Council, NST.

It is important to consider together all five elements of the NST and GRI regime, i.e. objectives, strategies, governance, funding, and management, in order to catch the characteristic features and interactions with external factors, such as technological, economic, social and political changes.

The NST and GRI regime change has been affected by economic and industrial policy, institutions, new ideas, interests, and political preferences. Therefore, it is necessary to know about such factors in greater detail in order to better understand the NST and GRIs.

South Korea's GRIs played a major role in the development of technology and talented human resources even before the nation's industries and universities were built up. As industries and universities evolved into elements of the system and network, the GRIs' role was diminished, but they played a different role inside the innovation system. Establishing various innovation systems and building capacity are very important, especially to countries in the early stages of the innovation system.

The objectives and strategies of the GRIs have focused too much on economic performance and growth, despite the expansion of their scope to include R&D aimed at solving social problems.

The current governance of GRIs is overly dominated by bureaucracy through the guidelines on their roles and functions and the related laws. In addition, the Research Council (NST) has no funding power.

It is necessary to formulate a more stable and discretionary funding system for the GRIs to replace the current PBS, which unusually focuses on short-term performance and project management. It is also crucial to introduce more strategic management of R&D and stimulate not only long-term fundamental research but problem-solving research.

The main aim of establishing the NST was to strengthen convergent research between the GRIs and other entities. As such, more places and programmes are needed to create more opportunities to develop the convergent technology.

The regional expansion of GRIs' organisations could play a role for the convergence of knowledge, technology, industry, and society with different stage of knowledge production model and system.

5. KAIST, an STI Specialised University

5.1 Introduction

Today, KAIST has a well-established reputation as Korea's best university in science and technology, in other words, as "the MIT of South Korea". This reputation rests not only on its major contribution to producing science manpower for industrial sectors, but also on its internationally-recognized research competitiveness. The success story of KAIST since its establishment in 1971 is part and parcel of Korea's late industrialisation, often dubbed as the miracle on the Han River. However, a number of important questions remain to be answered, such as "How did KAIST carve its path toward academic excellence while serving the nation's economic development?" or "What form of science policy did the government provide - especially during its formative period?" In a nutshell, what does a STI-specialised university mean for a nation-building country in East Asia? This chapter will address these questions after giving a brief overview of some notable aspects of KAIST and the role of universities in society.

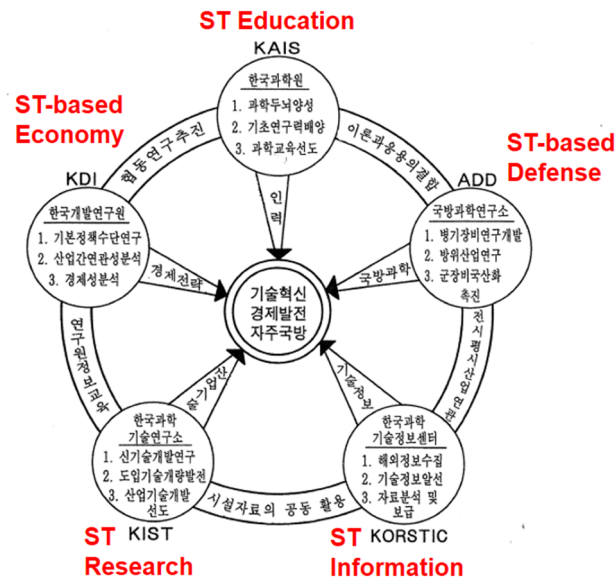
5.1.1 A Global Leader

KAIST is now collaborating with many other universities as a member of diverse academic networks in Asia and beyond, including ASPIRE (Asian Science and Technology Pioneering Institutes of Research and Education), the Association of East Asian Research Universities, APRU (including Stanford, UC Berkeley, U of Sydney, Peking U. and U of Tokyo), and the Members of the Global University Leaders Forum of the World Economic Forum. KAIST has formed global partnerships with 294 institutions from 65 countries, and runs dynamic student exchange programmes with 126 institutions in 33 countries. KAIST's overall position in terms of global leadership can be seen in a variety of world university rankings, i.e. 39th in 2020 according to the QS, with several departments ranked in the top 20, including the departments of electrical engineering, materials engineering, mechanical engineering, and chemistry.

5.1.2 A National STI Hub

From the start KAIST was conceived to serve as part of a national STI hub covering five key areas for development: education, economy, research, information, and defence. In the early 1970s the Korean government created an “R&D Park” in Seoul where five institutions would be located together to facilitate collaboration, including KAIST, which at that time was named KAIS, or the Korea Advanced Institute of Science. Their goals were three-fold: technological innovation, economic development, and self-defence. (See the diagram below for their functional correlations.) Later, KAIST was relocated to Daejeon, where more than two dozen government research institutes and a number of private research centres came to occupy a vast tract of land that had been set aside for R&D purposes. Since then, Daejeon has become widely known as “Science City”.

[Figure 2-13] Functional Relationships of the Seoul Research and Development Complex



Source: Park(2023)

5.1.3 Educating the nation's core manpower

The value of a STI-specialised university such as KAIST can be also seen in the places where its graduates end up working. Not surprisingly, a majority of KAIST degree holders (69,388

as of 2021) work in industry (45%), academia (31%), and government institutes (21%). It is worth noting here that nearly 25% of Samsung's R&D workforce are KAIST graduates, including top executives (9%). The same is true for other high-tech companies like SK hynix.

5.2 Why KAIST?

5.2.1 University and social reform

Since its emergence in twelfth-century Europe, the university has been an indispensable and robust institution in society due to its dual role of educating people and producing new knowledge. As such, the university has practical value in terms of maintaining the social order and responding to new challenges. The Middle Age universities came into being because of a growing need to train state administrators, the clergy, medical doctors, and commercial entrepreneurs. Law, medicine, and theology were studied for higher degrees, but to do that, basic teachings in language, mathematics, and philosophy had to be firmly implanted in undergraduate courses. While the text-based scholastic curriculum continued to prevail until the nineteenth century, university-trained people led the way in the Scientific Revolution (16-17 c.) and the Industrial Revolution (19-20 c.). Furthermore, new kinds of university emerged. The École Polytechnique, established in 1794 during the French Revolution, showed how the state could pursue sociopolitical reforms by completely changing the educational system. During Napoleon's empire period, the École Polytechnique functioned like a military academy with the famous motto of "For the homeland, science, and glory". Meanwhile, Berlin University (renamed later as Humboldt University), created in 1810 during the Prussian Reform Movement, heralded the ideal of the "research university" by emphasising original findings and theories in the humanities, sciences, and other disciplines. Undoubtedly, new and old universities formed an engine for the military, industrial, and economic expansion of Western countries.

5.2.2 The emergence of universities in East Asia

In East Asia, the importance of learning was, and still is, greatly emphasised. This is mainly due to the long tradition of running a merit-based administration with high-ranking bureaucrats selected through competitive examination. Yet it was not until the late nineteenth century that the university system began to emerge amid the need for social reform in response to the expansion of Western imperialism, which was manifested in the Opium War (1839-42) in China and the opening of Japan by American “Black Ships” (1853). While China was slow in adopting the Western-style education system (Peking University was established in 1898), Japan successfully transformed itself from the military government of the Edo (Tokugawa) period (1603-1867) to imperial rule under an emperor, in the period of transition known as the Meiji Restoration of 1867. Japan established nine imperial universities, including two in Seoul (1924) and in Taipei (1928), which were under its colonial rule at that time.

5.2.3 The brain drain problem after the Korean War

After the end of World War II in 1945, Korea was liberated from the Japanese colonial occupation. Tragically, it was immediately divided into two countries, North Korea and South Korea, with the onset of the Cold War, and the political tensions of that period led to the devastating Korean War (1950-53). In the meantime, Japan’s Imperial University in Seoul was transformed into Seoul National University (SNU), whose size was increased greatly by amalgamating several independent colleges and occupational schools. Understandably, SNU, though the best university in South Korea, lacked qualified professors to teach, and ambitious students decided to pursue higher degrees overseas, ultimately tending to settle there. This kind of brain drain was most noticeable in science and engineering, where basic laboratory setups were still in the making in Korea. To reverse the brain drain, the Korean government created a research institute, KIST (Korea Institute of Science and Technology), in 1966 with US aid in a bid to attract home elite Korean researchers from around the world. However, the space for them was limited. As such, an educational system needs to be established to train graduate-level students from inside.

Recognising the problem of sustainable supply of manpower in science and engineering, KIST proposed that a graduate school to be established within itself, but failed to persuade the Ministry of Education, which opposed the foundation of this kind of special university outside its purview.

5.2.4 Against the controversy

Another proposal for establishing a graduate school specialised in science and engineering came from GeunMo Chung, a professor at the Polytechnic Institute of Brooklyn, New York. Chung himself was a prime example of brain drain: after studying physics at Seoul National University, he went to Michigan State University where he earned his doctoral degree. While building his career as a plasma physics researcher, he received an opportunity to study the brain-drain problem in developing countries at the Harvard School of Public Policy in 1967. Two years later, Chung met with John A. Hannah, recently appointed as the administrator of the U.S. Agency for International Development (USAID) by President Richard Nixon. A former president of Michigan State University, Hannah welcomed Chung's idea for a graduate school – an example of real assistance, which is “not to give them fish, but to teach them how to fish” – and encouraged him to submit a concrete proposal.

Therefore, Chung's proposal to solve the brain drain problem in Korea, which was similar to KIST's idea in principle, came via a totally different path from USAID to Korea's Ministry of Economic Planning Board, the top decision-making branch in economic affairs within the government. In short, the establishment of an STI-specialised graduate school became a pressing issue for the nation's economic growth, not just that of education. Furthermore, with full support from President Park Jung-Hee, the proposal for creating the “Korea Advanced Institute of Science” (KAIST) prevailed despite the objections of the Ministry of Education.

The controversy was never solely confined within the government, however. Public debates ensued as some academicians voiced their strong opposition. A professor from Seoul National University, for example, argued that “the establishment of KAIS will hinder the development of universities in Korea”. Yet there were also strong supporters for a graduate

school idea from both overseas and inside the country. Chung also waded into this controversy by reiterating that its goal would be to “supply new knowledge for the industrial sector”, not to “compete with existing schools”.

5.2.5 Imagining the future

USAID played a pivotal role in not only providing financial support and overcoming administrative huddles but also laying out the future plan – by dispatching a high-profile investigative team led by Fredrich Terman, a former provost of Stanford University known as “the father of Silicon Valley” – and keeping this young institution on course by running a U.S. coordinating office to assist it for several years. GeunMo Chung himself was deeply involved in USAID’s activities to the extent that he was the main writer of the first draft of the document known as the “Terman Report”; and he also served as KAIS’s first vice-president in charge of recruiting highly competitive faculty members, mostly from abroad.

[Figure 2-14] Fredric Terman’s investigative team visiting the Ministry of Science and Technology in 1970



Source: KAIST Website⁸

* Terman stands on the front row, fourth from the left; the Korean standing beside him is Minister Kim Ki-hyung; GeunMo Chung is on the front row, far right.

⁸ <https://www.kaist.ac.kr/en/html/kaist/011701.html> (Accessed 18 Dec, 2023)

A few paragraphs towards the end of the Terman Report, submitted in 1971, stated the future vision as follows:

“By the year 2000 KAIS will be intimately identified with the industrial and technological development of Korea. KAIS alumni will be in leadership positions throughout Korean industry and in the Korean government. Concurrently, there will be continuing interplay between KAIS, government, industry, and educational institutions, which works greatly to the benefit of each of these components of Korean life.

Korea itself will have become a prosperous and modern albeit medium-sized nation, with a strong economy and a secure position in international trade. Its domestic prosperity, and its strength in international commerce will be based as much or more on merchandizing the products produced by trained minds as products representing the labor of trained hands.

Above all else, Korea will have become an inspiration and a model of “desire and expectation rolled into one for all other developing nations”.

5.3 Lessons

5.3.1 Building an institutional identity

Starting under the name of KAIS in 1971, this graduate school merged with KIST in 1981 to become KAIST with the hope of creating a synergy between the two premier institutes in South Korea. This “forced marriage” ended in disappointment with KIST becoming an independent entity once again in 1989. Meanwhile, KAIST merged with the Korea Institute of Technology (KIT), an ST-oriented undergraduate school, in 1989 and was relocated to Daejeon that year. Since then, KAIST has been a full-fledged university with undergraduate and graduate programmes, continuously growing and seeking out new possibilities. If “Creativity, Challenge, and Care,” the so-called three C’s, represent KAIST’s institutional identity, this identity is made for both the nation and the world, exactly as intended at the very beginning.

5.3.2 Conclusion: no fixed KAIST model

What lessons can we draw from the case of KAIST? It really depends on who the audience is. What is critically important, however, is to understand that KAIST is a historical product that has evolved alongside the changing sociopolitical circumstances. It is very difficult to pin down exactly what the KAIST model is. Nevertheless, it is fair to say that a great idea may not be enough. One needs to build an alliance, both domestically and internationally, to create a new kind of STI-specialised university and ensure its stable development.

6. Technology Transfer and Promotion

6.1 Definition of Technology Transfer

Technology Transfer (TT) is a multifaceted concept that encompasses a relatively wide range of definitions and interpretations, and which is vital for establishing a common ground of understanding in this field. Initially defined somewhat narrowly by Reisman in 1989, TT is seen as the process of conceiving and implementing novel applications for existing technologies. However, a more comprehensive view suggests that TT should be regarded as an extensive suite of processes. This broader definition includes all aspects pertaining to the origins and adoption of knowledge, experience, and equipment across diverse contexts, such as countries, organisations, and institutions. While it remains challenging to measure and analyse due to its implicit nature in the international trade of goods, services, and processes, and the difficulty in quantifying technology content, TT is nonetheless particularly instrumental in the manufacturing sector. In this context, it involves the utilisation of human, physical, and capital resources, leading to improvements in facilities, equipment, production, management, and marketing. This is achieved by providing the relevant knowledge, skills, information, and assistance related to TT, underlining its significance in advancing technological capabilities and organisational growth.

6.2 Technology Competitiveness in Developing Countries

Regarding technology competitiveness in developed countries, it is important to recognise that a small group of these nations are the primary sources of technological innovations that influence global market trends. In stark contrast, the majority of developing countries are often not actively involved in either innovating or adopting these advances. This disparity highlights a significant gap in the landscape of technology transfer (TT), offering insights into potential strategies to address the shortcomings and deficiencies related to TT among the less technologically advanced nations. Understanding this dynamic is crucial for developing

approaches to enhancing technological competitiveness and fostering greater participation in the global innovation ecosystem.

- 1) The great majority of technologies are imported from the developed countries.
- 2) R&D institutes lack technology infrastructure and testing facilities, and collaborative and co-working practices.
- 3) Technology acquisition is low, and efforts to adopt existing technologies are poor.
- 4) The number of skilled and trained personnel is insufficient to develop new technologies.

Before entering the mainstream, we will first look at the results of a comparison of the competitiveness of countries through the Global Innovation Index GII 2022 in order to gauge Ghana's current position. The Global Innovation Index (GII), an annual publication issued by the World Intellectual Property Organization (WIPO), offers a comprehensive analysis of innovation performance across 132 countries, utilizing 80 distinct indicators. In this index, Ghana was ranked 95th, highlighting specific areas of vulnerability in its innovation landscape. A closer examination of these indicators, as detailed in the 2022 GII Report, can provide valuable insights into the areas on which Ghana may need to concentrate its efforts. Understanding these vulnerabilities is key to formulating strategies for strengthening Ghana's position in future editions of the Global Innovation Index and enhancing its overall innovation capacity.

- Regulatory environment and cost of redundancy of dismissal
- Tertiary education (higher and vocational education) and R&D
- General infrastructure : electricity output
- Credit, especially domestic credit to private sector
- Knowledge-intensive employment
- Patent and high tech imports
- Software spending
- Mobile application creation

[Figure 2-15] Ghana's 2022 GII

GII 2022 rank

Ghana

95

Output rank	Input rank	Income	Region	Population (mn)	GDP, PPP\$ (bn)	GDP per capita, PPP\$
88	105	Lower middle	SSA	31.7	193.6	6,190
Institutions				Score/Value	Rank	
1.1 Political environment				46.9	100	
1.1.1	Political and operational stability*	55.9	79			
1.1.1	Political and operational stability*	65.5	74			
1.1.2	Government effectiveness*	46.4	78			
1.2 Regulatory environment				31.1	128	○ ◇
1.2.1	Regulatory quality*	44.9	73	◆		
1.2.2	Rule of law*	45.1	61	◆		
1.2.3	Cost of redundancy dismissal	49.8	128	○ ◇		
1.3 Business environment				53.5	[51]	
1.3.1	Policies for doing business*	53.5	53	●		
1.3.2	Entrepreneurship policies and culture*	n/a	n/a			
Human capital and research				18.5	101	
2.1 Education				44.5	82	
2.1.1	Expenditure on education, % GDP	○	3.9	82		
2.1.2	Government funding/pupil, secondary, % GDP/cap	○	18.9	59		
2.1.3	School life expectancy, years		12.0	91		
2.1.4	PISA scales in reading, maths and science		n/a	n/a		
2.1.5	Pupil-teacher ratio, secondary		14.5	68		
2.2 Tertiary education				10.7	108	
2.2.1	Tertiary enrolment, % gross		18.7	99		
2.2.2	Graduates in science and engineering, %		15.2	96		
2.2.3	Tertiary inbound mobility, %		1.0	88		
2.3 Research and development (R&D)				0.2	114	
2.3.1	Researchers, FTE/mn pop.	○	89.1	91		
2.3.2	Gross expenditure on R&D, % GDP		n/a	n/a		
2.3.3	Global corporate R&D investors, top 3, mn USD		0.0	38	○ ◇	
2.3.4	QS university ranking, top 3*		0.0	72	○ ◇	
Infrastructure				34.3	96	
3.1 Information and communication technologies (ICTs)				62.3	91	
3.1.1	ICT access*		72.4	93		
3.1.2	ICT use*		50.1	93		
3.1.3	Government's online service*		63.5	80		
3.1.4	E-participation*		63.1	82		
3.2 General infrastructure				16.6	116	
3.2.1	Electricity output, GWh/mn pop.		63.4	104		
3.2.2	Logistics performance*		24.2	97		
3.2.3	Gross capital formation, % GDP		18.6	103		
3.3 Ecological sustainability				24.1	77	
3.3.1	GDP/unit of energy use		15.0	25	● ◆	
3.3.2	Environmental performance*		27.7	124	○	
3.3.3	ISO 14001 environmental certificates/bn PPP\$ GDP		0.4	89		
Market sophistication				16.2	119	
4.1 Credit				1.9	130	○ ◇
4.1.1	Finance for startups and scaleups*		n/a	n/a		
4.1.2	Domestic credit to private sector, % GDP		11.5	125	○	
4.1.3	Loans from microfinance institutions, % GDP		0.1	54		
4.2 Investment				6.2	68	
4.2.1	Market capitalization, % GDP		13.5	69		
4.2.2	Venture capital investors, deals/bn PPP\$ GDP		0.0	55		
4.2.3	Venture capital recipients, deals/bn PPP\$ GDP		0.0	45		
4.2.4	Venture capital received, value, % GDP		0.0	61		
4.3 Trade, diversification, and market scale				40.6	98	
4.3.1	Applied tariff rate, weighted avg., %		10.5	121		
4.3.2	Domestic industry diversification	○	85.7	58		
4.3.3	Domestic market scale, bn PPP\$		193.6	70		
Business sophistication				23.1	88	
5.1 Knowledge workers				22.6	[86]	
5.1.1	Knowledge-intensive employment, %	○	9.6	108		
5.1.2	Firms offering formal training, %	○	40.1	33	●	
5.1.3	GERD performed by business, % GDP		n/a	n/a		
5.1.4	GERD financed by business, %		n/a	n/a		
5.1.5	Females employed w/advanced degrees, %	○	2.9	101		
5.2 Innovation linkages				25.7	52	● ◆
5.2.1	University-industry R&D collaboration*		44.8	62		
5.2.2	State of cluster development and depth*		52.8	42	●	
5.2.3	GERD financed by abroad, % GDP		n/a	n/a		
5.2.4	Joint venture/strategic alliance deals/bn PPP\$ GDP		0.0	79		
5.2.5	Patent families/bn PPP\$ GDP		0.0	101	○ ◇	
5.3 Knowledge absorption				20.9	106	
5.3.1	Intellectual property payments, % total trade	○	0.6	60		
5.3.2	High-tech imports, % total trade	○	2.8	127	○	
5.3.3	ICT services imports, % total trade	○	0.6	109		
5.3.4	FDI net inflows, % GDP		4.3	22	●	
5.3.5	Research talent, % in businesses		n/a	n/a		
Knowledge and technology outputs				11.3	103	
6.1 Knowledge creation				5.8	100	
6.1.1	Patents by origin/bn PPP\$ GDP		0.1	119		
6.1.2	PCT patents by origin/bn PPP\$ GDP		0.0	96		
6.1.3	Utility models by origin/bn PPP\$ GDP	○	0.0	75		
6.1.4	Scientific and technical articles/bn PPP\$ GDP		14.3	66		
6.1.5	Citable documents H-index		8.5	83		
6.2 Knowledge impact				16.0	105	
6.2.1	Labor productivity growth, %		1.8	39	●	
6.2.2	New businesses/th pop. 15-64	○	0.9	84		
6.2.3	Software spending, % GDP		0.0	121	○ ◇	
6.2.4	ISO 9001 quality certificates/bn PPP\$ GDP		0.6	115		
6.2.5	High-tech manufacturing, %	○	11.0	86		
6.3 Knowledge diffusion				12.0	96	
6.3.1	Intellectual property receipts, % total trade	○	0.7	26	● ◆	
6.3.2	Production and export complexity		21.6	98		
6.3.3	High-tech exports, % total trade	○	0.0	126	○	
6.3.4	ICT services exports, % total trade	○	0.4	108		
Creative outputs				16.3	[77]	
7.1 Intangible assets				16.4	[87]	
7.1.1	Intangible asset intensity, top 15, %		n/a	n/a		
7.1.2	Trademarks by origin/bn PPP\$ GDP		4.8	120		
7.1.3	Global brand value, top 5,000, % GDP		n/a	n/a		
7.1.4	Industrial designs by origin/bn PPP\$ GDP		5.2	22	●	
7.2 Creative goods and services				31.8	[22]	
7.2.1	Cultural and creative services exports, % total trade		2.5	7	● ◆	
7.2.2	National feature films/mn pop. 15-69		n/a	n/a		
7.2.3	Entertainment and media market/th pop. 15-69		n/a	n/a		
7.2.4	Printing and other media, % manufacturing	○	1.6	20	●	
7.2.5	Creative goods exports, % total trade	○	0.0	118		
7.3 Online creativity				0.4	113	
7.3.1	Generic top-level domains (TLDs)/th pop. 15-69		0.6	100		
7.3.2	Country-code TLDs/th pop. 15-69		0.0	124		
7.3.3	GitHub commit pushes received/mn pop. 15-69		1.0	101		
7.3.4	Mobile app creation/bn PPP\$ GDP		0.0	113	○	

Source: WIPO(2022)

6.3 Channels of Technology Transfer

Technology Transfer (TT) from developed countries plays a crucial role in bolstering the underdeveloped manufacturing capabilities of developing countries. However, this raises the question as to which are the most effective channels for facilitating TT. In countries with limited capital and technology, TT is significantly influenced by both formal and informal channels. Academically recognised formal channels predominantly include Foreign Direct Investment (FDI), which is usually sourced from advanced countries to developing ones, and technology-specific contractual agreements between developed and developing countries. These formal channels are essential for structured and organised transfers of technology. In contrast, informal channels encompass various methods not classified under the formal routes. Notably, Original Equipment Manufacturing (OEM) and Reverse Engineering are prominent in this category, especially in East Asian countries like Korea and Taiwan, as highlighted in the paper. These informal methods have been recognised as vital pathways for technology acquisition and adaptation in developing countries, offering alternative strategies for technological advancement.

[Table 2-3] Main Mechanisms of Technology Transfer

Form	Channel	Feature
Formal	Foreign Direct Investment (FDI)	Ghana : Resource-seeking FDI
	Foreign Licensing (FL)	
Informal	Original Equipment Manufacturing (OEM)	Korea : OEM and Reverse Engineering are predominant
	Reverse Engineering (Duplicative Imitation)	
	Technology Spillovers	

Source: Oh(2023)

In characterizing the main mechanisms of Technology Transfer adopted in Korea, particularly during its early stages of development, two key aspects emerge as critical:

- 1) **Predominance of Informal Technology Transfer:** During the initial phase of development, informal technology transfer played a far more significant role than formal methods. This type of transfer, often involving duplication or imitation, typically goes unreported due to its inherent nature. Informal methods were not only more prevalent but also crucial in laying the groundwork for technological development.
- 2) **Importance of Spillovers:** A substantial portion of the benefits derived from TT in Korea can be attributed to what are known as 'spillovers' – these include imitation, trade, and the movement of people. These spillovers, often uncompensated, contributed significantly to the transfer of technology, even though they are difficult to measure directly. The impact of these spillovers cannot be overstated as they played a fundamental role in the dissemination of technology across various sectors.

At this juncture, it is important to emphasise that much of the TT in Korea, particularly in the early stages of its industrialisation, was oriented towards original equipment manufacturing (OEM) and reverse engineering.

6.3.1 Technology spillovers

“Technology spillover” is one of the most important forms of indirect technology transfer. Technology spillover can be described as the informal technology market costs incurred by a party that exercises the spillover effect of stimulating technological improvements in a neighbor.

The path of technology propagation shows various effects, such as the front-to-back association effect, external effect, exhibition effect, and technical manpower training. It is generally agreed that there were three types of technology spillover:

- 1) **Demonstration effects** occur when local firms acquire knowledge or technology from foreign firms through imitation or reverse engineering.

- 2) Labor turnover occurs when former employees of a transnational corporation (TNC) are employed by local enterprises or establish their own firms, that is, through the physical movement of people within the local economy.
- 3) Vertical Linkages occur when local firms are engaged as suppliers in the TNC value chain. The requirement of meeting the necessary technical specifications and quality standards means that these local firms frequently need to improve their products and services, including by adopting improved technology and enhanced practices. Vertical linkages can also move from foreign suppliers to domestic buyers (forward linkages), whereby local capacities and competencies are developed to handle a technological advance.

Many scholars and review reports have confirmed that these informal channels of TS played an important role in stimulating TT in the early stage of industrialisation of developing countries.

In most developing countries, even policymakers involved in setting technology priorities are interested in advanced technologies, such as artificial intelligence and IT, which are currently leading the world. Indeed, most of them are trying to develop and secure such technologies in a short time.

One may also ask whether leapfrogging one's way to industrialisation is possible in the developing countries. A study by the World Bank and another report indicate that countries that fail to adopt old technologies have a disadvantage when they encounter new ones. In his book *The Third Wave*, Elvin Toffler agrees that an agricultural society cannot evolve into an information society without going through an industrial stage. This is because technologies are not suddenly generated but rather are similar to life forms in that they must pass through various growth stages, such as specific infancy, growth, and adulthood.

Therefore, it is important to realize that technologies are like organisms that must pass through a sequence of development processes in order to reach a particular stage. However, one can reduce the required time through a fast chase as a follower.

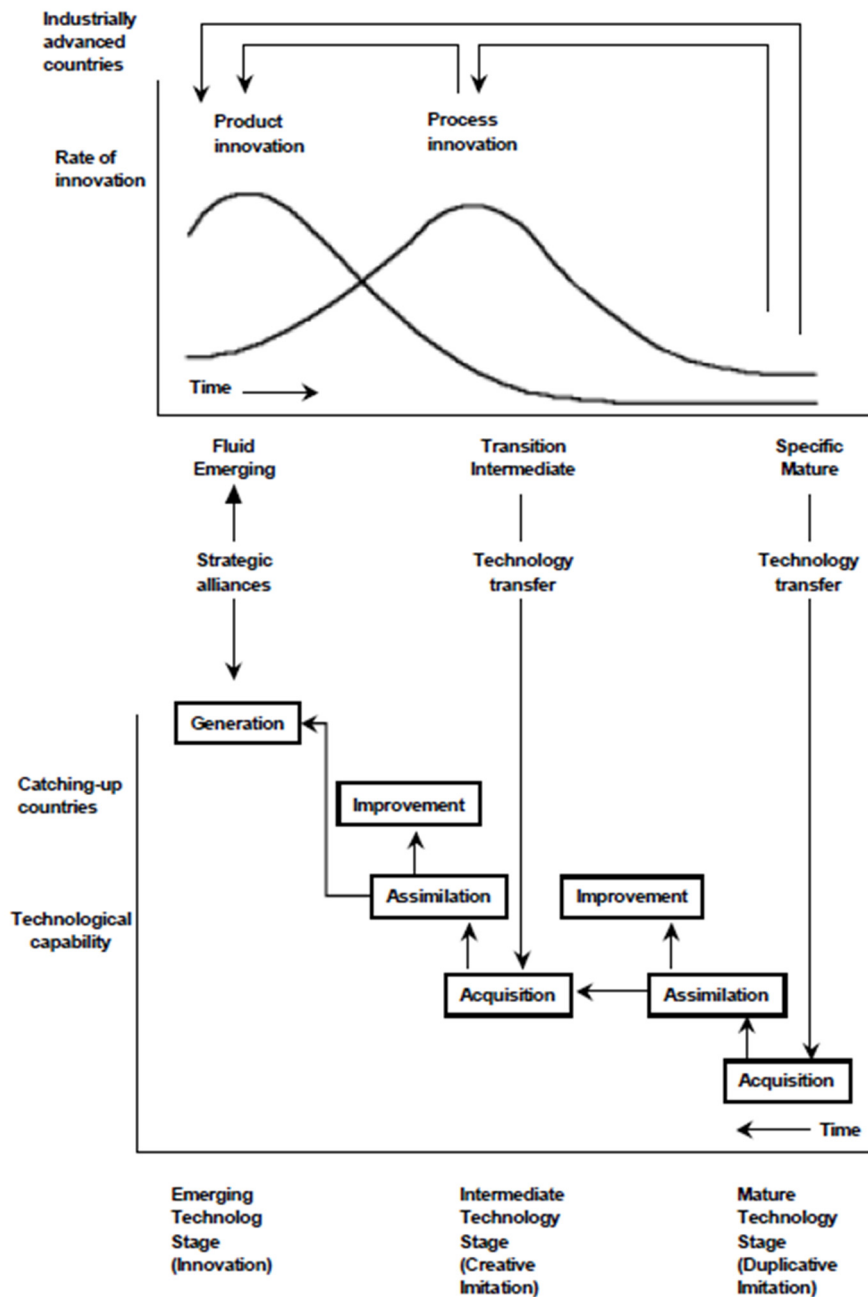
6.3.2 Technology Trajectory

Technology Trajectory refers to the evolutionary direction of technological advances that are observable across industries and sectors. Researchers have classified the development stages of late developing countries as duplicative imitation, creative imitation, and innovative stages. In advanced countries one can observe this kind of technology trajectory across industries and sectors.

In the figure below, the emerging technology stage belongs to innovation, the intermediate technology stage belongs to creative imitation, and the mature technology stage belongs to duplicative imitation. Thus, in certain “catching-up” countries one can observe reverse direction starting from the mature technology of advanced countries through imitation. Catching-up (developing) countries generally acquire mature foreign technologies and accumulate their own absorptive capability.

These observations clearly indicate that all the technologies to be transferred are neither high nor advanced in the developing countries.

[Figure 2-16] Stages of Technology Adoption and Development in Catching-Up Countries



Source: Oh(2023)

6.3.3 Features and character of technology transfer

The initiation of TT should be designed according to a clear realisation of the features and character of technology transfer.

- 1) It is essential to recognise that the adoption of technology is a slow process requiring sustainable development.
- 2) Specific and practical methodologies and tools are needed to promote adoption.
- 3) There is no single, ubiquitous approach. Therefore, prioritized initiatives are required depending on their needs and status.

Most developing countries are interested in adopting a policy of incentives to promote FDI, while the scope of the incentives must be extended further to include diverse policies in all areas with an incentivising nature in order to speed up and facilitate TT activities.

There are various well-known promotional measures for driving technology transfer. In this sense, incentives must be extended and changed to promoting measures in order to drive TT through a range of multilateral means.

- Incentives aimed at encouraging investment
- Reduced risk: fixed tariffs, stable exchange rate, guarantees, etc.
- More exchanges of information
- Alignment of interests in the market chain: public, enterprise, customs
- Clear, consistent, long-term market signals
- Entrepreneurship and education
- Integrated planning

Generally speaking, incentives are an effective measure for attracting FDI, yet there is also the question as to whether incentives are the best policy for attracting investment in the developing countries. It is generally known that the incentives used in the developing countries include working in business with large-scale investment, accelerating the

depreciation of fixed assets, cash-supporting incentives, and land price aid or real-estate supporting incentives.

On the other hand, business environments are more important for small-scale investment, as was mainly the case in the developing countries. This means that the business environment, including business constraints, market size, and labour potential, are more attractive than the specific incentives in the developing countries.

6.4 Korea's Experience of Technology Transfer

6.4.1 Economic & Industrial Policy

The direction of the macro-strategies to choose from is very important when establishing economic and industrial policies for a developing country in the early stages of development.

Looking back on the experience of the Republic of Korea, a representative example is as follows. Unlike other developing countries, Korea boldly chose an export-driven, open foreign policy despite its limited and inadequate national resources.

Therefore, during the initial industrialisation period of the 1960s, industrial policies were designated and concentrated on the products of light industries that could be exported at the national level.

1) 1960s :

Export-driven, open-door foreign policy

Policy and support oriented towards light industries

2) 1970s:

Self-supporting economy based on import substitution

Policy and support oriented towards the heavy chemical industry

Following the export-driven policy of the 1960s, a self-supporting economy based on import substitution was pursued in the 1970s, with policy and support centred on the heavy

chemical industry. This was also part of the effort to strengthen the internal quality of exports and develop domestic industries through the localisation of imported raw materials.

6.4.2 Korea's Experience in the early days of industrialisation

One of Korea's most important experiences was the very deliberate enactment of the Act for Promoting Six Key Industries in heavy chemical Industries by the Korean government. Through these promotion laws, Korea was able to strategically provide comprehensive measures and incentives by focusing on specific industries.

The enactment of intensive laws for these specific industries was a very unusual and effective measure almost never seen in other developing countries. Korea's experience in this respect demonstrates the importance of proper national intervention in the early stages of industrialisation of the developing countries to support technology improvements and advances in the selected industries.

The Korean government enacted a series of laws to promote the six selected key industries (1967-1970).

[Table 2-4] Laws Promoting Six Key Industries in Korea

Promotion Laws	Enacted
Law on Machinery Industry Promotion	30 July 1967
Law on Shipbuilding Industry Promotion	30 March 1967
Law on Electronics Industry Promotion	28 January 1969
Law on Steel Industry Promotion	1 January 1970
Law on Petrochemical Industry Promotion	1 January 1970
Law on Non-Iron Steel Refining Promotion	22 January 1970

Source: Oh(2023)

The main provisions of the laws designed to promote specialised industries included several key measures. First, they involved the establishment of a detailed promotion plan to guide and structure industrial growth. Alongside this, there was a provision for the formation of a government fund dedicated to providing long-term, low-interest rate loans, specifically targeted at supporting the six identified industries. Additionally, these laws called for the establishment of a promotional association, coupled with passage of the necessary regulations to govern and facilitate industry development. Finally, the legislation included measures for providing preferred incentives, with the aim of protecting and nurturing the industry concerned. These incentives were crucial for creating an environment conducive to the growth and sustainability of specialised industries, thereby ensuring that they received the support and protection they needed in order to thrive.

6.4.3 Establishment of Government-Funded Institutions

As Korea's industrialisation progressed, Korea started to establish government-funded research institutes backed by intensive government support in order to raise private technologies in its backward industries.

In 1966, South Korea established KIST (Korea Institute of Science and Technology) as its first, highly symbolic, national research institute and as a benchmarked model of Bell Labs in the United States. KIST adopted a strategy aimed at attracting advanced engineers who had settled in a foreign country after receiving a degree from abroad. Meanwhile, the Ministry of Science and Technology (MOST) was established in 1967 to support and control the nation's science and technology for the first time among the government ministries of developing countries. In 1971, the Korea Development Institute (KDI) was established to perform high-level research on establishing and consulting national development policy for the government.

In 1973, Korea enacted the Law on "Support of Specific Research Institutes" to promote research and development in specific industry sectors. Based on that law, sixteen government-funded institutes were established in succession.

6.4.4 Technology Strategy and Policy

Korea and Taiwan have pursued coherent technology strategies for many years, with the main goal of “indigenising” or localising capabilities related to their industrial production.

- 1) Assimilation Policy (1960s and 1970s): Assimilation before invention. This policy stimulated purposeful, experience-based learning using technology initially transferred from abroad, which was directed towards achieving internationally competitive production.
- 2) Invention Policy (1980s): Invention through capacity building. This policy was related to the acquisition of major innovation capabilities and their utilization in pursuit of major innovations, largely in the form of public technology infrastructure creation.

It so happens that these competent and consistent technology policies played a central part in the success of both countries.

6.4.5 Reverse Technology Trajectory

Catching-up (developing) countries acquire foreign mature technology and accumulate their own absorptive capability. The figure below shows the technology trajectory observed in early stages of South Korea’s industrialisation.

Republic of Korea in the 1960s and 1970s initiated a process of duplicative imitation by acquiring, assimilating, and improving from the mature technology stage depending on locally available capital, technology, and manpower.

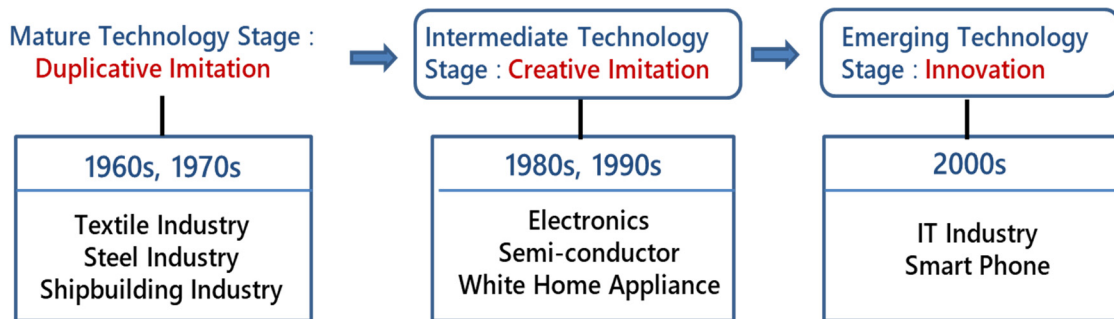
In doing this, Korea learned to indigenise the technology capabilities:

- through assimilation before invention; and
- through modification after experience-based learning.

South Korea progressed to the intermediate technology stage in the 1980s and 1990s through aggressive efforts to indigenise creative imitation.

Finally, Korea succeeded in moving to the emerging technology stage in the 2000s through competitive innovation in the international market.

[Figure 2-17] Technology Trajectory during the Early Stages of Industrialisation in Korea



Source: Oh(2023)

6.4.6 Technology Growth and Production & R&D Patterns

As can be seen from the table below, in Korea, where capital and technology were not available during the industrialisation stage of Korea (1960-1980s), it was possible to establish a strong foundation for economic growth by introducing turnkey plants from advanced countries and for the assembly and production of finished products with imported parts. Most of the turnkey projects linked to the introduction of foreign capital presented few opportunities for training domestic technical manpower, because the technicians already had dedicated engineering design, equipment and raw materials for the construction and operation of plants. In addition, FDI in free export zones made only a very limited contribution to TT.

On the other hand, OEM production of competitive products for export made a significant contribution to reducing the technology gap with developed countries in view of TT. Even though most small scale companies had taken successive steps towards OEM, ODM, and OBM, the Korean conglomerates (*chaebol*) did not follow the same path, but started their own brand businesses from the very beginning. In some respects, the Korean *chaebols* skipped the ODM stage in that they focused on final assembly while outsourcing most intermediate products.

According to the OECD (1996), “Significant changes took place in the late 1980s. Many Korean export industries switched to OBM production and entered the global market with their own brand. Most of these products were low-priced, low-quality standardised products. Korean companies recognized the importance of product differentiation and quality improvement long after they began exporting OBM.”

Building consumer loyalty and global brand awareness requires knowledge-based assets, or design capabilities, which are essential to developing products that can lead the market. Therefore, latecomers must acquire design and product development capabilities. However, it is not easy to gain product differentiation and design ability for innovation - a dilemma for Korean companies. The Korean *chaebol* realised that advanced foreign companies were no longer trying to provide their own schematics, and that they faced a stage of crisis in the dynamic path towards technology development.

In order to create a unique and long-lasting competitive advantage, we had to acquire design capability and realise that only then could we continue to receive orders from multinational companies. However, simply maintaining a domestic network or international subcontracting would not make it possible to easily acquire the required design capability. The Korean *chaebol* overcame this crisis by cross-subsidising huge R&D investments among its affiliates; and when that proved insufficient, a consortium was formed with the government.

[Table 2-5] Technology Development and Production Processes in Korea's Industrial Growth

Period	Base Formulation Period : Socialization (1960s –Mid 1970s)	High-Growth Period : Internalization (Mid 1970s – Mid 1980s)	Switch-Over Period : Externalization (Mid 1980s –Late 1990s)	Re-Launch Period : Consolidation (After 1998 – Present)
Growing Industry	Light Industry	Heavy-chemical, Television Automobile	Electronics, Steel, Semi-conductor Industry	Advanced Industry (IT, BT, NT)
Production & R&D	Assembled Product with Imported Parts : OEM dominant No R&D Activity	OEM/OBM : High Localization Rate of Parts	OEM/ODM : Product Innovation Process Innovation Active R&D	Self-Development of Technology : World New Novelty Leading R&D

Source: Oh(2023)

■ Socialisation Stage (1960s - mid 1970s); Localisation through Joint Ventures

In addition, efforts were made to strengthen the absorption capacity and to improve the technology level of indigenous enterprises that supplied parts by specifying the localisation rate of the parts to produce and a system for foreign capital investment and joint ventures.

In 1967, Hyundai Motor Company started out as a joint venture with Ford of America. Because Hyundai had no technology, it was actually a type of assembly and production (SKD; Semi Knock Down) of Ford's vehicles. Ford's headquarters in the United States appointed Hyundai Motor and a Ford UK subsidiary as its partners, and the Ford Cortina (1598cc), 20M (1985cc medium car), and D-750 truck and bus were selected as the first assembly models. In May 1968, Hyundai began to build a plant capable of producing 3,500 units on 660,000 m² of land in Ulsan.

When the Korean Automobile Industry Protection Act, enacted in 1962, stipulated that more than 21% of automobile parts should be produced by domestic companies, Hyundai appointed over fifty producers in Korea - including battery, electric wire, cooling system components, spring, glass and sheets - in coordination with Ford.

This case shows the positive effects of the proactive legislative measures planned in advance by the government to boost the development of Korea's industries.

■ **Internalization Stage (mid 1970s - mid 1980s)**

From the point of view of technology acquisition in Korea, we can identify the sequence of learning patterns running from the internalization stage to the externalization stage of the pursuit of technology.

1975: Hyundai succeeded in its pursuit of imitation by developing a "Pony" based on a licensing agreement with Mitsubishi.

■ **Externalization Stage (mid 1980s - late 1990s)**

1980: Samsung's Silicon Valley R&D team developed a 256K DRAM superior to those of Japanese companies.

1984: Hyundai developed its own "Alpha" engine from 1984 to 1992 after Mitsubishi refused to relocate its engine design technology.

■ **Consolidation Stage (post 1998 - present)**

This step is a genuine advance towards innovation that involves the creation of new technology development paths or a path-pioneering chase.

1) Autonomous R&D and Private-Public R&D : Public-Private R&D Consortium

First, the CDMA Development R&D Consortium led by ETRI, a government research institute, that developed and commercialised the world's first CDMA wireless communication system is a good example of path creation chase. A second example is the development of digital TV by Samsung and LG through a public-private R&D consortium in 1993. Government institutes like ETRI and KETI played a positive role in the development of TDX, DRAM, CDMA, and Digital TV.

2) Joint Development: Complementary Assets and Absorption Capacity

Joint development began with Qualcomm, a US venture company. In addition, a collaborative venture between GI (General Instrument) and Samsung was also formed as a

result of GI's need for a prototype of digital TV to be developed by its partners, specifically hardware for GI's R&D activities. Horizontal cooperation with companies in developed countries is possible only when the capabilities of second-tier companies are bound in a complementary relationship with that of the first-tier companies.

For large or risky R&D projects or for projects requiring advanced design skills, various forms of R&D consortium have served as an effective means of technology catch-up. The successful experience of the public-private R&D consortium in Korea, ranging from TDX and DRAM to CDMA and digital TV, illustrates the positive role played by government and government research institutes in the technology chase of latecomers.

6.4.7 Technology Transfer Promotion Act

The national law that has laid the legal foundation for technology transfer and commercialisation in Korea is the "Act on Promoting Technology Transfer and Commercialisation.

- Activation of the technology transfer path: Fair transfer of development technology from P to P, guarantee of developer's rights, obligation of technology market opening by local governments, provision of support and incentives.
- Acquisition of resources to promote technology transfer: National budget, public funds, financial support targets and payment regulations for TT.
- Establishment of a technology evaluation system and nurturing of professional personnel: Expansion of the technology market, building up of human resources, implementing agency of Korea technology exchanges.
- Government deliberation and coordination for the Annual TT Promotion Plan: Private transfer of public technology, promotion of private technology, promotion of investment in technology and commercialisation.

Among the other relevant laws were the Special Act for the Development of Special R&D Zones, the Special Act on the Promotion of Venture Companies, and the Act on the Promotion of Industrial Education and Industry-Academia Cooperation.

6.5 Features of Korean Experience on Technology Transfer

After the Korean War (1950-53), South Korea started out on the path to industrial development with virtually no industrial base, such as infrastructures and technologies, few natural resources, and scarcely any accumulated national wealth.

Korea had only human resources as a tool to develop the country in the initial stage of industrialisation. Therefore, absorptive capability building turned out to be a very important asset.

A comprehensive analysis of more than ten different sectors in South Korea reveals that informal transfer mechanisms, including reverse engineering, original equipment manufacturing (OEM), and spillovers, have been instrumental in enhancing the country's competitiveness in the international market. These informal methods provided a significant advantage, especially in the early stages of industrial development. Contrasting with a limited reliance on formal transfer mechanisms, which can be likened to an all-court pressing strategy in sports, Korea's approach to technological development was highly selective. This selectivity proved successful due to the dynamic interplay between specialisation in exports and import substitution. Furthermore, this strategy was aligned with the rapid acquisition of technology, making it an essential vehicle for developing technological capabilities and boosting domestic production. The combination of these strategies not only facilitated Korea's technological advancement but also played a vital role in establishing Korea as a significant player in the global market.

6.6 Conclusion

So far, Korea's experience has been examined in terms of TT, which is now recognised as the most important factor in industrial development based on manufacturing.

A developing country that lacks technology and capital should endeavor to raise and nurture native-owned enterprises even amid difficult conditions in which it is wholly dependent on foreign capital investment in order to maintain sustainable development for the future,

while the native-owned enterprises should focus on building their capacity for technology transfer and technology development.

In addition, native-owned enterprises are able to digest the mature technology of developed countries, which means that the stage of technology imitation can be shortened to the stage of their own technology development; and the government should have a long-term policy, without exerting any pressure on private indigenous-owned companies to develop their own designed products. As such, it is necessary to systematically provide predictable but consistent long-term incentivised measures that can be implemented and customised in various ways.

There is a well-known proverb of the Native American Omar: *"It takes all the efforts of the village to raise one child."*

Having recalled the proverb above and similarly emphasised the importance of native-owned enterprises in Ghana, I would now like to finish this presentation. A path-pioneering chase to the innovation stage with Ghana's own domestically designed products in the international market cannot be achieved in Ghana without native-owned enterprises.

Therefore, one could adapt the aforementioned proverb to say that *"It takes all the efforts of the nation to raise a native-owned enterprise equipped with ownership, management rights, and technological capability."*

However, no matter how good the policies and practices proposed in the present situation are, in order to actualize them in Ghana it will be necessary to emphasise the importance of the leaders' will to practice and the strong enforcement power of the government.

7.

Grant Management for R&D Commercialisation

7.1 The Grant Management Process of the National Research Foundation of Korea (NRF)

7.1.1 Brief history of the NRF

The National Research Foundation of Korea(NRF) is the foremost funding agency in Korea, unwaveringly supporting academic research across diverse disciplines, from the humanities and social sciences to the realms of science and engineering. Established in 2009 through the merging of three funding agencies (KOSEF/KRF/KICOS), the NRF boasts a rich legacy spanning more than forty-six years, and has secured its status as the largest funding agency in Korea.

The NRF's noble mission is to propel the advancement of knowledge and enhance the quality of life in Korea by nurturing creative research and supporting talented researchers. Its vision extends beyond national borders, as it aspires to emerge as a global leader fostering a robust research ecosystem. Guided by our core values of “Creativity, Openness, Responsibility, and Eminence,” which form the acronym "CORE", the NRF navigates diverse endeavors and engagements with its stakeholders.

At present, a cohort of approximately 700 dedicated full-time employees propels the NRF's mission forwards. The agency includes 21 programme managers in major research fields, the architects of innovative programmes, and about 1,100 members of the Research Board who are intricately involved in recommending project evaluation reviewers.

Looking at the financial landscape, the fiscal year 2023 beckons with a budget of approximately US \$7.7 billion. This allocation encompasses all key domains, with a substantial 40% directed towards academic research and university funding, followed by 27% for basic research in science and engineering. The remaining sum will be used to support

national strategic R&D programmes, emblematic of the NRF's monumental standing as the world's second-largest funding agency, trailing only the NSF's budget for 2022.⁹

7.1.2 Grant Management Process

R&D activities are categorized into basic research, fundamental R&D, and business. Basic research pursues creative research, mostly bottom-up based, with the focus on TRL 3 to 5; whereas, in the case of fundamental R&D, national strategic R&D programmes are implemented with top-down and middle-up approaches with the focus on TRL 5 to 6. The results of research are papers and patents fall into the domain of basic research; original patents, technology transfers, pilot products, and licensing into the domain of fundamental R&D; and technology commercialisation, royalties, and job creation into the domain of business.

7.1.3 R&D Planning

As regards the grant management procedure, the first step is the establishment of an implementation plan by the government and the NRF. The next steps consists in the planning of new programmes by the NRF and experts, while the next concerns the proposal and selection of projects, cooperative agreements between the government and the NRF, and between the NRF and research institutions. After this, the NRF provides grants to the selected research institutions, and the R&D activities start. During the project, monitoring and evaluation of its stages and progress are conducted by the NRF and experts. The evaluation of R&D results and performance management are handled by the NRF and the government/NRF, respectively.

To establish the implementation plan, policy research and analysis of recent trends are necessary. After that, the draft implementation plan will be prepared through discussions with MSIT and MOE. As for the selection of the target programme of planning, a technology demand survey will be conducted, and open suggestions by experts and people will be

⁹ https://www.nrf.re.kr/cms/page/main?menu_no=98 (Accessed 18 Dec 2023)

included. Finally, to carry out the R&D planning, a feasibility study will be implemented. Preparations for the request for proposals constitute the next step.

7.1.4 Proposal Submission

In the stage of submitting a proposal, a request for proposals (RFP) will be announced at the NRF homepage so that the project managers at universities and institutions will be acknowledged. In order to submit a proposal to the NRF by researchers, they have to first log in to the R&D Information System, select the desired academic research field, upload their proposal and the required documents, and obtain the approval of their respective institutions. Next, the NRF conducts an eligibility check by confirming the researchers' qualifications, checking the requirements of the proposals, and identifying any duplication of the proposals.

7.1.5 Review Process

For the review process, the NRF chooses from the NRF D/B recommended by the Review Board, Core Review Board, and Programme Managers (ad-hoc base). Experts with the appropriate expertise are then selected, taking into account potential conflicts of interest. For the online review, each proposal is reviewed by 35 reviewers in the related field, and the results are monitored by the Review Committee. A blind review is also used for some programmes. For the panel review, each panel is composed of 5-15 reviewers who assess 5-30 applications. They can exchange views from a broad perspective, and decide on the recommendations for funding.

The general evaluation criteria include the following: the creativity and originality of the project; the excellence of the participating researchers; the validity of the project's goal and methodology; justification for the required funding and project duration; the expected outcomes and their impacts; the validity of the self-evaluation results; the necessity of international cooperation; and the impact of university-industry cooperation. Each criterion will be differentially applied to each programme.

Once all the review processes have been completed, the NRF notifies each institution of the results of the review and also posts them on its website. Objections to the review results should be filed within two weeks of their announcement. Principle Investigators (PIs) are required to change the original proposal to reflect the results of the review and to upload a new proposal to the eRND system. The signing of a cooperative agreement between the PI/Institution and the NRF is the final stage before the NRF provides grants to the institutions online.

7.1.6 Project Implementation and Reporting to the NRF

The signing of a cooperative agreement between the PI/Institution and the NRF is the final stage before the NRF allocates grants to the institutions via online means.

The PIs need to follow the national R&D management regulation and research ethics manual. During and after project implementation, they need to upload the research results to the eRND system of the NRF, including papers, patents, technology transfers, and other outputs. The NRF conducts random on-site auditing, or internal auditing is conducted by the universities and institutions themselves.

The PIs must submit annual reports, final reports, and financial reports. They can use grants with an R&D card issued by the NRF. When they need to change a research plan, they can undertake minor corrections themselves or report to the NRF for major corrections that require a change to the cooperative agreement.

7.2 LINC Program

7.2.1 Outline of the LINC Programme

Now, allow me to introduce the Leaders in Industry-University Cooperation, or LINC programme. Established in 2004, this programme has evolved under different names over the years. It comprises two types of programme: Type 1, known as the Advanced University-

Industry Cooperation programme, and Type 2, known as the Tailor-made Department for Industrial Demand programme.

The primary goals of the LINC programme are to foster university-industry cooperation, address employment mismatches between universities and companies, and facilitate the co-development of universities and local industries. Once selected, the LINC programme receives robust support of US\$301 million annually for five years. As of now, 75 universities have been chosen for Type 1, and 59 universities for Type 2.

The programme operates under three key strategies: The first strategy emphasises the expansion of university-industry cooperation (UIC), involving the autonomous concept of UIC, changes in faculty assessment systems, and dual ways of cooperation between universities and industries. Next, the second strategy focuses on diversifying programmes, encompassing knowledge transfer and start-ups, culture-arts-contents-service initiatives, one-stop services, and specialization in local industries. Lastly, the third strategy aims to reinforce sustainability through education aimed at solving social problems, institutional reforms and incentives, and collaboration with small and medium-sized enterprises (SMEs) and local industries.

7.2.2 Characteristics of the LINC Programme

A crucial characteristic of the LINC programme is its emphasis on specialisation in areas closely connected to local industries and businesses. It is closely aligned with the conditions of local industries and the mid-to-long-range development plans of the host universities.

In modeling university-industry cooperation, the programme has reformed the faculty assessment system, such as recruiting more faculty members with UIC expertise. In addition, the Industry-University Cooperation Foundations located at universities serve as hubs for UIC, while start-up education centres and industry field training centres have also been established.

The academic curricula are tailored for local industries, with family companies involved in their development, as well as in the selection of students, the production of educational

materials, and even the dispatch of company staff for teaching purposes. The programme has transformed the university education culture by introducing site-centred education, capstone design, and entrepreneurship lectures. Policy measures support the common utilisation of equipment, exchanges about the experience of cooperation with companies, and frequent meetings with family companies.

Various services are provided to industries under this programme, including the operation of the University-Industry Cooperation Council for the resolution of technical problems, connections to job creation, support for talented personnel, re-training for company personnel, joint research projects, technology transfer, and opportunities for new grant applications.

The LINC programme has two main components: the manpower development programme and the support programme. The former programme includes capstone design, on-the-job training, academic operations cluster education, entrepreneurship education, and creative workspaces and studios, while the latter includes technology development challenges, industry employee education, corporate advisory, joint equipment centres, and industry-academia cooperation councils.

The performance of the LINC programme has been outstanding. The substitution rate of faculty evaluation by UIC has increased from 70% before the programme's introduction to 92.5% afterwards. The numbers of fieldwork students and capstone design students have also increased significantly. The number of academic curricula based on industrial demand, entrepreneurship lectures, family companies, and technology transfer projects has also experienced substantial growth, showcasing the programme's effectiveness.

7.3 BRIDGE Programme

7.3.1 Outline of BRIDGE Programme

The BRIDGE programme, an acronym for “**B**yond **R**esearch and **I**nnovation **D**evelopment for **G**ood **E**nterprise,” has been developed with the objective of enhancing the contributions

of university entrepreneurs, generating additional value from outstanding technologies, and promoting the progression of university technology development. Its primary focus is to respond effectively to industrial demand and establish a virtuous cycle system for technical commercialisation of the results of university research.

The programme aims to expand the practical use of university assets by promoting collaboration between universities, local governments, and companies, and promoting the transfer and commercialisation of medium and large-scale technologies. Strategically, BRIDGE contributes not only to revitalising valuable medium- and large-size technology transfers and strengthening the social role of universities, but also to driving regional specialised industries and supporting university-technology-commercialisation organizations.

The budget for FY 2022 was US\$17.2 million, supporting twenty-four BRIDGE centres. The programme components span several domains. First, the programme supports the discovery and commercialisation of university-owned creative assets, analysis of the technology market environment, and identification of the final demand of technology for enterprises. Second, it assists R&D planning by designing additional directions for promising technologies and commercialisation strategies, including the planning and activity expenses for winning national projects. Third, it supports business feasibility analysis, such as the discovery of customers, environmental analysis, and the overall design of business management. Fourth, it supports the commercialisation and development of prototypes that are judged likely to be put to practical use, covering 30% of the labour costs for dedicated personnel involved in technology transfer and commercialisation.

7.3.2 Commercialisation Process of the BRIDGE Programme

The commercialisation process of the BRIDGE programme starts with the identification of ideas, properties, and technologies, followed by the scanning, evaluation, and screening thereof. Subsequently, IP portfolio design and business modeling lead to technology feasibility studies, the development of prototypes, and the creation of pilot products. The final steps involve technology transfer, start-ups, joint ventures, sales, and potential funding or business transfers.

7.3.3 Performance Indicators of the BRIDGE Programme

The programme's performance indicators - including the development of new prototypes, new start-ups, and IP portfolios, the conclusion of licensing agreements, and the recruitment of specialists in technology commercialisation - have been remarkable.

Two types of evaluations, peer review and panel discussion, are conducted to select a centre. The evaluation criteria for quantity aspects include professional technology transfer, organisational factors (number of personnel for professional TT and technology commercialisation, capability of TT & TC orgs.), and capacity and performance in TT and TC. The quality factors include the vision, the plan, the willingness for TT and TC of the host university, the performance management plan, professional TT and TC organisation, the strategy for R&D commercialisation, and the infrastructure for R&D commercialisation of the host university.

7.3.4 Case Study : Kangwon National University

A case study on Kangwon National University exemplifies the success of the BRIDGE programme. The university's centre boasts well-established organisations, including a start-up support centre and a technology management centre, technology holding companies, a knowledge asset team, and CTO and TT teams.

Through in-depth interviews, 1,234 properties were identified by the patent agents. These underwent a thorough survey based on market analysis, technology evaluation, and business value analysis. The university established nine R&BD study groups in specific fields, maintaining close cooperation with the companies concerned through such activities as operating a technology call centre, technology matching, problem-solving, technology transfer, and close consulting. Noteworthy successes include the commercialisation of automatic CPR equipment, a bio-material to improve sarcopenia, and potato milk, achieving large-scale technology transfers worth US\$2 million in 2018 and US\$8 million in 2022.

7.4 Policy Implications

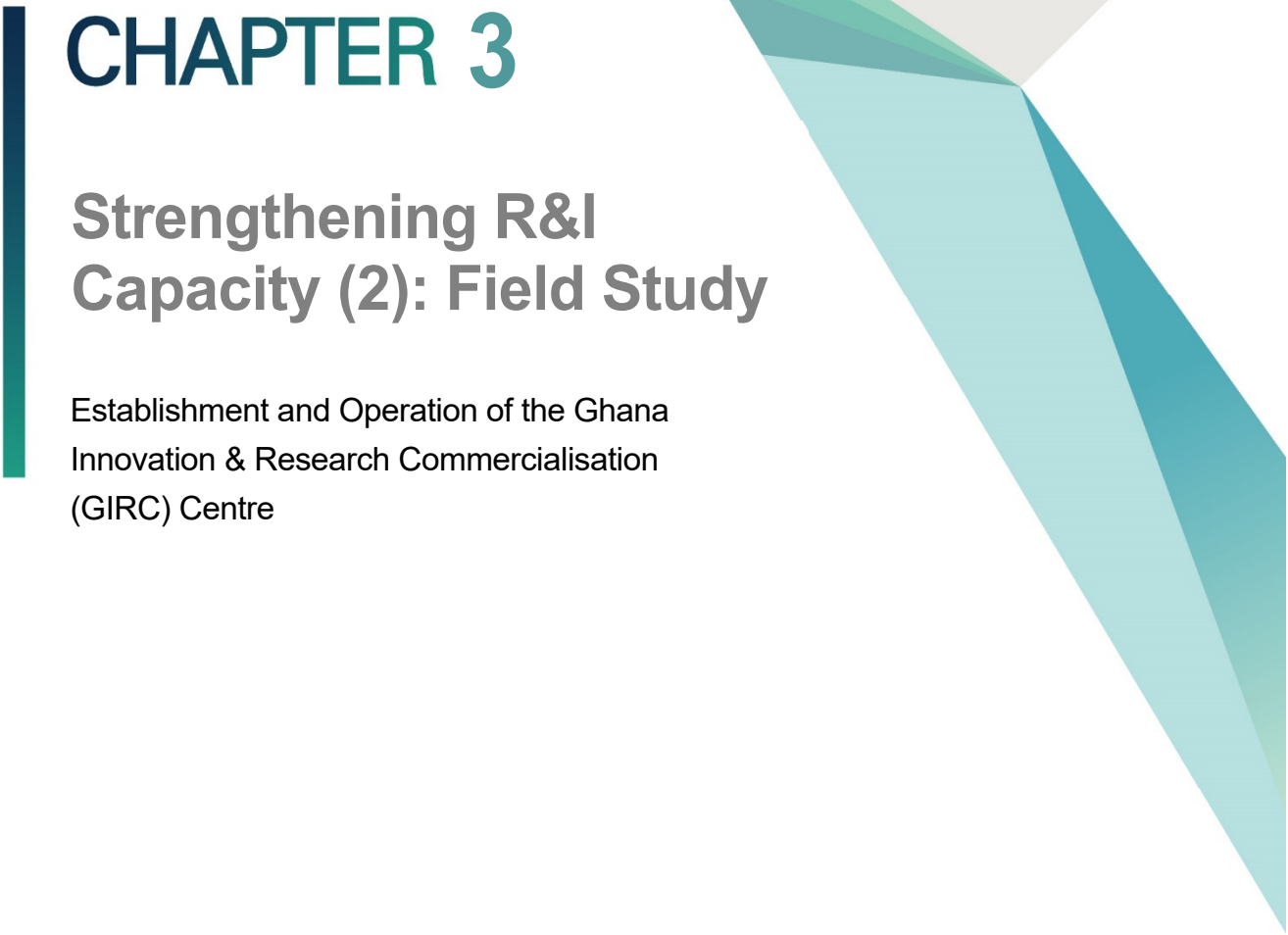
This section aims to delineate the policy implications associated with the grant management system of the National Research Foundation of Korea (NRF). The entire grant management process is a result of Korea's long history of research and development (R&D), management culture, and practices. The planning of the R&D programme is intricately linked to the national R&D agenda, influencing the programme implementation plan and, fundamentally, R&D projects. Despite the comparatively short timeframe from analysing new research trends to producing R&D programmes, i.e. 16 months for large-scale programmes and 10 months for small-scale ones, efficient program management, including a peer review period of 2-6 months, has been established. Importantly, an online R&D management system was developed early on and is now shared by all innovation actors.

Turning now to the R&D commercialisation programmes, significant policy mechanisms have evolved over time, incorporating different aspects within the legal institutional framework and various types of University-Industry Cooperation (UIC) programmes. These mechanisms adapt to the needs of industry, reflecting feedback from academia, industry stakeholders, and societal changes. Stable and long-term support has contributed to establishing a long-term institutional framework for cooperation, accumulating know-how and experience in collaborating with industry, and fostering a co-working system among academic faculty members and the research facilities of host universities. Cultivating specialists for technology commercialisation, establishing Industry-University Cooperation Foundations and Technology Holding Companies, and generating royalty income have all played pivotal roles in the sustainable development of university and industry cooperation.

The LINC and BRIDGE programmes have had a significant impact upon the culture of cooperation between academia and industry, bringing about changes in the academic assessment system and fostering an atmosphere conducive to collaboration with industry within the academic community. The small grants provided for producing pilot products under the BRIDGE programme have acted as attractive incentives for faculty members to engage in collaborative projects with industry and develop more prototypes.

Notably, the connection to local industry has promoted university and industry cooperation by ensuring clear goal-setting to solve the technical problems of companies, developing curricula and education materials, strengthening trust between university and industry, encouraging proactive industry participation, and contributing to job creation.

In conclusion, these policy implications underscore the importance of strategic alignment with the national agenda, efficient programme management, long-term support mechanisms, and evolving policy mechanisms to adapt to the changing landscape of academia, industry, and society.

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CHAPTER 3

Strengthening R&I Capacity (2): Field Study

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre

Chapter 3. Strengthening R&I Capacity (2): Field Study

1. Global Knowledge Exchange and Development Centre(GKEDC)

The Global Knowledge Exchange and Development Centre (GKEDC) serves as a platform for exchanges of knowledge and learning, focusing on South Korea's remarkable socio-economic development since it regained its independence in 1945. It showcases South Korea's transformation from a poor, agricultural country into a major global economic player, while overcoming huge challenges like the Korean War. The centre highlights the contributions of ordinary citizens to this growth, organises exhibitions, runs educational programmes, and provides insights into economic issues including the Fourth Industrial Revolution. The GKEDC also collaborates internationally, sharing its development model with the world and fostering global prosperity and exchanges of knowledge.¹⁰

The GKEDC's permanent exhibition is a detailed walkthrough of Korea's economic development. It starts with a Prologue sketching Korea's seventy-year economic history. The Economic Development Hall narrates Korea's effort to overcome such crises as the Korean War (1950-53) and the foreign exchange crisis of 1997. The exhibition is divided into a series of zones, each one focusing on a different era and a different theme, including post-war reconstruction, rapid industrial growth, correction of economic imbalances, and modern challenges. It covers a variety of topics ranging from the country's initial struggles and reconstruction efforts to the challenges of modernisation and globalisation, offering a

¹⁰ <https://www.gkedic.go.kr/page/1183/detail?menuId=1056> (Accessed 18 Dec 2023)

comprehensive view of Korea's tumultuous economic journey. Meanwhile, the Future Innovation Hall contemplates the Fourth Industrial Revolution while the Industrial Development Hall showcases the global achievements of Korean industry, and the Human Hall celebrates some of the outstanding individuals who have contributed to Korea's growth.¹¹

¹¹ <https://www.gkedc.go.kr/page/1307/detail?menuId=1272> (Accessed 18 Dec 2023)

2. Korea Institute of Science and Technology (KIST)

The Korea Institute of Science and Technology (KIST), established in 1966, has a rich history as South Korea's first science and technology research institute. Over the past fifty years, KIST has been a central axis in the nation's scientific advances and technological development. The visit to KIST offered an opportunity to delve into the institute's evolution from its inception to the present day, while the tour of KIST included a comprehensive overview of its history, starting with its establishment and including details about its geographical location and operational aspects. A significant part of the visit was dedicated to viewing exhibitions that showcased items developed by KIST in the past. These exhibits provided a timeline of the institute's growth, highlighting the major research milestones over the years. KIST's ongoing research and technological developments show that it currently focuses on various specialised fields, including the work of the Brain Science Institute, the Biomedical Engineering Research Institute, and the Green City Technology Research Institute. In the Dr. Choi Hyung-seop Memorial Exhibition Room, Dr. Choi, the first director of KIST and a former Minister of Science and Technology, is introduced as someone who played a crucial role in the establishment of the Daedeok Research Complex and numerous other research institutes. His contributions, together with his definition of KIST as "an institute where the lights never go out," symbolise the enduring spirit of innovation so characteristic of KIST. This could provide unique insights into the journey of KIST from its early days to its current status as a leader in science and technology research in Korea. The institute's ongoing dedication to innovation and its influential role in national development were both clearly evident.¹²

¹² <https://eng.kist.re.kr/eng/aboutus/history.do> (Accessed 18 Dec, 2023)

3. Korea Advanced Institute of Science and Technology (KAIST)

The Korea Advanced Institute of Science and Technology (KAIST) is the most prestigious, leading research-oriented university in South Korea. It was founded in 1971 as South Korea's first research-oriented science and engineering institution, with the aim of fostering the elite human resources in science and engineering needed by the nation. KAIST's main campus is located in Daejeon, 160 kilometres south of Seoul, the capital city of Korea, where its College of Business is located. The university comprises a diverse and comprehensive academic framework, including five colleges, seven schools, thirteen graduate schools, and twenty-seven departments. This structure allows KAIST to offer a wide range of programmes and specialisations, catering to diverse interests and fields within the vast domain of science and technology. The alumni network of KAIST is substantial, totalling some 69,388 individuals. This network is spread across various levels of academic achievement, including 19,457 holders of a Bachelor of Science (BS), 35,513 holders of a Master of Science (MS), and 14,418 holders of a Doctor of Philosophy (PhD). Such a robust alumni community reflects the university's long-standing history and its significant impact on the academic and professional world.

KAIST currently has a student population of 11,059, distributed across different levels of study: 3,665 undergraduate (BS) students, 3,229 Master's (MS) students, 1,396 Doctoral (PhD) students, and 2,769 students enrolled on joint MS/PhD programmes. This diverse student body demonstrates KAIST's role as a comprehensive research university, catering to a broad spectrum of academic interests and research pursuits. The university employs a total of 1,574 faculty and staff members, which includes 666 faculty members and 908 staff members. Reflecting its global reach and international collaboration efforts, KAIST has a notable international population of 137 international faculty members and 929 international students. This international presence not only enriches the cultural diversity on campus but also enhances the global perspectives and learning experiences for each and every member of the KAIST community. Overall, these details paint a picture of KAIST as a dynamic, globally recognised institution with a deep commitment to advancing science and technology through comprehensive education and research.¹³

¹³ <https://www.kaist.ac.kr/en/html/kaist/01.html#0114> (Accessed 18 Dec, 2023)

4. Pohang Iron and Steel Co., Ltd.(POSCO)

POSCO, which celebrated its fiftieth anniversary in 2018, is a testament to South Korea's remarkable journey towards economic development and industrial modernisation. The company emerged from the strong national aspiration to build a better country, undeterred by its limited capital, technology, and steelmaking experience. Founded in the aftermath of the Korean War, POSCO's success is often hailed as a miraculous achievement in the global steel industry. Its founding philosophy was centred on producing high-quality steel to bolster Korea's economic growth and lead its industrial modernisation.

The company's progress is attributed to the innovative vision and devout commitment of its management and employees, including such figures as Honorary Chairman Park Tae-joon. Their unwavering passion and dedication were pivotal in building a strong nation through steel production. POSCO's growth was also supported by its stakeholders, including the government, shareholders, suppliers, customers, partners, and local communities, all of whom have contributed to its fifty-five year journey.

POSCO's half-century history is marked by both challenges and milestones, embodying not just the growth of a single company but also the rapid development of Korea's steel industry, which played a crucial role in the country's industrialisation and modernisation. This history is a tribute to all those who made sacrifices and showed a profound commitment to the company's achievements.

Looking ahead, POSCO aims to build on its fifty-plus years of experience to lead not only the steel industry but also emerging business areas like energy materials. While it still adheres to its founding philosophy of producing high-quality steel to strengthen the country, the company seeks to evolve by focusing on becoming a responsible corporate citizen, pursuing both economic and social values and enhancing the quality of life through its contributions to Korea's economic development.¹⁴

¹⁴ <https://www.posco.co.kr/homepage/docs/eng7/jsp/company/posco/s91a1000019c.jsp> (Accessed 18 Dec, 2023)



CHAPTER 4

Cooperative Partnership with Korea (1): Workshop

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre

Chapter 4. Cooperative Partnership with Korea (1): Workshop

1. Cooperative Partnership Workshop

The inaugural workshop of the cooperative partnership was conducted during the visit of the Science and Technology Policy Institute (STEPI) and the Korean research team to Ghana. Following the initial workshop, Ghana's Ministry of Environment, Science, Technology, and Innovation (MESTI) reviewed the draft outcomes and responded with the "Notes of Intent". This response was aimed at facilitating deeper discussions and furthering the collaborative efforts initiated during the first workshop. Further, this step underscored MESTI's commitment to advancing the partnership and exploring more in-depth cooperation between Korea and Ghana in the field of science and technology innovation.

Subsequently, in July 2023, the second workshop was held in an online format. This workshop focused on exploring specific themes and programmes that could foster cooperation between Korea and Ghana, with an emphasis on strengthening the collaborative network for future projects. Key attendees included representatives from Ghana's Ministry of Environment, Science, Technology, and Innovation (MESTI), along with those from other relevant sectors and institutions. This initiative marked a strategic effort to enhance bilateral ties in the realm of science and technology innovation.

2. Potential Project Concept Notes

2.1 [Project Proposal] Testbed for a Smart Factory by TIPA

Concept Note for the TIPA-MESTI Project

Introduction of a test bed for a smart factory in connection with the industrial transformation of Ghana and professional manpower training for its smooth and efficient operation

April 2023



I. Project Background

- The Ghanaian government has announced its long-term national vision, 'Ghana Beyond Aid', based on high value-added manufacturing and service industries in a production- and export-dependent economy, and suggested a strategy for advancing the economy
- Digital technologies are also considered crucial for attaining the "Ghana Beyond Aid Agenda," which emphasises the maximisation of resource/revenue mobilisation, reduction of corruption, investment in agriculture and industrialisation, and efforts to boost productivity
- To accelerate the digital transformation of each industry, the Ministry of Environment, Science, Technology and Innovation (MESTI) has requested the application of Korea's digital transformation paradigm to the country's manufacturing industries.
- TIPA (Korea Technology and Information Promotion Agency for SMEs) has accumulated a great fund of experience in establishing over 30,000 smart factories in South Korea.

II. Project Overview

1. Project Title

- Introduction of a smart factory test bed centre in connection with the industrial transformation of Ghana and professional manpower training for its smooth and efficient operation

2. Project Period

- 2025–2027 (3 years)

3. Project Budget

- TIPA: Operational expenses of facilities (including a smart factory test bed centre etc.), and training programmes such as education, conferences, etc. (TBD)

- MESTI: Provision of space for a centre for the smart factory test bed(if possible), and administrative support for setting up the centre (in kind)

4. Project Objectives

- To enhance the level of digital transformation so as to support the digitalisation of the manufacturing sites of Ghanaian companies
- To expand cooperation and exchanges of policies, technologies, and business related to digital transformation between the two countries
- To revitalise the related industries by expanding the manufacturing base, such as support for smart process improvement of Ghanaian companies

III. Project Activities

1. Smartisation Process Improvement

- TIPA: Dispatch/diagnosis of related experts for smart factory
- MESTI: Finding and diagnosing Ghanaian demand companies
(Finding companies in the Accra Industrial Complex, Ghana, dispatching/diagnosing related experts to the local area, etc.)

2. Establishment of a smart factory test bed centre

- TIPA: Establishment of a centre to support the construction and efficient operation of a smart factory
- MESTI: Provision of a space for the smart factory (Needs to negotiate)

3. Operating of education and training programmes

- TIPA: Conducting local/invited training for smart factory-related policy makers and operating personnel, focusing on theoretical and practical education
- MESTI: Selection of smart factory demand companies and related ministries/institutions officials

IV. Expected Outcomes

1. For Ghana

- Promotion of the digital transformation of manufacturing companies
- Fostering of smart factory experts and strengthening of capabilities

2. For Korea

- Securing information on the digital transformation market in Ghana and opportunities for entry
- Utilisation as a base for establishing a network between Korean and Ghanaian companies

V. Preparation Process

1. Second half of 2023

- TIPA: Design education and training programmes and derive detailed specifications for the construction of a pilot plant.
- MESTI: Conduct a feasibility study for the construction of a smart factory test bed
- TIPA & MESTI: Hold detailed discussions on how to establish and operate the smart factory test bed centre

2. First half of 2024

- TIPA: Submit all official documents including PCP etc. to MOFA, South Korea
- MESTI: Make it done with entire official documents for ODA from MOF, Ghana

3. Second half of 2024

- TIPA: Secure approval of the ODA project via formulation of the government budget and deliberation and resolution by the National Assembly

2.2 [Project Proposal] Support for the foundation of Ghana Digital Manufacturing Centre and Reinforcing Ghana Digital Competency by KTPA

Concept Note for the KTPA-MESTI Project

Support for the foundation of Ghana Digital Manufacturing Centre and Reinforcing Ghana Digital Competency

April 2023



<Contents>

- (Project Title) Support for the foundation of Ghana Digital Manufacturing Centre and Reinforcing Ghana Digital Competency (DRAFT)
- (Project Duration/Budget) 2026-2029 (4 years) / US\$ 7 million (Need to consult with Ghana)
- (Objectives) Establish the Ghana Digital Manufacturing Centre, nurture experts and foster IT businesses in the digital field in order to promote the digital transformation of Ghana's manufacturing industries.

I. Background to and necessity of the project

1. Background

- Manufacturing industry in Ghana accounts for about 24.5% of total GDP, and industrial production is rising at 7.8% due to the government's industrialisation policies.
- However, high production costs are a bad condition for manufacturing, and there is a vicious cycle in which Ghanaian manufacturers' production costs increase further due to their lower price competitiveness compared to imported goods.

2. Necessity

- There is a pressing need to reduce manufacturing production costs through digital manufacturing in order to overcome the vicious cycle of increasing manufacturing costs in Ghana.
- It is also necessary to increase the productivity of Ghana's manufacturing industries and reduce their transaction costs by supporting their digital transformation, as this will contribute significantly to improving their access to global markets while boosting economic growth.
- Due to its lack of digital skills and difficulty in adopting new technologies, Ghana requires support for a soft landing so as to enable its integration into the digital economy through the promotion of ODA projects.

II. Project Overview

1. Project Scope

- Establishment of a master plan for the digital manufacturing support centre
 - (Establishment of medium- and long-term plans) Support the establishment of medium and long-term plans for construction and operation of the Centre
 - (Policy exchange) Support the establishment of a policy-setting system for digital manufacturing in Ghana by operating the centre and holding a policy exchange seminar
 - (Creating a foundation dissemination) Organise and operate a steering committee for the Ghana Digital Manufacturing Centre
 - Construct the infrastructure of the Ghana Digital Manufacturing Centre
 - Carry out remodelling of the Ghana Digital Manufacturing Centre
 - Install equipment including 3D modeling, 3D printing, laser cutter, and SW solutions
 - Competency training for digital manufacturing
 - (Nurture IT experts and IT professional instructors) Nurture and operate mentoring consultants for manufacturing companies so as to facilitate the digitisation of Ghana's manufacturing industries
 - Foster digital manufacturing companies
 - (Support for the digitalisation of manufacturing industry) Support the digitalisation of existing manufacturing industries and implement scale-up and acceleration (growth promotion) programmes
 - (Support for foundation of digital companies) Support the foundation of self-sustaining digital companies by running a start-up selection and fostering programme
- * Digital company: A business that provides IT solutions to improve industrial efficiency by incorporating IoT into production/management, and that helps companies which still use traditional production methods

2. Commitments of two countries by sector

Sector	KTPA	Ghana
Establishment of a master plan for the digital manufacturing support centre	• Survey and analyse the status of Ghana's manufacturing industries and the conditions for digital transformation • Support the establishment of a master plan for the policy and operation of the centre • Establish the centre, organisation, budget, and personnel structure • Establish the operation plan and self-reliance plan	• Organise and deploy dedicated project personnel for project execution • Establish the master plan as a regulation
Infrastructure construction	• Investigate the demand for equipment • Listing of required equipment • Construct buildings for the centre (Remodeling) • Installation of digital equipment	• Provide business sites • Secure a building that can be built as the Ghana Digital Manufacturing Support Centre • Cooperation in administrative work, such as tax exemption support for customs clearance of equipment
Competency building for digital manufacturing	• Investigate demand for digital manufacturing • Design and operate IT technology education and digital manufacturing consultant training programmes	• Select candidates for digital competency building • Co-host a digital competency building programme
Foster digital manufacturing company	• Develop and operate corporate support programs (Mentor-mentee connection for scale-up, holding pitching/IR, networking events and technical exchange events, etc)	• Select beneficiary companies • Co-host a corporate fostering programme
Common	• Share Korea's experience of industrial development • Analysis of local condition in Ghana through the dispatch of Korean Experts	• Cooperation in administrative affairs, such as visa issuance and tax exemption at customs clearance

3. Expected Outcomes

- (Ghana) Support economic growth by promoting digital transformation of Ghana's manufacturing industry
 - (Short-term effect) Improve manufacturing productivity through digitalization of manufacturing companies
 - (Long-term effect) Strengthen the digital manufacturing ecosystem by production of digital professionals
- (Korea) Korean digital companies can get opportunities to expand to **West African market**
 - Create a bridgehead for domestic digital solution companies to expand to West Africa by enhancing mutual understanding and strengthening the foundation for cooperation between Korea and Ghana

4. Preliminary tasks for implementing the project

- (Phase 1) Derive consensus on project (~3rd quarter of 2023)
 - Mutual review and consultation of the project proposal
 - Confirm local environment and conditions for the project
- (Phase 2) Confirm willingness of local recipient institutions to cooperate (~4th quarter of 2023)
 - Review of candidates for Ghana's recipient institutions, and discuss matters related to projects and strengthen networks with Ghanaian officials
 - Selection of ministry and recipient institutions in charge of projects
- (Phase 3) Pre-Feasibility study and completing PCP (2nd~3rd quarter of 2024)
 - Conduct a pre-feasibility study by dispatching Korean experts
 - Discuss the size, the details of the support, and the commitments of two countries for the project
 - Calculate project budget and discuss the duration of project
 - Prepare PCP and consult with diplomatic offices

- (Phase 4) Deliberation by the Ministry of Strategy and Finance of Korea and the Ministry of Foreign Affairs of Korea (4th quarter of 2024 ~ 3rd quarter of 2025)
 - Submit PCP, official document from the recipient institutions to Embassy of the Republic of Korea in Ghana(4th quarter of 2024)
 - Submit the finalized PCP-official letter from the recipient institutions of Ghana to the Ministry of Foreign Affairs (1st quarter of 2025)
 - Deliberation by the Ministry of Strategy and Finance of Korea (2nd~3rd quarter of 2025)
- (Phase 5) Implementing the Project (2026~2029)
 - Execute project with signing the Record of Discussion

2.3 [Project Proposal] Ghana Maker Space for Nurturing Digital Transformation Enterprises by GBTP

Project Concept Note for

Ghana Maker Space for Nurturing Digital Transformation Enterprises

November , 2023



I. Project Background

- The Ghanaian government announced an economic development roadmap for enhancing productivity and competitiveness in the manufacturing industry.
- To accelerate the productivity and competitiveness of each industry, the Ministry of Environment, Science, Technology and Innovation (MESTI) requested the support for capacity building of innovative manufacturing enterprises.
- Korea is one of the countries with the highest level of competitiveness in the field of digital transformation, possessing various cases of digital transformation in government & enterprises and numerous outstanding ICT companies.
- GBTP(Gyeongbuk Technopark) provides various support programs for enhancing innovative technology based enterprises in Ghana.
- Based on the cooperation between specialized government agencies responsible for digital transformation in both countries, GBTP is planned to support establish maker space to nurture digital transformation enterprises .
- To enhance the bilateral cooperation, innovative maker space that plays as a bridge of a technical cooperation between two countries is needed.
- Through collaboration among the national Government, Academia, the Private Sector and a Municipal Government in Ghana, the project will be used to lay the foundation and nurture technological and innovative enterprises in Ghana based on the Public-Private-Partnership model. It will also promote commercialization of research output and innovation through Industry-University-Research (I-U-R) collaboration.

II. Project Overview

1. Project Title

- Ghana Maker Space for Nurturing Digital Transformation Enterprises

2. Project Period

- 2025 – 2028 (4 years)

3. Project Budget

- Korean Side : Operational expenses (including personnel, remodelling, interior design, utilities, equipment, office equipment, etc.), technology education, business networking programs and for operating cooperation programs such as seminars, conferences, and exhibitions; TBD
- Ghana Side: Land and building for establishing centre, Recruitment of training subjects, Provision of office space, and administrative support for customs clearance of supporting equipment and other support for setting up the centre.: in kind

4. Project Objectives

- To foster technological innovative enterprises in Ghana based on the public-private partnership model and strengthening capabilities for sustainable growth of innovative companies
- To expand cooperation and exchange of policies, technologies, and business related to digital transformation between the two countries.
- Support for commercialization through Industry-University-Research Centre-Government network

5. Korean Partner Agency

- Gyeongbuk Technopark (GBTP)

6. Ghanaian Partner Institutions

- MESTI/GIRC-Centre
- KNUST
- Selected Ghanaian private sector institutions from the Association of Ghana industries (AGI)
- Juaben Municipal Assembly (JMA) – Ashanti Region

III. Project Activities

1. Establishment of innovation centre

- Technology training for technology-based enterprises, prototype manufacturing, establishment of test bed facilities and other amenities

1) Space design and remodeling of Innovation Centre

- ① Support for space design and working design for innovation centre construction
- ② Support for Innovation Centre Remodeling

2) Construction of Equipment

- ① Support Equipment for Innovation Centre
- Machine Makerspace : Equipment and facilities support for Machine & Automobile Maker Space
 - Robot Lab : Equipment and facilities support for Robot Lab
 - 3D Printing Lab : Commercialization design using 3D printers, Support equipment to manufacture prototype,
 - IoT Lab : IoT Device Test, Test App-based equipment support, PC, display, CAM for video conference, speaker, camera, software, etc.

- Office equipment : Office furniture, Office computer, Complexer, Shredder, Seminar room, Equipment to construct conference room, etc.
 - ② Support equipment installation and utility construction
 - Support for the construction of optimized facility environment for the operation of constructed equipment
- Support for facilities such as Internet system, electricity, ventilation, dust collector, heating and cooling system, complementary/access control system, substructure, etc.
 - ③ Equipment management training for equipment operation and maintenance
 - Training in using equipment
 - Equipment maintenance and management training, maintenance/management manual establishment

2. Capacity strengthening for innovative Enterprises

1) Support for strengthening technology capabilities of innovative Enterprises

- Equipment operation training: Training in using Robot, Grone, 3D Printing, IoT, Makerspace equipment
 - Training in manufacturing prototype: Training in manufacturing prototype that embody ideas
 - Technology seminar and workshop: Seminar for innovative technology trends and information, Technology workshop
- Support for technology transfer by cooperation of Industry-University-Research Centre: Technology consulting through technology experts of Industry-University-Research Centre cooperation, and technology transfer

2) Training for professional entrepreneur

- Entrepreneurship
- Establishing a commercialization strategy

- IP strategy
- Business model development
- Investment attraction strategy
- Incorporation/Finance/Accounting/Human resources affairs

3. Invitational training program

- Invitational training for operation personnel of innovation centre
- Invitational training for innovative enterprises

4. Support for public-private partnership type commercialization

1) Support for investment attraction programs

- IR Round(Seed, Pre Series A, Series A)
- Support for accelerating program
- Support for demo day and investment attraction

2) Public-private partnership networking event

- Support for B2C, B2B business meet-up
- Support for enterprises networking
- Support for global networking

IV. Expected Outcome

1. For Ghana

- Establishment of Innovation Centre to strengthen capabilities of innovative enterprises
- Strengthening the one-stop support capabilities of operating personnel
- Capability strengthening for enterprises based on innovative technologies
- Support for commercialization through Industry-University-Research Centre-

Government network Promotion of digital transformation for manufacturing companies

2. For Korea

- Securing information on the digital transformation market in Ghana and opportunities for entry
- Exploring new markets and securing new production bases
- Exchange of technical personnel between the two countries and create joint R&D results

V. Process for Preparation

1. Second half of 2023

- GIRC-Centre & GBTP : Detailed discussions on how to establish/operate the centre
- MOU : Between GIRC-Centre and GBTP to proceed new project
- Complete final Project Concept Paper

2. First half of 2024

- Proceed Feasibility Study in Ghana
- Submission Official Request Letter of Ghanaian Government(~until february 2024)

3. Second half of 2024

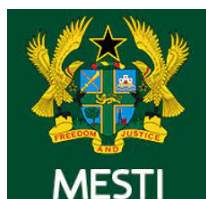
- Budget reviewing of Korean Government

2.4 [Project Proposal] Technologies for Food Processing & Food Technology by JDCC-JBTP

Concept Note for Korea JBTP and Ghana MESTI/GIRC Centre/CSIR Food Research Institute Project

Technologies for Food Processing & Food Technology

May, 2023



I. Project Background

- The main industries of Ghana in Africa are agriculture, mining, petroleum, manufacturing, tourism and services.
- Jeonbuk Technopark has been thinking about projects that can be cooperated among the main industries including agriculture and manufacturing for industrial development in Ghana, Africa.
- When the Korean delegation visited Ghana, Africa last March, Jeonbuk Technopark (JBTP) proposed smart farm, food processing, and AR/VR technology as a cooperative project with Ghana.
- Among the projects proposed by JBTP, MESTI in Africa presented technological cooperation in the food processing industry that fits the industrial conditions of Ghana as the main cooperation project.
- Accordingly, at the request of Ghana in Africa, JBTP intends to propose new detailed projects such as technology development cooperation in the field of food processing and establishment of pilot trial food processing infrastructure.

II. Project Overview

1. Project Title

- Technologies for Food Processing & Food Technology : Pilot factory establishment for food processing production and demonstration support, and R&D cooperation for advancement of food tech technology

2. Project Period

- 2024 – 2028 (5 years)

3. Project Budget

- Minimum 5M USD

4. Project Objectives

- Through the establishment of a multi-purpose pilot plant for food processing support,
- Food processing production/quality/packaging technology support
- AIoT/data-based food processing advancement technology development between 2 nations
- Fostering experts to enhance food processing manufacturing and engineering capabilities

III. Project Activities

1. Establishment of 'food-only pilot plant' to support pilot production of companies

- Equipment and facility support from product development research to prototype production and small-volume consignment production
- Washer, separator, sterilizer, blender, mixer, quick freezing system, multi-purpose food processing system, cutter, shredder, capsule filling machine, stick packaging machine, sealer, etc. are required.



- Facilities are different depending on the type of processed food, such as wet or dry food, and a lot of cost is required to build all facilities. Therefore, it is desirable to build facilities based on the food processing sector favored by SMEs in Ghana.
- Others, sites and buildings, water supply, power supply, communication facilities, etc. must be additionally built.
- The establishment site should be discussed with the Ghanaian government, but it is preferable to be located adjacent to the Ghana Food Research Institute.
- Additional review is needed for building ISO22000 basic facilities for HACCP, GMP and international food safety certification
- Depending on the production item, it can be divided into a general-purpose line and

a specialized line. When building a specialized line, a lot of cost is required.

- The general-purpose line is built with wet and dry zones according to the production process of general processed food
- Establish support services and procedures

Trial production support

- Production by utilizing advanced equipment (e.g. trial product development, process improvement)*
- * Trial production by using the latest and specialized equipment

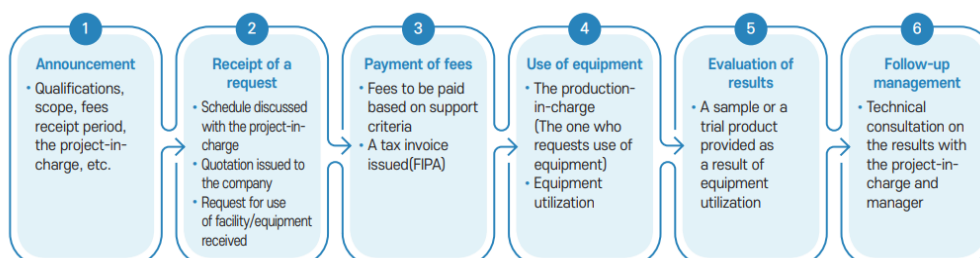
Consignment processing

- Support for consignment small quantity batch production for ingredients or trial products to enter a market*
- * Consignment processing and production of health functional foods can be performed only by GMP-certified companies.

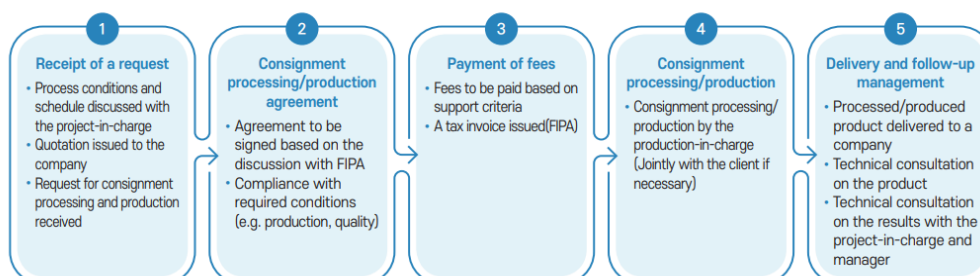
2. AIoT/data-based food processing advancement technology development

- The fields of research cooperation for advancement of food processing technology are as follows.

Procedures(support for a trial product)

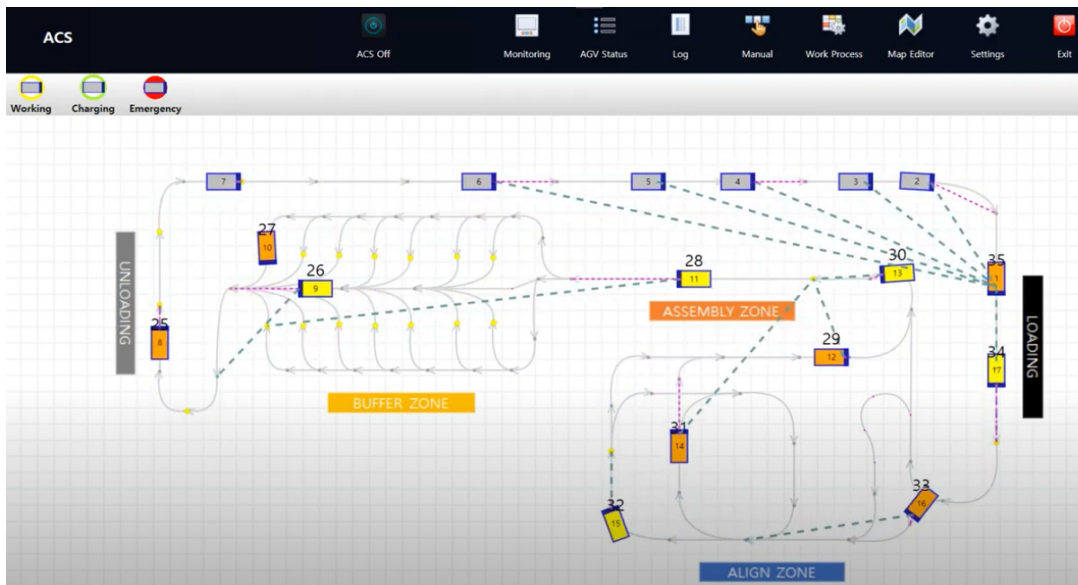


Procedures(support for consignment processing/production)

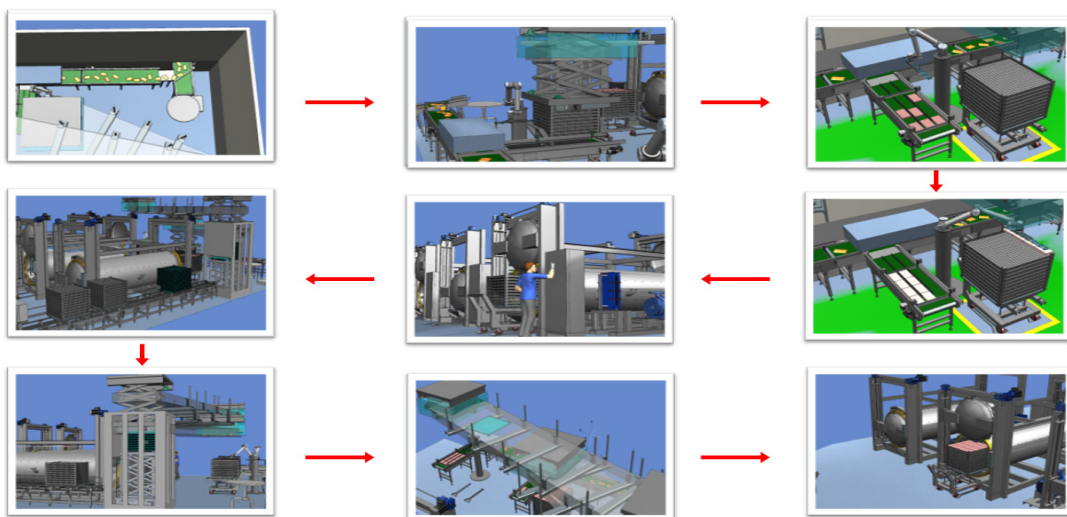


2023 K-Innovation Partnership Program with Ghana

- Development of manufacturing standard process according to processed food target
- Process analysis and system analysis for each process for processed food
- Development of preliminary simulation techniques for manufacturing processes using cobots, etc.



- Development of smart factory manufacturing standard platform



- Smart factory big data system development and data standardization, ML application and AI platform development, CPS (Digital Twin) development and establishment

3. Fostering experts to enhance food processing manufacturing and engineering capabilities

IV. Expected Outcome

1. For Ghana

- Access to abundant and locally sourced raw materials can lead to cost efficiencies and increased competitiveness in the food processing sector.
- By promoting technology development and pilot factory construction, companies can tap into this market potential and cater to the changing consumer preferences for high-quality, convenient, and value-added food products.
- Ghana has taken steps to improve its investment climate, making it attractive for foreign companies to invest in technology development and pilot factory construction.
- By leveraging the local talent pool, companies can build a skilled workforce for the food processing manufacturing sector.
- Ghana serves as a regional hub for technology transfer and knowledge dissemination in West Africa.

2. For Korea

- (Access to New Markets) By partnering with Ghana research institutes or companies, Korean companies can gain access to new markets in Africa. This can help to diversify their customer base and expand their business.
- (Increased Collaboration) Joint research can lead to increased collaboration and cooperation between Korean and Ghanaian researchers and institutions. This can help to strengthen ties between the two countries and lead to further collaborations

in other areas.

- (Knowledge and Skill Transfer) Through joint research, Korean companies can transfer their knowledge and skills in food technology to Ghanaian researchers and companies. This can help to develop a skilled workforce Ghana and contribute to the development of the local food processing industry.

V. Process for Preparation

1. 2nd Half of 2023

- Detailed Survey on food processing business demand in Ghana
- Details of joint research project project plan and completion of RFP
- Discussion on role division by project content
- Prior consultation on project implementation methods and procedures

2. 1st Half of 2024

- Consultation with relevant organizations in the project implementation process, such as MESTI in Ghana, and the Korean Embassy in Indonesia
- Korea's Ministry of Science and ICT announces MESTI cooperation project in Ghana
- Submission and selection process of cooperation project execution plan hosted by JDCC in Korea

May differ slightly depending on the type of cooperative project such as ODA

3. 2nd half of 2024

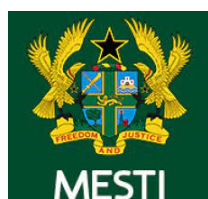
- Promotion of business project agreement
- Determination of detailed work promotion method between Korea and Ghana
- Division of duties between institutions

2.5 [Project Proposal] Cooperation on Grants Management, Innovation Challenge Management and Intellectual Property Rights Management by NRF

Concept Note for the NRF-MESTI Project

Cooperation on Grants Management, Innovation Challenge Management and Intellectual Property Rights Management

November, 2023



I. Project Background

- The Ministry of Environment, Science, Technology and Innovation (MESTI) in the Republic of Ghana has established the Ghana Innovation Research and Commercialisation Centre (GIRC-Centre) in 2023. The major purpose of the GIRC-Centre is to realise the Ghanaian government's vision of driving the nation's socio-economic development based on science, technology and innovation.
- On the occasion of Korean S&T delegation's visit to Ghana from 1-6 April 2023, Dr. Jong-Deok Kim, Education Director of the National Research Foundation (NRF), presented the NRF's ODA programmes to the Ghanaian government officers who attended the workshop held in the MESTI conference room on 4 April 2023.
- Considering the NRF's rich experience of grant management over more than four decades, MESTI suggested collaboration with the NRF in the following three areas: (1) the management of grants, including fund management, (2) the management of innovation challenges, and (3) the management of intellectual property rights.
- The GIRC Centre is a newly established government agency of Ghana which has no experience in managing grants or operating research and innovation programmes. On the other hand, the NRF has more than forty-six years of experience in managing research grants, evaluating projects, planning research programmes, managing project performance, and managing on-line projects, etc. As such, there is a huge potential for collaboration in many areas of grant management and institutional operations between the NRF and the GIRC Centre. Indeed, the experience and know-how of the NRF will be very helpful to the successful operation of the GIRC Centre in Ghana in the future.
- The NRF welcomed MESTI's suggestion of collaboration in the three main areas. More detailed discussions on the scope, methodologies, period and implementation process for cooperation on these issues should follow between the two parties in the near future.
- During the capacity building workshop for the Ghanaian government delegation held in Seoul on 5-6 September 2023, Dr. Jong-Deok Kim gave a presentation on grant

management for R&D commercialisation with the focus on the grant management process, case studies on LINC and BRIDGE programmes, and policy implications.

- On 8 September 2023, the Ghanaian government delegation led by Mr. Tommy Oliver Boachie paid a visit to the NRF HQ in Daejeon City, Korea. The delegation met with Dr. Yung Chul Kim, Secretary-General of the NRF and several NRF officers. Dr. Jong-Deok Kim, Education Director of the NRF, introduced the NRF's role as a funding agency, its current status, its major directorates and their activities, and recent policy issues.. Meanwhile, Mr. Oliver Boachie expressed his sincere gratitude for the warm hospitality shown by the NRF and the dedication and strong commitment to cooperation of Dr. Jong-Deok Kim. Dr. Youngchul Kim welcomed the NRF's cooperation with the Ghanaian government in establishing and managing the GIRC Centre.
- Ghana's government has expressed its willingness of cooperate with the NRF by sending the modified Concept Notes for the NRF-MESTI project: "Through this project, the NRF will conduct training and share its knowledge in order to improve the grant management and operational capability of the GIRC Centre, based on the NRF's wealth of experience and best practices in managing grants, innovation challenges, and intellectual property rights."

II. Project Overview

1. Project Title

- Cooperation on Grant Management, Innovation Challenge Management, and Intellectual Property Rights Management between MESTI/GIRC Centre and the NRF

2. Project Period

- 2024-2026 (3 years)

3. Project Budget

- NRF: The NRF is a funding agency, not a research institution, and cannot act as a principle investigator(PI) of Korean government funding. Therefore, the NRF may

need a third-party organisation like STEPI, which can facilitate the cooperation between Korea and Ghana as a principal investigator(PI), secure the budget for operational expenses, including exchanges of staff and the organising of seminars and workshops (TBD).

- MESTI/GIRC-Centre: Operational expenses for exchanges of staff and the organising of seminars and workshops and the provision of office space (if possible). (TBD)

4. Project Objectives

- To improve the grant management and operational capability of the GIRC Centre by studying the experience and best practices of grant management, innovation challenge administration, and intellectual property rights management of the NRF.
- To achieve the establishment of the GIRC Centre by benchmarking successful models of the R&D management process, ranging from the NRF's planning and implementation of new programmes to performance management.
- To contribute to the self-reliant growth and development of the GIRC Centre as a full-fledged funding agency in Africa within ten years.

III. Project Activities

1. Organisation of staff meetings

- MESTI/GIRC-Centre and NRF: Both parties need to organise a staff meeting to discuss in detail the scope, methodologies, timetable and implementation process of cooperation on the three subject areas.

2. Organising seminars and workshops

- NRF: Composition of experts on the three areas of cooperation among programme managers and staff members.
- MESTI/GIRC Centre: Composition of managers and staff who will attend the seminars and workshops.
- MESTI/GIRC Centre and NRF: Holding of seminars and workshops on grant

management, including fund management, innovation challenge administration, and intellectual property rights management.

3. Consultation on general grant management and operational programmes

- NRF & GIRC Centre: Continuous consulting and information sharing on general grant management and operational programmes between the two organisations.

IV. Expected Outcomes

1. For Ghana

- Acquisition of knowledge and know-how from NRF on management, innovation challenge administration, and intellectual property rights management
- Successful implementation of the operational programmes of the GIRC Centre by establishing the right grants management and R&D administration practices

2. For Korea

- Strengthening of the cooperation between the funding agencies of Korea and Ghana.
- Upgrading of South Korea's brand image in the field of international ODA programmes.

V. Process for Preparation

1. Second half of 2023

- On 8 September 2023, the Ghanaian government delegation visited the HQ of the NRF in Daejeon City, South Korea. Dr. Jong-Deok Kim, Education Director of the NRF, introduced the NRF's role as a funding agency, its current status, its major directorates and their activities, and recent policy issues. According to the demand of MESTI/GIRC Centre and the discussions with the NRF, follow-up activities will be arranged to provide training and knowledge sharing to improve the grant management and operational capability of the GIRC Centre, based on the NRF's experience and best practices in grant management, innovation challenge

adminiation, and intellectual property rights management.

2. First year (2024)

- Organisation of seminars and workshops(May)
- Visit by NRF staff to Ghana for on-site consulting and advisory on the three areas of cooperation (September)

3. Second year (2025)

- Organisation of capacity building workshop(May)
- Visit by NRF staff to Ghana for on-site consulting and advisory on grant management and institutional operation(September)

4. Third year(2026)

- Organisation of seminars and workshops(May)
- Visit by NRF staff to Ghana for on-site consulting and advisory on grant management and institutional operation(September)



CHAPTER 5

Cooperative Partnership with Korea (2): Field Meeting

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre

Chapter 5. Cooperative Partnership with Korea (2): Field Meeting

1. Science and Technology Policy Institute (STEPI)

STEPI, or the Science and Technology Policy Institute, is a South Korean research institute that plays a pivotal role in shaping and advising on the country's science and technology policies. Established in 1987 with the aim of driving national development through science and technology innovation, STEPI conducts in-depth research and analysis of current and emerging trends in the science and technology sector. It serves as a think tank that provides strategic insights and policy recommendations to the government, assisting in the formulation of effective and forward-looking science and technology policies.

The Centre for Science and Technology Policy (CSTP), established under KAIST, was rebranded as STEPI in 1993 and aligned with KIST. It laid the groundwork for science and technology policy research, acting as a resource for government bodies, and analysing global trends and methodologies for national R&D investment assessments. STEPI played a pivotal role in enhancing R&D project planning and evaluation, formulating long-term strategic visions, and fostering an S&T innovation system to position Korea alongside the developed nations. Additionally, STEPI's efforts included research on S&T security, the execution of national S&T projects, and providing educational programmes on S&T policy for developing nations.

In 1999, STEPI was positioned under the National Research Council for Economics and Social Sciences which, in 2005, expanded its scope to include the humanities, thus being rechristened as the National Research Council for Economics, Humanities, and Social Sciences (NRC). With this transition, STEPI honed its focus on science and technology policy,

championing a system of national innovation. It dedicated its efforts to discovering science and technology solutions for societal challenges, cultivated regional innovation frameworks, and pursued stronger global science and technology partnerships.

Steering towards the horizon, STEPI is actively engaged in shaping the future through S&T policy innovation. The institute is delving into comprehensive research on regulatory reform and interdisciplinary studies, fostering an entrepreneurial culture and nurturing the startup landscape. It is broadening partnerships with emerging nations via specialised S&T ODA organisations. Additionally, STEPI is crafting forward-thinking national R&D innovation strategies attuned to the current times of significant change, and is charting future developmental trajectories grounded in Fourth Industrial Revolution technologies.

As of 2023, STEPI's organisational structure comprises three divisions and three centre within its research sector. These include the Division of R&D Strategy Research, the Division of Strategy Research for Future Innovation, and the Division of Global Innovation Strategy Research, as well as the Research Centre for S&T HR Policy, the Korea Space Policy Research Centre, and the Centre for Public R&D Programme Evaluation. Their main roles and responsibilities are as follows.

The Division of Research on R&D Strategy formulates strategic S&T innovation policies with a systemic view, and develops data-driven research infrastructures. The Division of Strategy Research for Future Innovation devises policies for S&T-led growth, supports the emergence of advanced technologies and innovation-centric industries, and promotes the scaling up of innovative companies.

The Division of Research on Global Innovation Strategies investigates cooperative strategies with international bodies on global S&T matters, implements policies for sustainable S&T-driven global development, and undertakes various international S&T cooperative initiatives.

The Centre for Public R&D Programme Evaluation researches and applies methodologies for feasibility studies in state R&D projects, with the aim of enhancing these systems. The Korea Space Policy Research Centre is dedicated to fostering a conducive environment for the

space industry, expanding research and expert networks, and supporting space technology policies.

Finally, the Research Centre for S&T HR Policy focuses on policy innovation for S&T institutions, conducts surveys on S&T graduate programmes and the S&T workforce, and develops strategies for the global circulation of S&T personnel. It also explores the development of S&T vocational guidance programmes aimed at securing and nurturing talents in emerging industries and technologies.

One of the prime projects of the Division of Research on Global Innovation Strategies at STEPI is the K-Innovation Partnership Programme, an ODA initiative established by the Korean Government with the focus on STI development consulting. Since its inception in 2014 and rebranding in 2021, thirty projects across thirteen countries have been executed under this programme.

The K-Innovation Partnership Programme encompasses a wide range of operational projects across various countries, each tailored to the specific S&T needs of the region concerned, including the following: Operational Programmes of the ANAS HTP in Azerbaijan; S&T Planning and Technological Forecasting for Prioritised Sectors in Peru; Policy Consultation on the Design and Implementation of R&D Support Policy for the Improvement of the Science, Technology and Innovation System in Indonesia; Project Consultation and Pilot Implementation for the Development of the National R&D Management System and R&D centre of Excellence in Cambodia; and the Master plan for the Establishment and Operation of the Ghana Innovation & Research Commercialisation Commercialisation Centre, etc.

The achievements of Cambodia are particularly notable across three phases and show the ideal path of the K-Innovation Programme: During the first phase (2018-2019), it provided policy consultation, leading to the draft of Cambodia's "STI Policy 2019-2025" and the establishment of the Ministry of Industry, Science, Technology, and Innovation. This phase also saw the official adoption of the "STI Policy 2020-2030". The second phase (2020-2022) focused on building up capacity and implementing the Technology Roadmap, resulting in

the official adoption of six national technology roadmaps and the inaugural National STI Day. Finally, the ongoing third phase (2023-2025) aims to develop a national R&D management system and centre of excellence in Cambodia, with the expected outcomes including the creation of a roadmap and the establishment of the relevant systems (Kim, 2023).

2. National Research Foundation of Korea (NRF)

The National Research Foundation of Korea (NRF) is a prominent South Korean organisation dedicated to the advancement of knowledge and technology. Established by the government, its primary mission is to support and foster academic research across a range of disciplines. The NRF plays a crucial role in funding and managing research initiatives, ranging from basic science to applied research in fields like engineering, biotechnology, and the humanities. This support extends not just to individual researchers and projects, but also to research institutions and universities across the nation. By promoting and investing in research and development, the NRF aims to enhance South Korea's competitiveness in the global arena. Furthermore, the foundation is instrumental in cultivating a vibrant academic and scientific community, encouraging collaboration among researchers, and fostering the next generation of innovators and scholars. Through these efforts, the National Research Foundation of Korea significantly contributes to the nation's economic growth and technological advancement, consolidating its standing in the global scientific community.¹⁵

¹⁵ https://nrf.re.kr/cms/page/main?menu_no=98 (Accessed 18 Dec, 2023)

3. Korea Technology and Information Promotion Agency for SMEs (TIPA)

The Korea Technology and Information Promotion Agency for SMEs (TIPA) is a specialised agency in South Korea dedicated to supporting small and medium-sized enterprises (SMEs) in the field of technology and information. Its primary objective is to bolster the competitiveness of SMEs by facilitating access to advanced technologies and promoting the adoption of information technology solutions. Established in 2002, TIPA plays a key role in bridging the technological gap between large corporations and smaller businesses, offering various services such as technology transfer, technical consulting, and financial support for technology development. The agency also provides resources for training and skills development, helping SMEs to keep abreast of the latest technological trends and innovations. Through these initiatives, TIPA aims to create a more equitable and dynamic business ecosystem in South Korea, where SMEs can thrive and contribute significantly to the national economy. By empowering these smaller enterprises with cutting-edge technology and information, TIPA is instrumental in fostering innovation and driving sustainable growth in the South Korean business sector.¹⁶

TIPA offers tailored support via two different tracks: Track 1) enterprise-led R&D; and Track 2) collaboration with academia and research institutions based on their level of innovation. During the early stage, Track 1 focuses on companies which have potential growth capacity and assists them with technology ideas, and companies for re-challenge, etc., while Track 2 focuses on the solution of technical difficulties by using the infrastructures of universities and research institutes. In the leading stage, R&D led by enterprises focuses on selecting excellent companies, developing new products, and entering new markets, whereas R&D for collaborating with I-U-R promotes open innovation and commercialisation. In the maturation phase, the emphasis is on pairing private investment with R&D for first-track

¹⁶ <https://tipa.or.kr/> 98 (Accessed 18 Dec, 2023)

companies in order to maintain a competitive edge, while the second track supports global cooperative technological development and the advancement of leading and general-purpose technologies. TIPA guides SMEs through strategic technology roadmaps and provides comprehensive IP-R&D support, including training, the development of strategies, and assistance with commercialisation.

TIPA enhances the ability of SMEs to generate technology patents through IP strategic consulting from the very start of R&D projects. This initiative involves collaboration with IP Institutes such as the Korea Invention Promotion Association (KIPA) and the Korea Intellectual Property and Strategy Agency (KIPSA), offering IP training and consulting tailored to R&D efforts, known as IP-R&D. The IP-R&D support process begins with online pre-training, followed by basic and in-depth training phases, alongside strategy consulting for establishing IP-R&D linkages. This process culminates in the provision of support for actual patent application and registration, ensuring a robust IP foundation for the participating SMEs. In 2022, the IP-R&D programme achieved notable results, including the development of ninety-four technological innovation projects, forty projects for commercialisation through a tech-bridge, support for thirteen promising companies under the government's "Green New Deal" initiative, and the advancement of eight projects aimed at commercialising overseas original technologies.

TIPA's support has made a significant contribution to creating new jobs, boosting sales and export growth, and enhancing the technological advances of SMEs. In the current global landscape - which is marked by rapid advances in science and technology, such as artificial intelligence, big data and cloud computing - the imperative for innovation and the adoption of new technologies is more critical than ever before. TIPA prioritises support for technological innovation and the digital transformation of innovative enterprises and startups. Moreover, TIPA is dedicated to sharing its expertise and experience with developing nations, aiming to foster a global environment of mutual technological advancement and innovation (Cho, 2023).

4.

Gyeongbuk Technopark(GBTP)

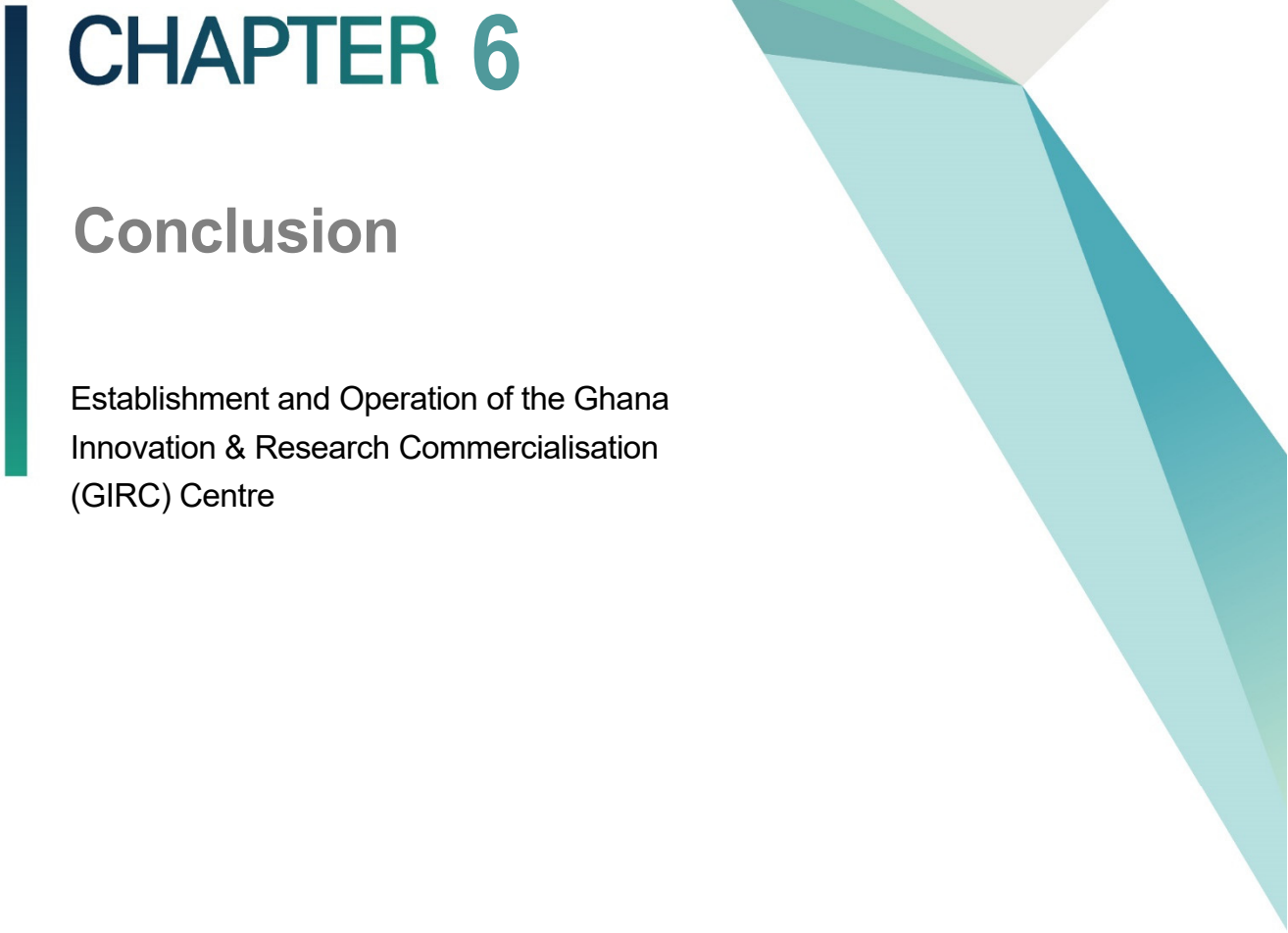
Gyeongbuk Technopark (GBTP), located in Gyeongsangbuk-do province in South Korea, is a dynamic hub for technological innovation and industrial development. It has played a critical role in nurturing small and medium-sized enterprises (SMEs) through collaboration across industry, academia, and government by supporting technology development, building cooperative networks, and establishing and utilising infrastructures among other initiatives. Celebrating its 23rd anniversary, the park has facilitated significant achievements, assisting over 5,700 companies and leading the rise of 39 global small giants and 48 local star companies.

Amidst the evolving global automotive trend towards eco-friendliness and efficiency, the Future Automotive Advanced Materials Molding Centre was inaugurated within Gyeongbuk TechnoPark in 2020. This initiative targets the development of lightweight, high-efficiency automotive materials to meet the increasingly stringent fuel efficiency and environmental regulations. Occupying a state-of-the-art facility covering 3,921.8 square meters, the centre is dedicated to research and testing in advanced materials application.

The establishment of the centre was motivated by the pressing need for the automotive industry to adapt to shifting market demands, including the transition towards environmentally-friendly vehicles and the integration of advanced materials into automotive manufacturing. The centre aims to catalyse a paradigm shift among automotive parts manufacturers towards business restructuring and innovation.

Focusing on body parts, moving parts, chassis, and interiors, the centre is equipped with cutting-edge equipment for testing material properties, durability, and environmental impact. Its strategic goals encompass enhancing equipment operation, diversifying research areas based on corporate demand, and bolstering long-term corporate growth. The centre's commitment is towards fostering exceptional local companies, encouraging company-centric projects, and facilitating the commercialisation of breakthrough products.

The Mobility Centre is equipped with state-of-the-art facilities and infrastructure, enabling extensive testing and refinement of new mobility technologies. Through these efforts, GBTP's Mobility Centre not only contributes to the local economy but also positions Gyeongsangbuk-do as a leader in the future of transportation technology, aligning with South Korea's broader goals of innovation and economic development in the high-tech sector. By partnering with local businesses and leveraging advanced research and development, the centre aspires to drive sustainable progress in the automotive industry, while remaining aligned with global trends and environmental standards (Gyeongbuk Technopark, 2023).

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing right, and below it, several overlapping triangles in shades of teal and light blue, also pointing right.

CHAPTER 6

Conclusion

Establishment and Operation of the Ghana
Innovation & Research Commercialisation
(GIRC) Centre



Chapter 6. Conclusion

This project was carried out as the third-year (2023) activities of the three-year programme titled “Establishment and Operation of the Ghana Innovation & Research Commercialisation Commercialisation (GIRC) Centre”. A key objective of the third-year activities is to explore cooperative partnerships between Korea and Ghana as an important means of strengthening capacities and securing external funding for the successful implementation of the operational programmes specified in the GIRC Centre’s master plan developed in the second year.

The Capacity Development Workshop was held to learn from Korea’s R&I policies as an effort to strengthen the operational capacities of the GIRC Centre. The workshop was soon followed by a Field Study, which provided the Ghanaian participants with opportunities to visit Korean STI agencies. In addition, the Cooperative Partnership Workshop involving five tentative partnership institutions (STEPI, NRF, TIPA, GBTP, and JDCC) was held to discuss future cooperative partnerships between the two countries. Following the workshop, a Field Meeting was held to discuss practical cooperative measures with these potential partners.

During the project, five ‘Project Concept Notes’ were proposed for future cooperation between the five potential Korean STI agencies and their Ghanaian counterparts, and concrete measures for enhancing cooperation between the Korean and Ghanaian partner agencies were discussed. The five project agendas proposed in the process of the project are expected to serve as a foundation for operating the GIRC Centre and building the national innovation system in Ghana.

As aforementioned, to ensure the successful establishment and operation of the GIRC Centre, the following initiatives need to be addressed as the final suggestions of this third-year project, which concludes the three-year programme from 2021 to 2023.

First, at the national level, the governance structure of the GIRC Centre ('GIA' in the future) and the legal systems required to support its governance need to be immediately mapped out and established. For example, decisions need to be made swiftly on whether to establish and operate the GIRC Centre as a subordinate body under a government ministry or as a wholly independent incorporated entity. For this purpose, the so-called 'Stakeholders' Exchange Meeting' needs to be established and held to support the governance-related discussions and implementation. In addition, legislative efforts also need to be exerted to support the operation of the GIRC Centre when it is established as an independent incorporated entity.

Second, the successful operation of the GIRC Centre will also require stable budgetary support once its governance structure has been finalised. This implies the need for efforts to institutionalise funding from the NRF to financially support the establishment and operation of the GIRC Centre, which has already been approved by the National Assembly. For example, political efforts need to be exerted to ensure a stable supply of funds from the NRF for the operation of the GIRC Centre.

Lastly, full-time employees who will practically run the GIRC Centre need to be secured, and their capacities will need to be developed once the centre's governance structure and funding scheme have been secured. Currently, MESTI has secured the physical infrastructure for establishing the GIRC Centre by designating a separate space for the centre within the premises of INSTI and opening the centre's website. While a small group of MESTI employees is currently supporting the establishment of the GIRC Centre, full-time employees, including the director of the centre responsible for its daily operation, have yet to be secured. Therefore, full-time employees including the director need to be immediately recruited and their capacities for building and running the centre's operational programmes

need to be developed and strengthened as soon as the centre's governance system and funding mechanism are finalised.

In parallel, efforts need to be made at the global level to secure global funding for establishing and operating the GIRC Centre. First of all, ODA grants and concessional loans need to be secured through intensive efforts aimed at building close cooperative networks with various international development agencies and banks (e.g. UNDP, AfDB, World Bank) and/or ODA support groups in other countries (e.g. SGCI, DFID, GIZ, KOICA).

Second, efforts need to be exerted to acquire knowledge on establishing and operating the GIRC Centre. For example, the operational know-how of the leading global STI agencies (e.g. IIA in Israel, TIA in South Africa, and NRF in South Korea) needs to be learned through close networking with these institutions. Additionally, partners who can support the successful operation of the centre need to be secured through networking with global U-R-I institutions (e.g. NST and SEWB in Korea).

Lastly, one of the most important factors for the successful operation of the centre is the active exploration of opportunities for cooperative partnerships with the five aforementioned tentative cooperation partners (STEPI, NRF, TIPa, BGTP, and JDCC) identified in the third year of the project. As part of the project, five sets of Project Concept Notes have been proposed and concrete cooperative measures between the centre and its cooperation partners have been discussed. Extensive partnership efforts need to be exerted so as not to nullify the cooperative opportunities identified by the project.

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